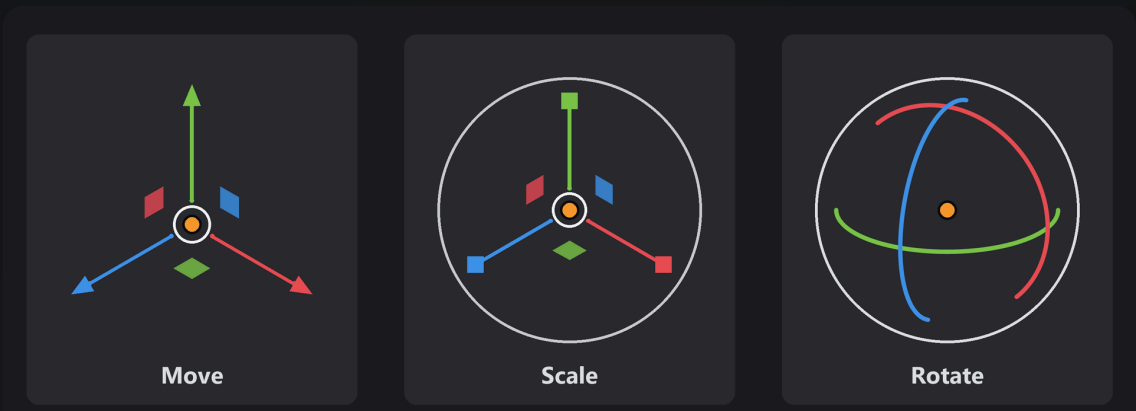
Blendon

Blender-style Scene view, gizmos and pie menus for the Unity Editor



User Manual

Version 1.0.0

Unity 6000.0 (Unity 6) or newer

DRYGIEL

Table of Contents

1 Introduction

1.1 Editor-only by design

1.2 No scripting required

1.3 How to use this manual

2 Requirements

3 Installation and setup

3.1 Step 1 - Import the package

3.2 Step 2 - Choose a keyboard layout

3.3 Step 3 - Open the settings window

3.4 Step 4 - Verify the installation

3.5 Step 5 - Work through the in-Editor tutorial

3.6 Choosing a starting preset

4 Navigation

4.1 Orbit Selected - Middle Mouse

4.2 Quick Roll - Alt + Middle Mouse, flick

4.3 Camera Pan - Shift + Middle Mouse

4.4 Zoom - Scroll Wheel

4.5 Orientation Gizmo

4.6 Numpad Views

5 Transform gizmos

5.1 Shared gizmo settings

5.2 Pivot Point

5.3 Move Tool - W

5.4 Rotate Tool - E

5.5 Scale Tool - R

5.6 Transform Tool - Y

5.7 Grab - G / R / S

5.8 Vertex snapping - V / Shift + V

6 Modal numeric input

7 Scene utilities

7.1 Snap To Floor - End

7.2 Frame Selected - Numpad .

7.3 Isolate View - Numpad /

7.4 Navigation History

7.5 Box Select - Left Mouse drag

8 Pie menus

8.1 Built-in pies

8.2 Draw Mode pie - Z

8.3 Toggle Wireframe - Shift + Z

8.4 Add Object pie - Shift + A

8.5 Snapping pie - Shift + S

8.6 Orientation pie - Comma

8.7 Pivot Point pie - Period

8.8 Reset pie - Slash

8.9 View pie - Backtick

8.10 Tools pie - Q

8.11 Custom pie menus

8.12 Shared pie settings

9 Context menu

9.1 The Editor's own menu

9.2 Context menu settings

10 Scene tutorial

10.1 Shortcut tips

11 Modifier keys

12 Default shortcuts

13 Settings window

13.1 Overview page

13.2 Defaults and overrides

13.3 Keyboard page

13.4 Keyboard presets

14 Data and files

15 Troubleshooting

16 Uninstalling

17 Support

1 Introduction

Blendon rebinds and extends the Unity Scene view so that it behaves like Blender. You orbit around your selection, pan and zoom the way Blender does, replace the Move, Rotate, Scale and Transform gizmos with Blender-style ones, type exact values in the middle of a drag, drag a box that selects everything it touches, reach for utilities such as isolate view and snap-to-floor, open radial pie menus on a single key, and right-click for a searchable menu that still holds every entry Unity's own menu has.

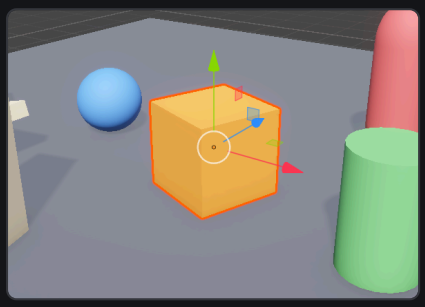


Figure 1.1 The Move gizmo, drawn the Blender way.

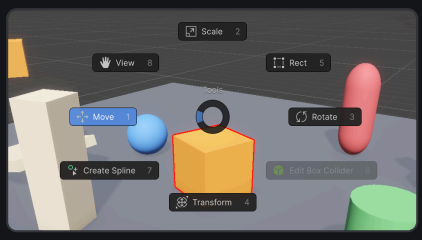


Figure 1.2 The Tools pie menu, opened on Q.

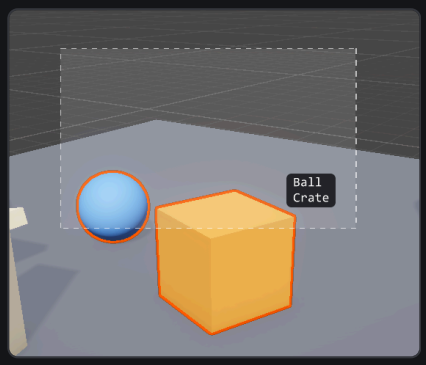


Figure 1.3 Box Select picks everything the box touches.

Every feature is optional and fully configurable from a single settings window. Nothing is forced on you, and nothing is written to your project until you ask for it.

1.1 Editor-only by design

Blendon is an Editor-only extension. It adds nothing to player builds, contains no runtime components, makes no network requests, collects no analytics and installs no additional packages. Its assemblies are compiled for the Editor platform only, so no part of Blendon can end up in a shipped game.

1.2 No scripting required

Blendon is used entirely through the Unity Editor interface: the Scene view, the settings window and Unity's own Shortcut Manager. You never have to write, modify or reference a single line of code to use any feature in this manual, and Blendon exposes no public scripting API for you to call. For that reason this manual contains no script reference and no code samples.

1.3 How to use this manual

If you have just installed Blendon, read Installation and setup and then work through the in-Editor Scene tutorial. Everything after the setup chapter is reference material: each chapter covers one group of features and can be read on its own, in any order.

Throughout this manual, menu paths are written as **Tools -> Blendon**, and keys are written as **Shift + Z**. Where a default binding is given, it can always be changed; see Settings window. Screenshots show the Editor's dark skin with Blendon's default settings unless the caption says otherwise.

2 Requirements

- **Unity 6000.0 (Unity 6) or newer.** Everything introduced after Unity 6.0 (`EntityId` , the separate angle and scale snap toggles, the Grid handle orientation, the movable Scene grid, overlay groups) is routed through a compatibility layer, so Unity 6.0 and the newest release behave identically wherever the Editor itself allows it.
- **Any render pipeline.** Blendon draws through the Editor's own Handles and gizmo systems and does not depend on Built-in, URP or HDRP.
- **Windows, macOS or Linux.** Modifier labels are platform-correct: macOS users see Cmd and Option rather than Ctrl and Alt.
- **A numeric keypad with NumLock enabled** is required only for the optional numpad viewport navigation described in Numpad Views. Every other feature works without one.

3 Installation and setup

This chapter is the step-by-step setup guide. Follow steps 1 to 5 in order; the whole process takes about two minutes.

3.1 Step 1 - Import the package

1. In Unity, open **Window -> Package Manager**.
2. Select **My Assets** from the packages dropdown at the top left.
3. Find **Blendon** in the list, then click **Download** and, once it finishes, **Import**.
4. Leave every file selected in the import dialog and click **Import**.

Blendon works from wherever it lands: directly under `Assets/` , inside a subfolder of your own, or as a package under `Packages/` . Nothing in the plugin references a fixed path, so you are free to move the folder afterwards.

Wait for Unity to finish compiling before continuing. The progress bar in the bottom right corner of the Editor disappears when compilation is done.

3.2 Step 2 - Choose a keyboard layout

On its first load Blendon asks a single question, in a dialog titled **Blendon - keyboard presets**: whether it should apply the Blendon preset, taking the Blender keys (G, R, S and the middle mouse button) and moving a handful of Editor commands one modifier aside. The dialog says how many Editor shortcuts would move before you decide.

Answer	Result
Apply Blendon preset	Blendon keeps its Blender-style bindings. Editor commands that wanted the same keys move to a free binding instead. This is the recommended choice if you are coming from Blender.
Not now	Nothing is written and your Editor is left untouched. Blendon's features are still installed and you can make the choice later.

Nothing is written to disk until you answer. Both keyboard presets stay available on the Overview page of the settings window, shown below: **Blendon**, and **Unity**, where the Editor keeps every key untouched and Blendon's gestures move instead (orbit to **Ctrl + Middle Mouse**, quick roll to **Ctrl + Shift + Middle Mouse**, pan to **Alt + Middle Mouse**, vertex snapping to **Alt + V**, and so on). See Keyboard presets for the full comparison.

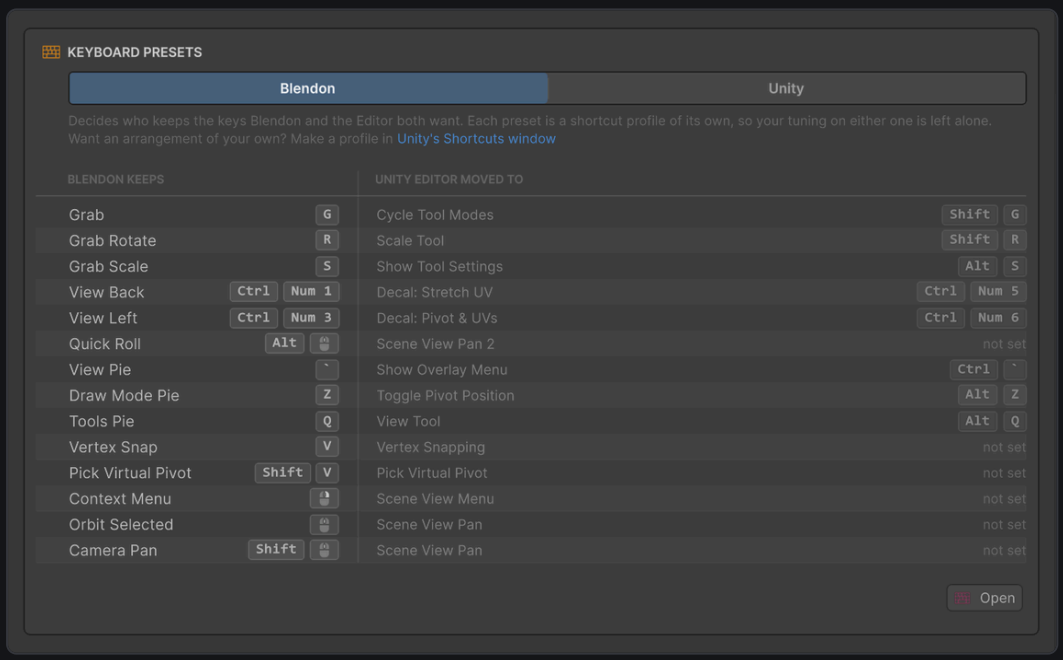


Figure 3.1 The Keyboard Presets panel on the Overview page. The left column lists the keys Blendon keeps, the right column where each displaced Editor command moved to.



Figure 3.2 Blendon preset: Blendon keeps the key, the Editor's command moves one modifier along.

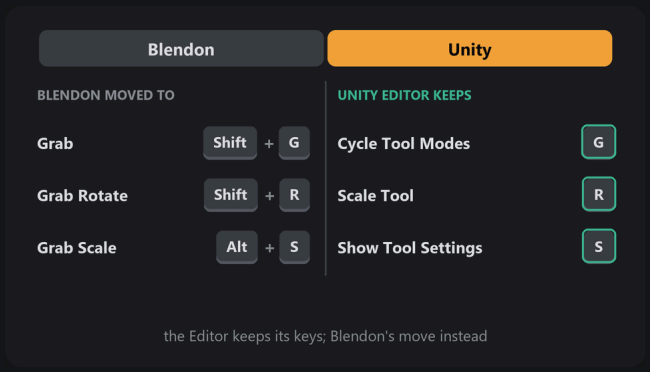


Figure 3.3 Unity preset: the Editor keeps its keys, and Blendon's move instead.

Blendon places its keys on a Unity shortcut profile of its own, so selecting **Default** in **Edit -> Shortcuts** undoes every key change at once.

3.3 Step 3 - Open the settings window

Open **Tools -> Blendon**.

This one window contains every setting Blendon has. The sidebar groups all features into **Get Started**, **Navigating**, **Transform Tools**, **Scene Tools** and **Keys & Menus**. The window opens on the Overview page, which lists every feature with its current key and an on/off toggle.

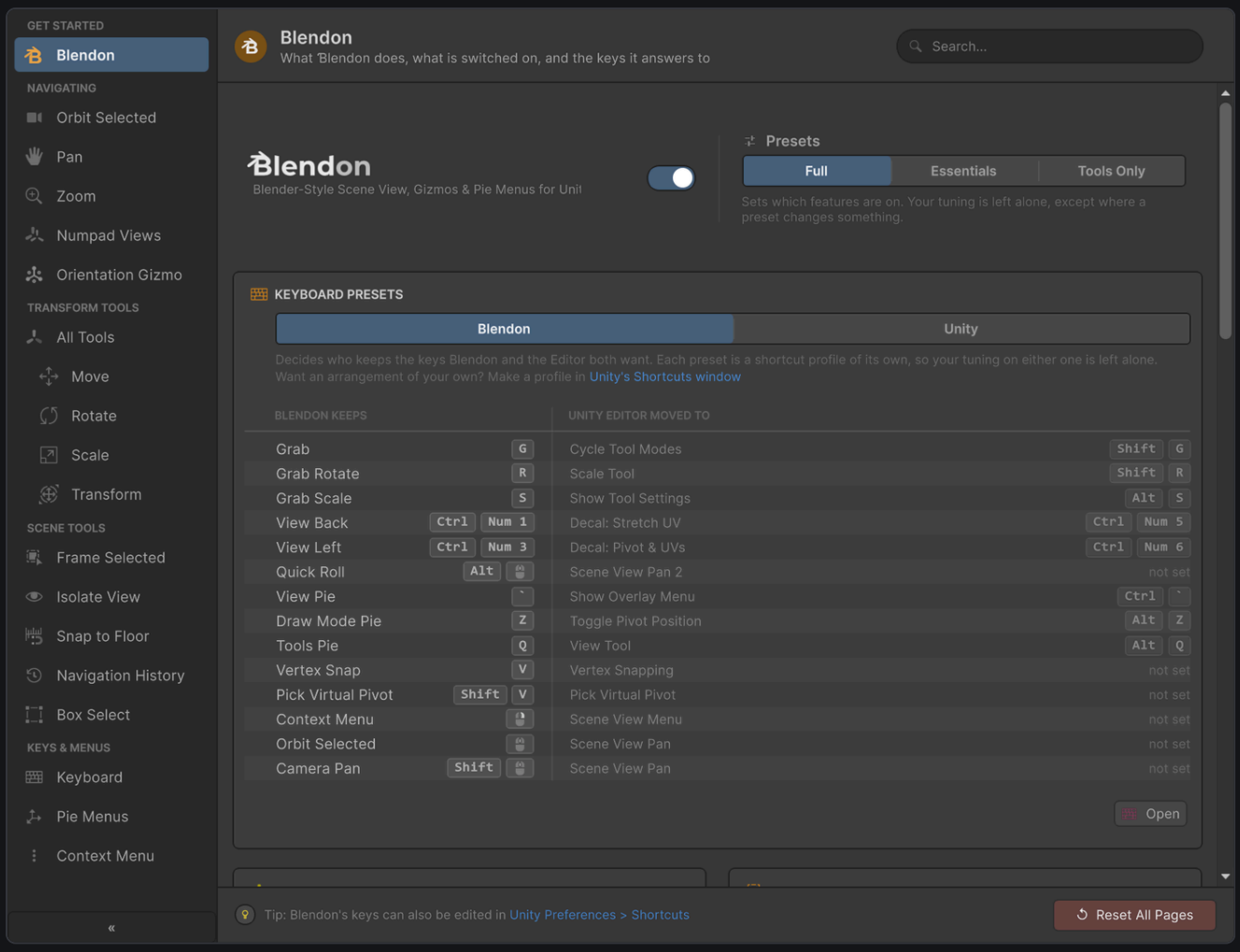


Figure 3.4 The settings window on its Overview page: the sidebar on the left, the master switch and feature presets at the top, and the Keyboard Presets panel below.

3.4 Step 4 - Verify the installation

Open any scene with at least one object in it and confirm the following:

1. **Navigation.** Select an object, then hold the **Middle Mouse Button** in the Scene view and drag. The view should orbit around the selected object rather than around the Scene view's own pivot.
2. **Gizmos.** With the object still selected, press **W**. The Move gizmo should be drawn in the Blender style: three colored axis arrows, plane handles, a white screen-space ring and a center dot, as in Transform gizmos.
3. **Grab.** Press **G** and move the mouse. The object should follow the cursor immediately. Press **Esc** or right-click to cancel, and the object returns to where it started.

If any of these does nothing, open **Tools -> Blendon** and check that the feature's row on the Overview page is switched on and that its key is not highlighted as clashing with a Unity command. See Troubleshooting.

3.5 Step 5 - Work through the in-Editor tutorial

A floating checklist card appears in the Scene view on first use and walks you through Blendon's gestures in six chapters. A task ticks off only once you actually perform the gesture, not by reading about it.

This is the fastest way to learn the plugin, and it is described in full in Scene tutorial. If you dismissed the card, you can restart the tutorial at any time from the Overview page of the settings window.

3.6 Choosing a starting preset

The Overview page offers three presets for a fresh install, under **Presets**:

Preset	What it enables
Full	Everything on, with every value reset to the shipped defaults. Custom pie menus are kept.
Essentials	Navigation and the transform gizmos, with their grab and vertex-snap keys. The numpad views, pie menus, context menu and scene tools stay off, and the Editor keeps the keys they would have taken.
Tools Only	Just the Move, Rotate, Scale and Transform gizmos, with their grab and vertex-snap keys. Everything else stays off, and the Editor keeps every key it would have taken.

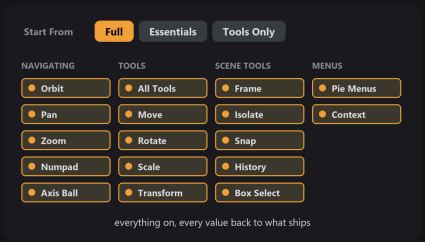


Figure 3.5 Full: every feature on, every value back to what ships.

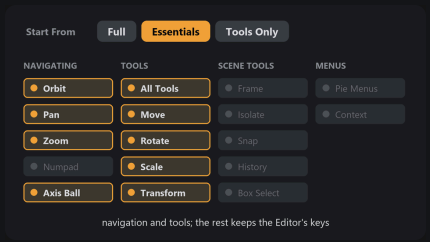


Figure 3.6 Essentials: navigation and the transform gizmos.

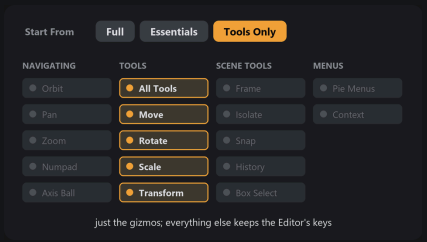


Figure 3.7 Tools Only: just the four gizmos.

Pick **Tools Only** or **Essentials** if you want to adopt Blendon gradually, and switch individual features on from the Overview page as you go. A preset only switches features on and off; your tuning is left alone, except where a preset changes something on purpose.

4 Navigation

Blendon moves the Scene view's camera the way Blender does: around the selection, anchored on the point under the cursor, and without ever leaving the mouse.

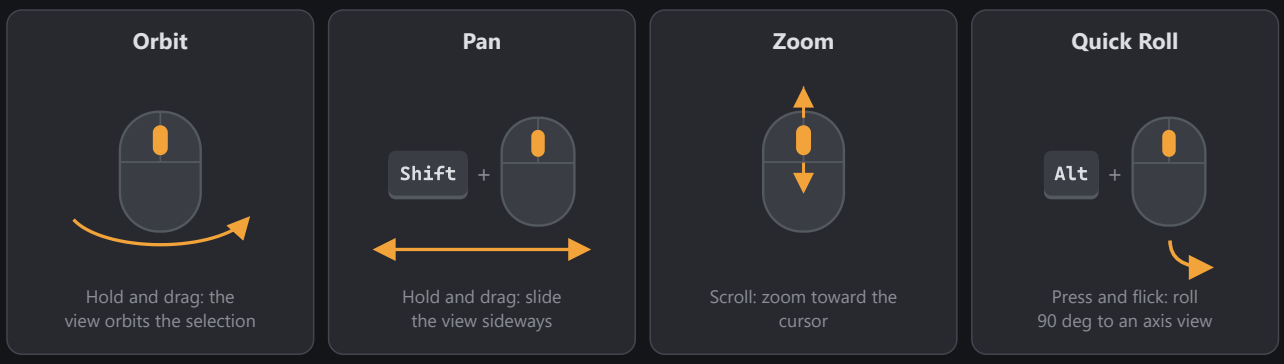


Figure 4.1 The four mouse gestures at a glance. All of them are rebindable, and the Unity keyboard preset moves them to Ctrl, Ctrl + Shift and Alt instead.

4.1 Orbit Selected - Middle Mouse

Hold the **Middle Mouse Button** and drag to orbit the Scene view around the current selection. With multiple objects selected, the view orbits their common center.

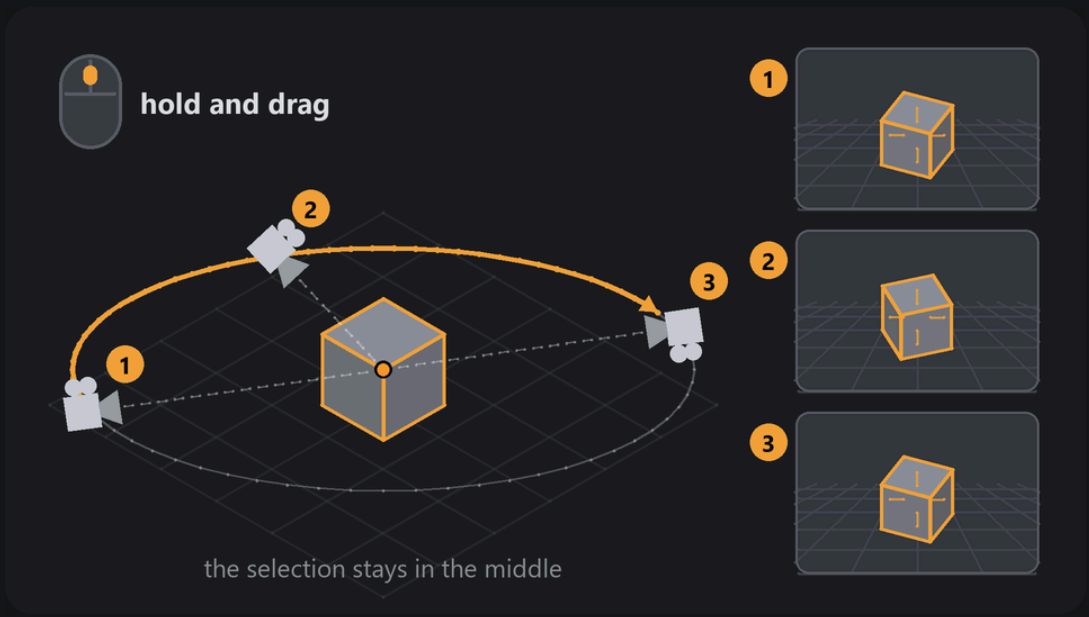


Figure 4.2 Hold the middle mouse button and drag. The camera circles the selection, which stays in the middle of the view.

- Separate horizontal and vertical sensitivity, in degrees per pixel.
- **Depth.** With nothing selected, orbit around the scene point under the cursor instead of the Scene view's own pivot. **Show Pivot Dot** marks that point while you orbit.
- **Snap View to World Axis.** Hold the snap modifier (**Alt** by default) in the middle of an orbit to lock the view to the nearest world axis and switch to orthographic; release it to return to a free perspective orbit. The **Threshold** angle, 12 degrees by default, decides how close the view has to be before it snaps.
- **Right-click or Esc** while dragging cancels the orbit and restores the original view.

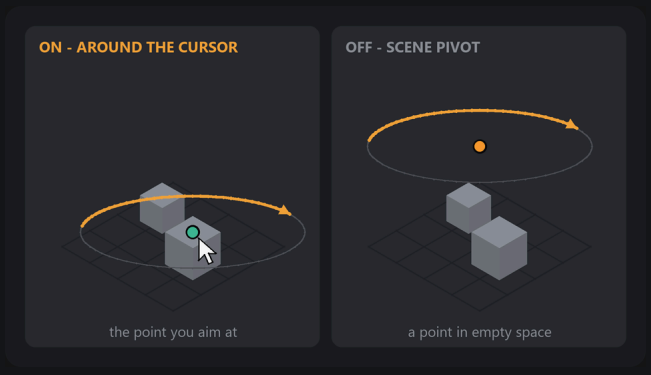


Figure 4.3 Depth: with nothing selected, the view turns around the point you aim at.

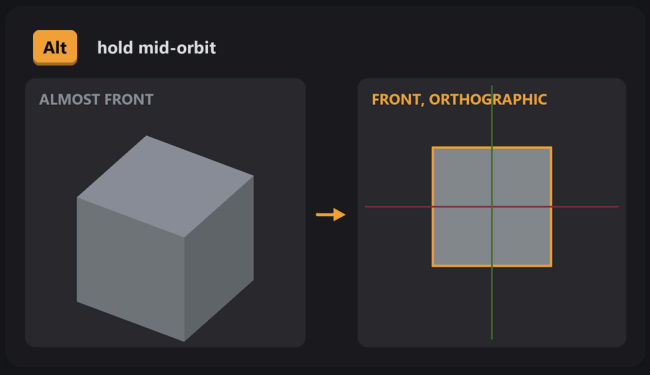


Figure 4.4 Hold Alt mid-orbit and the view snaps to the nearest axis, orthographic.

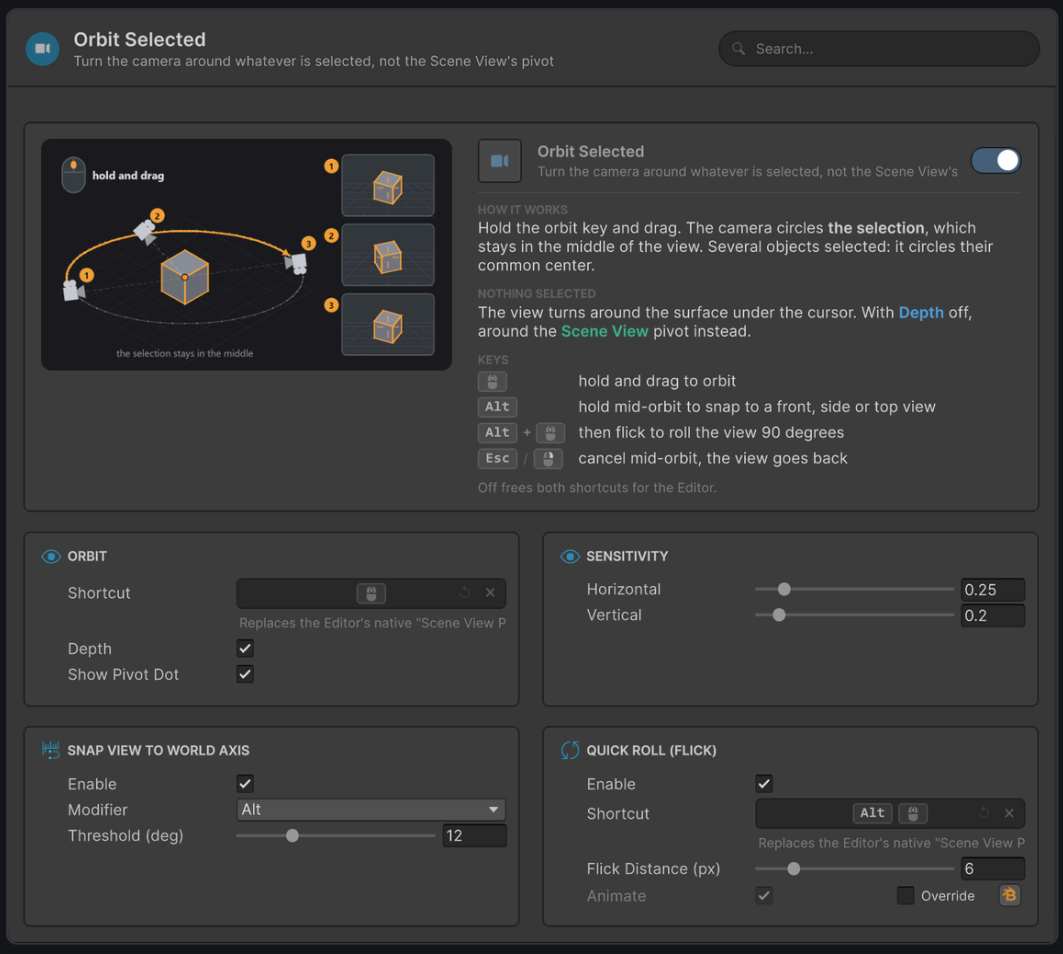


Figure 4.5 The Orbit Selected page. Its header card sums the gesture up, and Quick Roll lives here too, since it is the same middle-mouse gesture.

4.2 Quick Roll - Alt + Middle Mouse, flick

Hold **Alt**, press the **Middle Mouse Button**, then flick a short distance in any direction to roll the view 90 degrees that way and drop into orthographic. The view lands squarely on the nearest named axis view regardless of where the flick started.

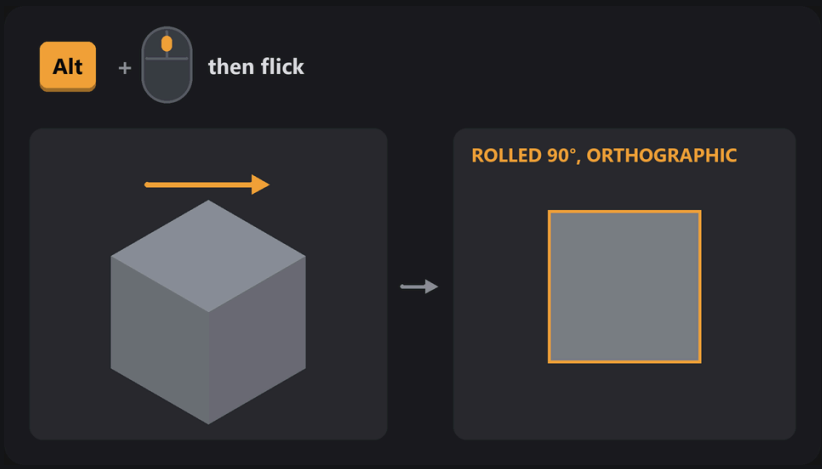


Figure 4.6 Alt, then a short flick of the middle mouse button: the view rolls 90 degrees onto an orthographic axis view.

- **Flick Distance.** How far the mouse must move before the roll commits, 6 pixels by default.
- **Animate.** The roll can be animated instead of cutting straight to the target.
- Quick Roll can be disabled on its own while the rest of Orbit Selected stays active.

4.3 Camera Pan - Shift + Middle Mouse

Pan the Scene view with **Shift + Middle Mouse Button**. Unity reserves Shift inside the Scene view shortcut context, so Blendon handles this input directly rather than through the Shortcut Manager.

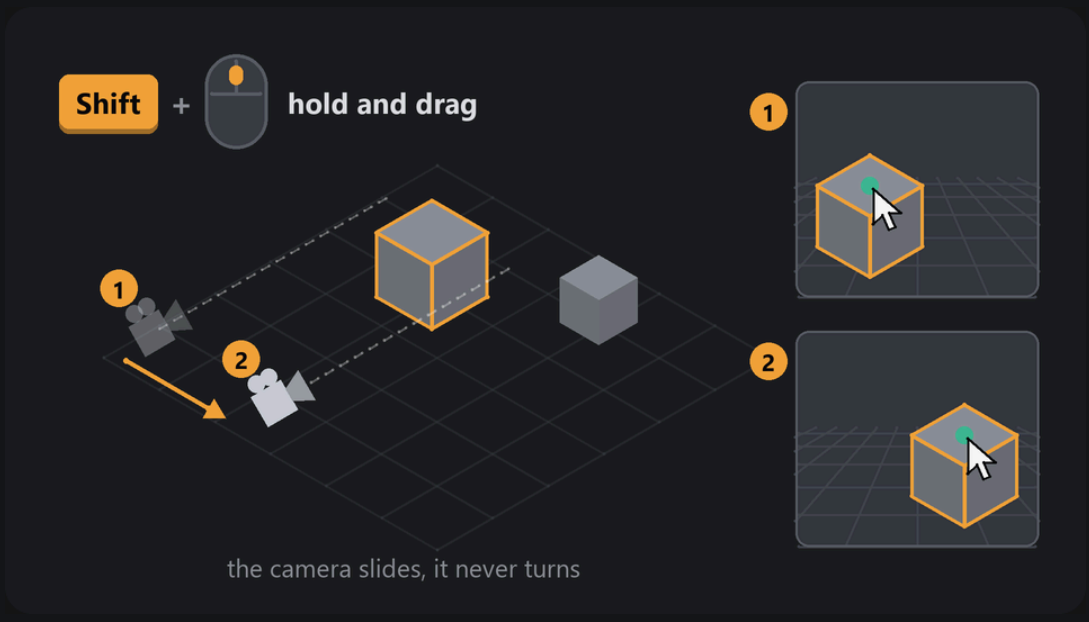


Figure 4.7 The camera slides sideways without turning, and the point you grabbed stays under the cursor.

- **Depth.** Keep the exact world point under the cursor anchored while panning.
- **Invert Drag.** Reverses the drag direction.
- **Cursor Wrap.** Lets a pan run past the edge of the view; see Defaults.

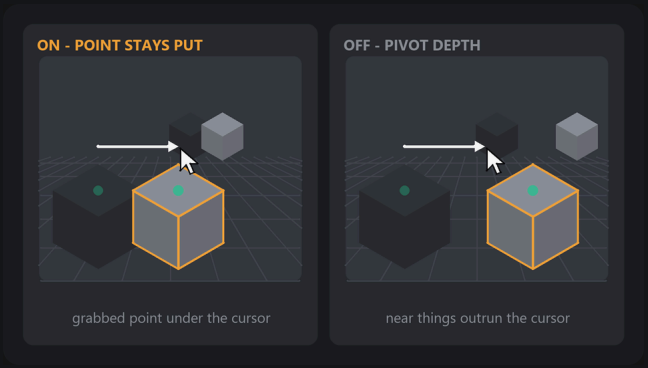


Figure 4.8 Depth on: the grabbed point stays exactly under the cursor, near or far.

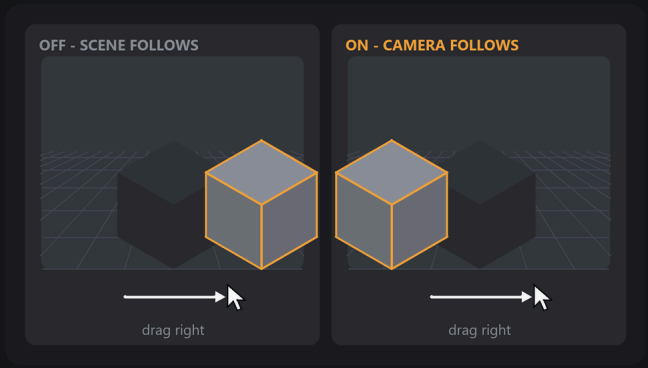


Figure 4.9 Invert Drag: the camera follows the cursor instead of the scene.

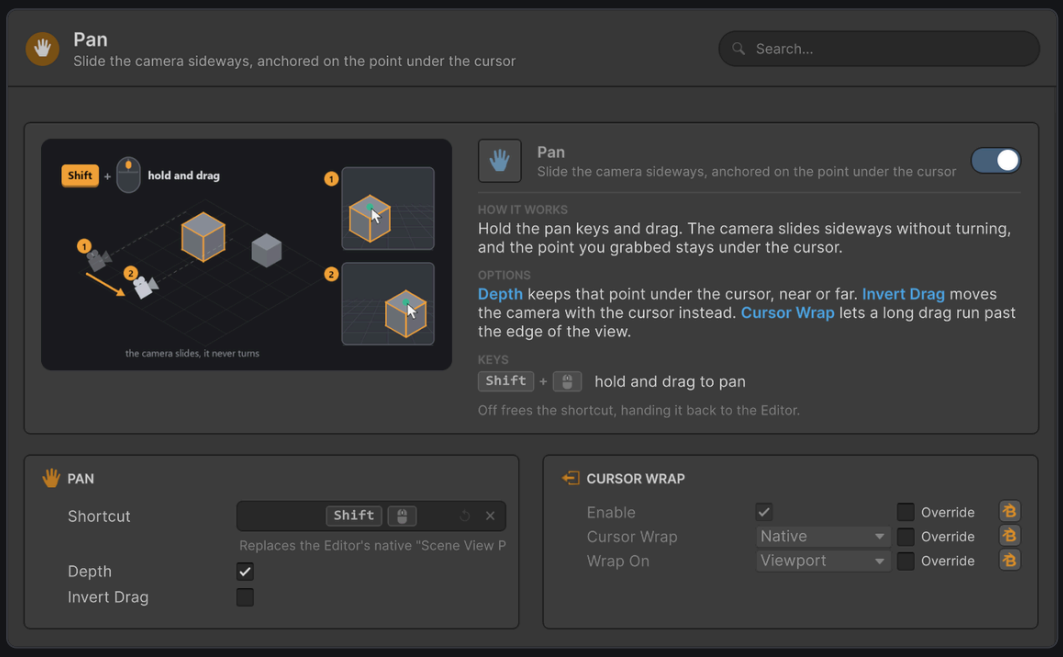


Figure 4.10 The Pan page. The Cursor Wrap panel reads the shared default until you tick one of its Override boxes.

4.4 Zoom - Scroll Wheel

Replaces Unity's scroll-wheel zoom with Blender's, mirroring Blender's *Preferences -> Navigation -> Zoom Method*:

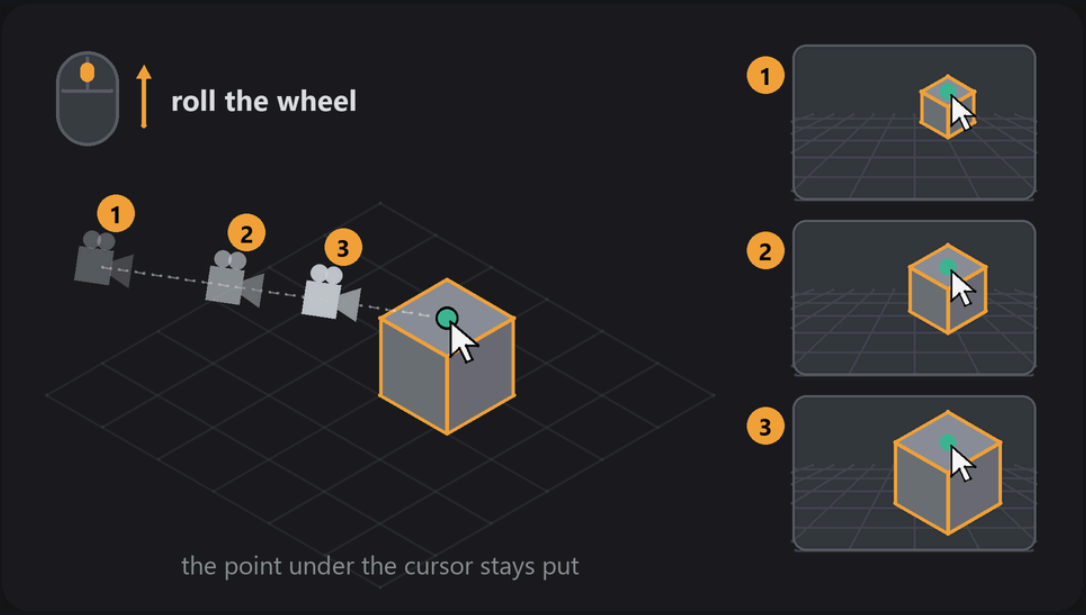


Figure 4.11 Roll the wheel over the Scene view. The point under the cursor stays put while the camera closes in.

Method	Behavior
Scale (default)	Stepwise multiplicative scale of the view size around the pivot.
Dolly	Physically moves the viewpoint along the view axis, keeping the view size constant.
Continue	Continuous and magnitude-sensitive: faster scrolls zoom proportionally more.

- **Zoom To Cursor.** Keeps the scene point under the cursor stationary.
- **Sensitivity** is adjustable and **Invert Direction** reverses the wheel.
- **Keyboard Zoom. Numpad +** and **Numpad -** zoom in fixed steps, 20% of the view size by default, toward the point under the cursor when Zoom To Cursor is on and the cursor is over the view. A zoom-out step is the exact reverse of a zoom-in step. Keyboard zoom has its own switch, which hands both keys back to the Editor when it is off.

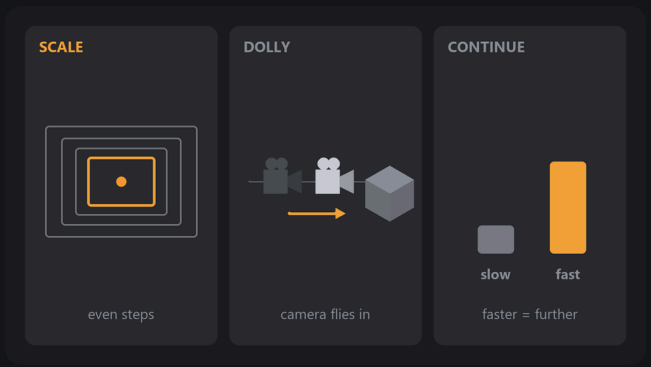


Figure 4.12 The three zoom methods: even steps, a camera that flies in, or speed-sensitive.

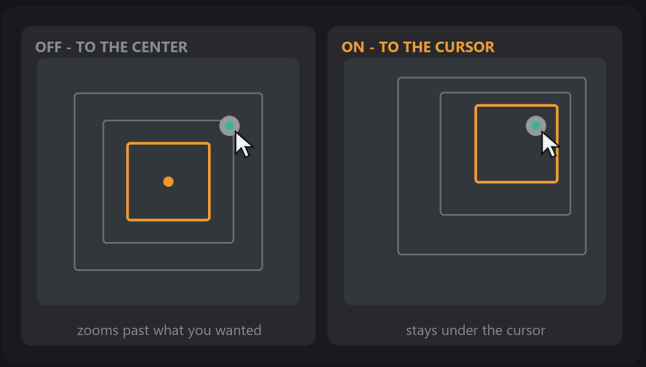


Figure 4.13 Zoom To Cursor keeps whatever you point at under the cursor.

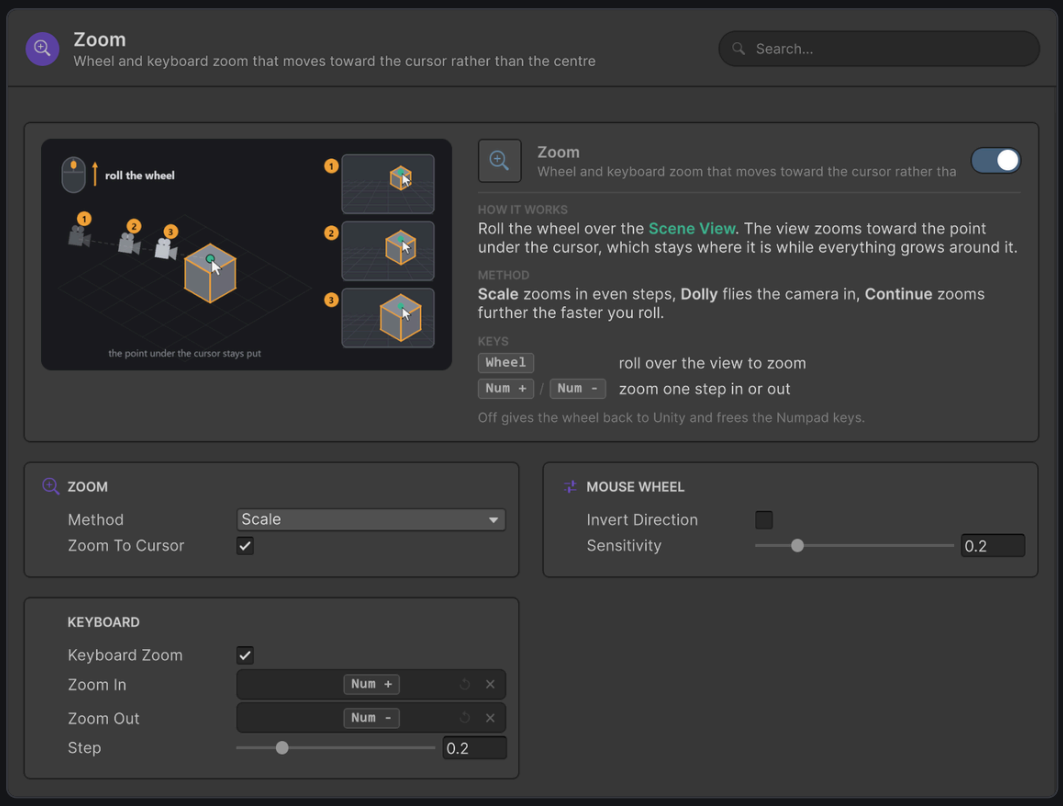


Figure 4.14 The Zoom page.

4.5 Orientation Gizmo

A Blender-style axis-ball overlay in the corner of the Scene view, in place of Unity's own scene gizmo.



Figure 4.15 The Orientation Gizmo at rest.

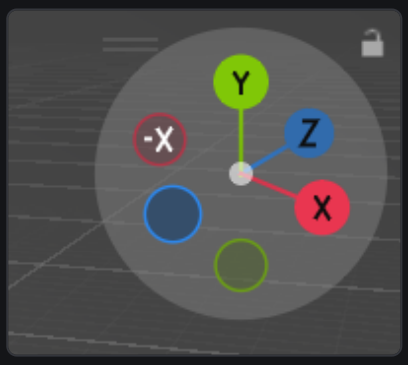


Figure 4.16 With the cursor over it: a backdrop, and the label of the negative handle under the cursor.

- Click an axis handle to snap the view to it, optionally animated instead of jumping.
- **Drag to orbit.** Dragging inside the gizmo orbits the view with the Orbit Selected sensitivity.
- **MMB toggles projection.** Middle-click the gizmo to toggle perspective and orthographic, or enable **Center circle** for a dedicated button that does the same.
- X, Y and Z axis letters, with optional always-on **Negative letters** and a **View Name** label.
- Adjustable size, opacity, padding and line thickness, plus configurable **Hover Backdrop** and **Depth Fade** colors.
- **Hide Unity's gizmo.** On by default, so the two never overlap.

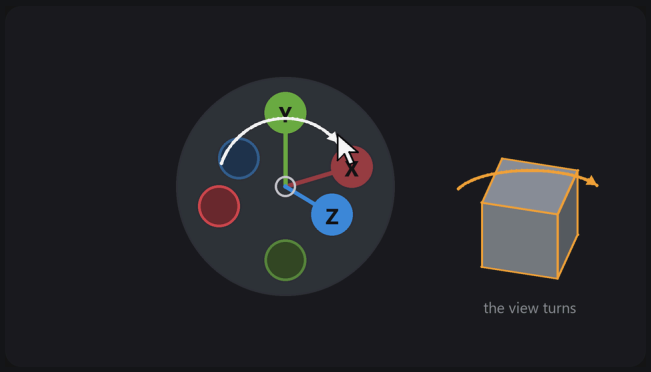


Figure 4.17 Drag inside the gizmo to orbit the view.

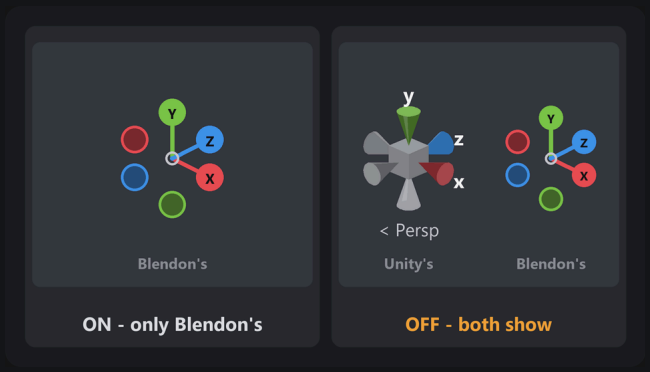


Figure 4.18 Hide Unity's gizmo: on by default, so only Blendon's is drawn.

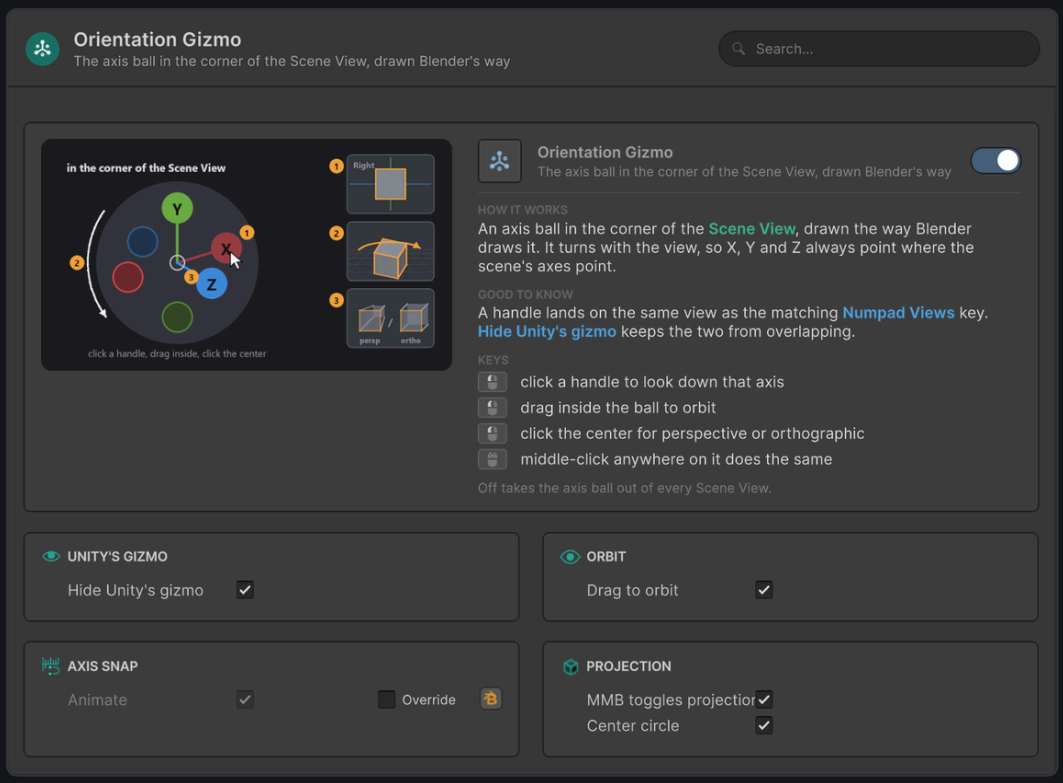


Figure 4.19 The Orientation Gizmo page.

4.6 Numpad Views

Blender's numpad viewport navigation, rebuilt on Unity's Shortcut Manager. NumLock must be on; the settings page warns inline when it is off.

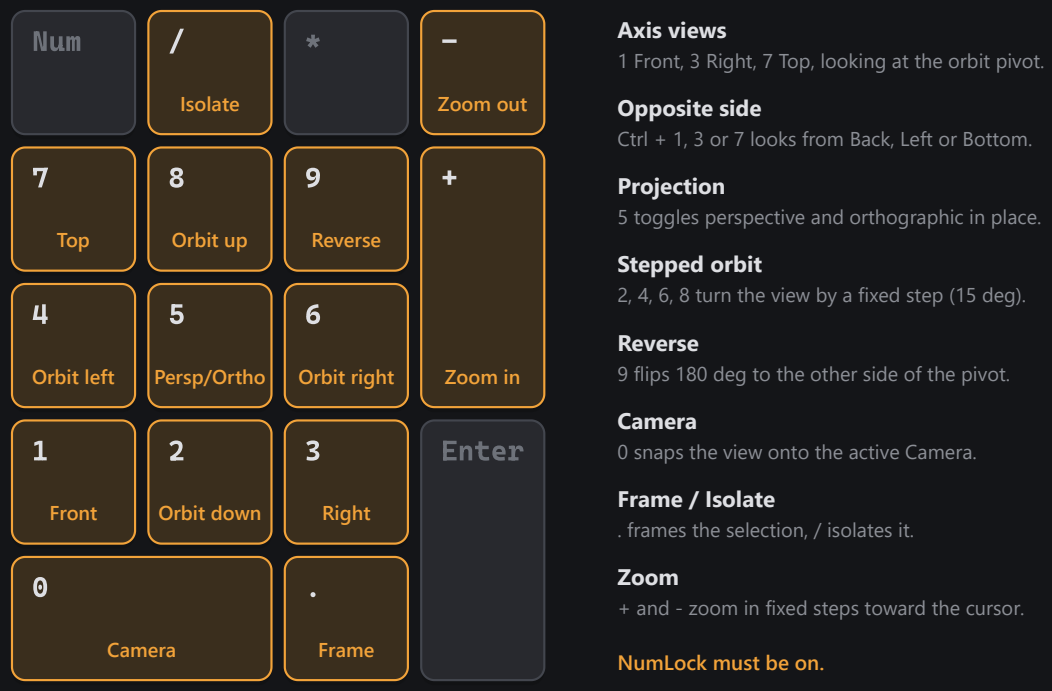


Figure 4.20 Every key of the numeric keypad, as Blendon binds it by default.

- **Axis views. Numpad 1**, **3** and **7** look down -Z, +X and +Y at the orbit pivot. Holding **Ctrl** with the same key looks from the opposite side. This table is shared with the Orientation Gizmo, so a numpad press and a handle click land on the same pose.
- **Toggle projection. Numpad 5** switches between perspective and orthographic in place, the same toggle as the gizmo's center circle.
- **Orbit steps. Numpad 2**, **4**, **6** and **8** turn the view a fixed number of degrees around the pivot, 15 by default. With Orbit Selected enabled, the view turns around the selection instead.
- **Reverse view. Numpad 9** flips 180 degrees to the opposite side of the pivot.
- **Align to camera. Numpad 0** snaps the Scene view onto the active Camera component: the selected camera first, then `Camera.main`, then any enabled camera. **Match Field of View** optionally copies the camera's field of view as well.
- Frame Selected and Isolate View keep their own bindings. This page mirrors them for reference rather than shadowing them with duplicate shortcuts.
- Every jump can be animated instead of cut to.

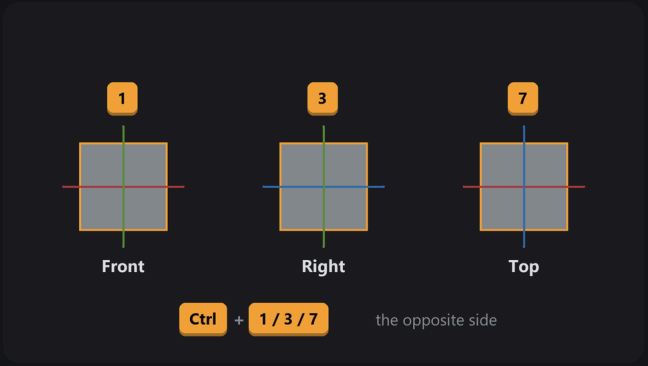


Figure 4.21 Numpad 1, 3 and 7 look from the front, right and top; with Ctrl, from the opposite side.

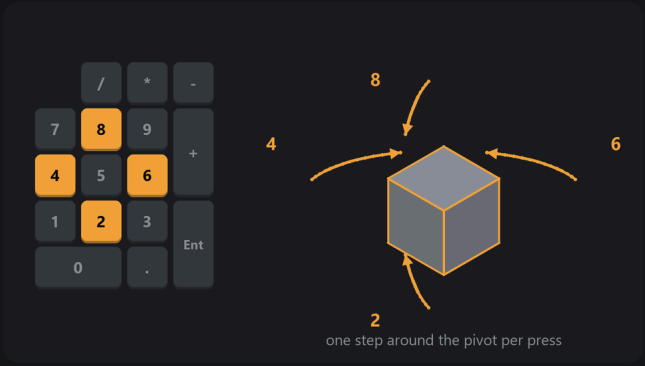


Figure 4.22 Numpad 2, 4, 6 and 8 turn the view one step around the pivot per press.

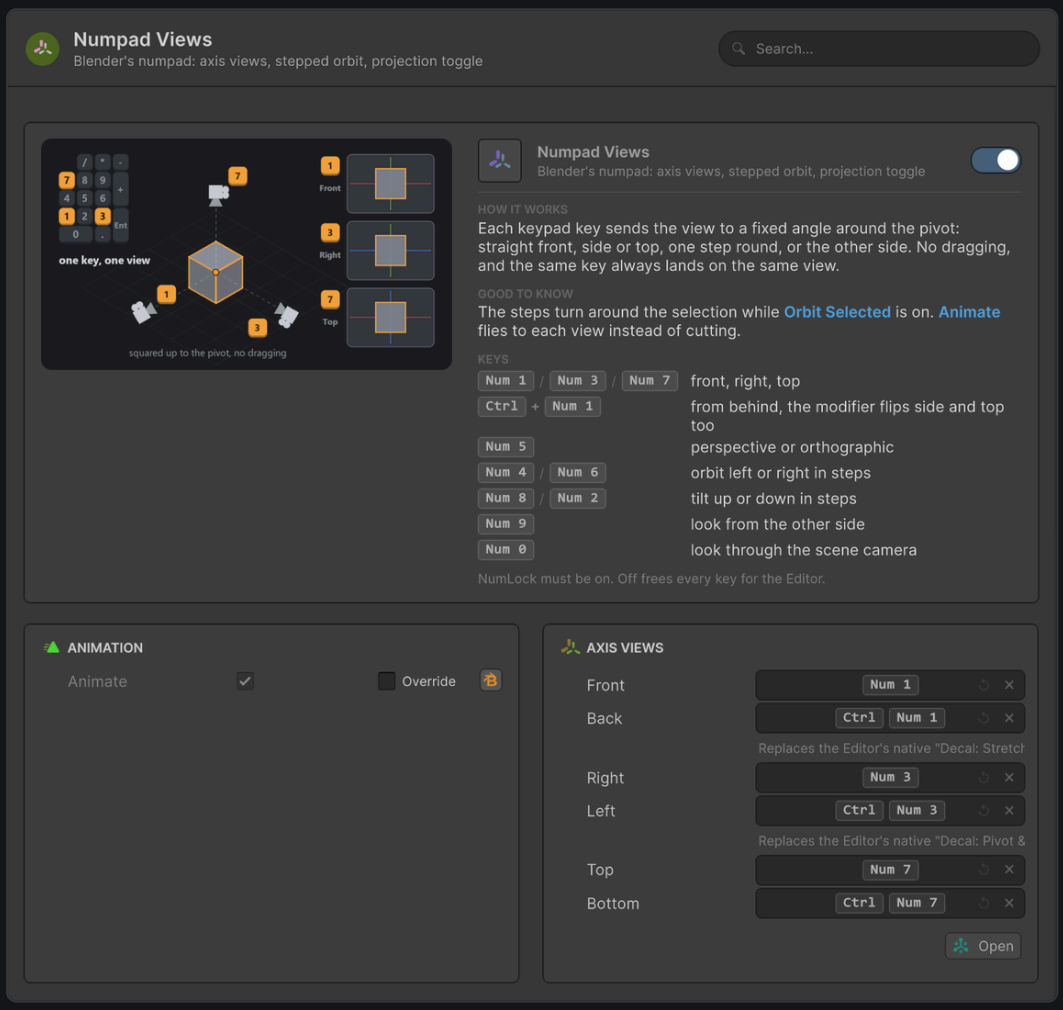


Figure 4.23 The Numpad Views page. Each binding can be changed here or in Unity's Shortcuts window.

5 Transform gizmos

Blendon replaces Unity's **Move (W)**, **Rotate (E)**, **Scale (R)** and **Transform (Y)** gizmos with Blender-style ones. Each can be enabled or disabled independently, and a disabled tool falls back to Unity's own gizmo. Move, Rotate and Scale can also be triggered straight from the keyboard without clicking a handle; see

Grab.

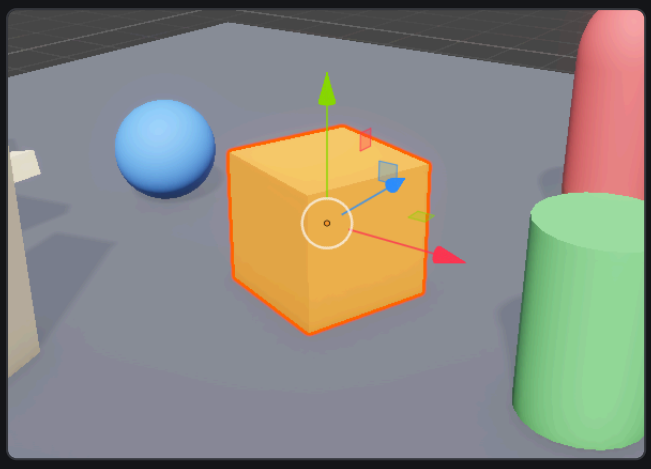


Figure 5.1 Move (W): axis arrows, plane handles, a white screen ring and a center dot.

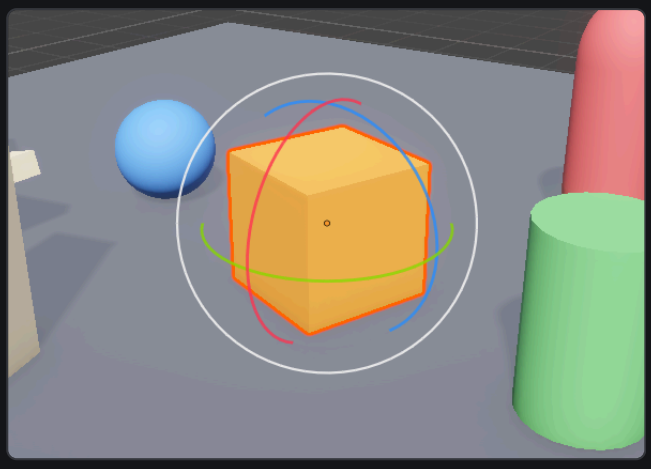


Figure 5.2 Rotate (E): axis rings, an outer view ring and a shaded trackball.

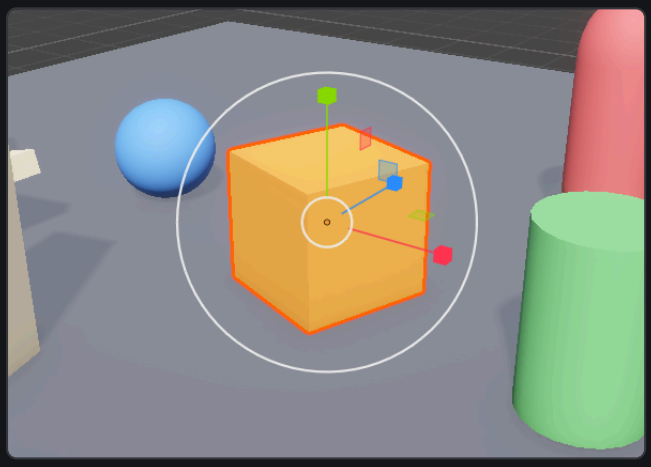


Figure 5.3 Scale (R): box-tipped axes, plane handles and a uniform-scale ring.

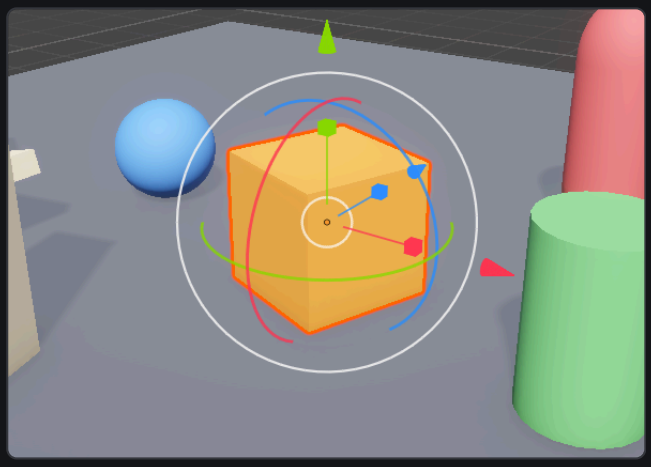


Figure 5.4 Transform (Y): all three nested into one gizmo.

All four gizmos:

- Honor Unity's **Pivot/Center** and **Local/Global** toggles, and Blendon's own Pivot Point modes.
- Support multi-object selection, and are fully undoable.
- Have their own vertex snapping (hold **V**) and work with Unity's grid snapping (hold **Ctrl**).
- Support **right-click to cancel** an in-progress drag, restoring the pre-drag transforms.
- Show a live readout of the exact change while dragging, and accept typed values; see Modal numeric input.

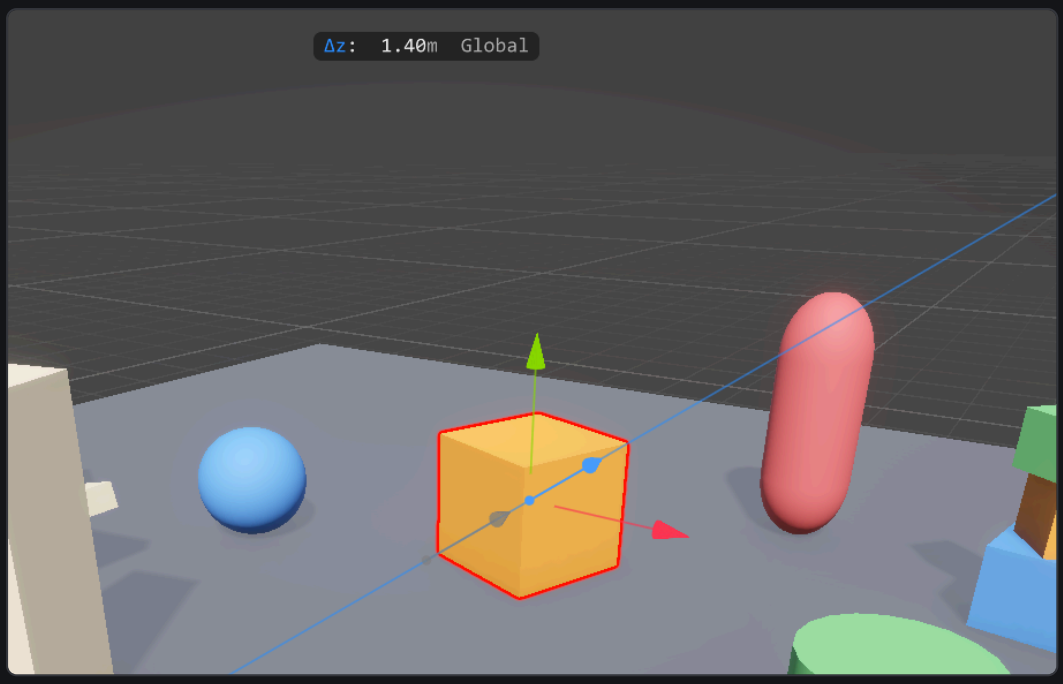


Figure 5.5 A Move drag along Z: the constraint line runs the length of the axis, a ghost dot marks where the drag started, and the readout at the top shows the exact distance.

5.1 Shared gizmo settings

One settings page, **All Tools** in the sidebar, drives everything the gizmos have in common.

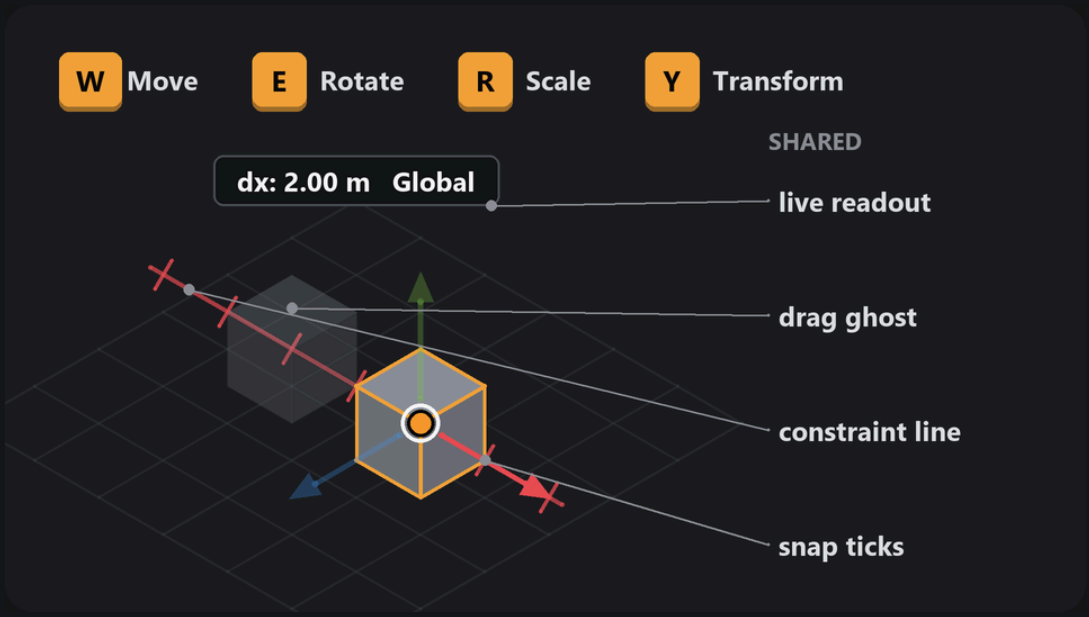


Figure 5.6 What every drag shares, whichever tool runs it: the live readout, the drag ghost, the constraint line and the snap ticks.

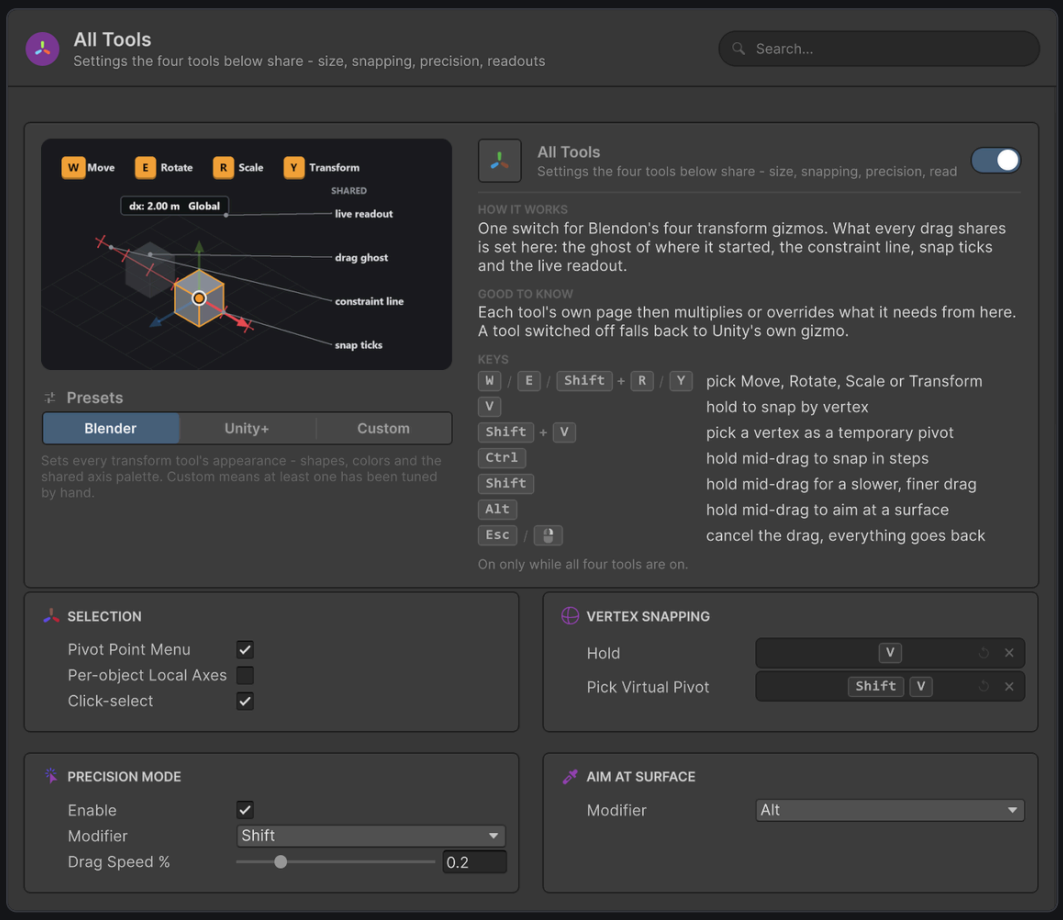


Figure 5.7 The All Tools page. Each tool's own page then multiplies or overrides what it needs from here.

- **Presets.** Applies the **Blender** or **Unity+** appearance preset (shapes, sizes, colors and the shared axis palette) to every transform gizmo at once, from a shipped, read-only preset file. This control appears on All Tools and on the Move, Rotate, Scale and Transform pages alike, since one click reaches all of them regardless of where it is clicked. It reads back as **Custom** as soon as any value has been hand-tuned away from either preset.
- **Resolution, scale, opacity and color contrast.** Global multipliers. Each gizmo then has its own layout settings on top of these.
- **Pivot Point Menu.** Turns Unity's Pivot/Center dropdown into Blender's pivot point menu; see Pivot Point.
- **Per-object Local Axes.** Off by default, which is Unity's way: in Local space every selected object follows the active object's axes. On, it matches Blender: each object moves, turns and scales along its own axes, still about the shared pivot point.
- **Click-select.** A plain click on a handle, with no drag, selects whatever GameObject is under the cursor rather than just reselecting the handle's own object.
- **Fade-out angle.** Axis handles fade as they approach a grazing angle to the camera.
- **Axis colors**, outline color, and pivot dot size, thickness and color.
- **Drag ghosts.** A translucent marker of the drag-start state, with a configurable tint.
- **Constraint line.** A full-extent line along the dragged axis or plane, with adjustable thickness, opacity and contrast.

- **Snap Ticks.** While Ctrl-snapping, tick marks are drawn along the dragged handle at every snap increment. Length and gap are configurable.
- **Live Readout.** A Blender-style info box showing the exact delta while dragging, for example `dz: 1.40m Global`. **Snap to Cursor** pins it next to the cursor instead of the top of the view.
- **Axis labels.** An optional X, Y or Z label near the dragged handle's tip.
- **Precision Mode.** Hold the precision modifier (**Shift** by default) to slow the drag down (20 percent speed by default) and halve the snap increment. With precision mode switched off, the modifier does neither.
- **Aim at Surface** (**Alt** by default). Drives Move's surface snap and Rotate's look-at.
- **Vertex Snapping.** The **Hold** and **Pick Virtual Pivot** keys; see Vertex snapping.
- **Active view only.** An active drag is exclusive to the Scene view it started in. Turn this off to mirror the dragged gizmo into every open Scene view.
- **Cursor Wrap.** Wraps the cursor to the opposite edge during any Blendon drag (camera pan, orbit, and every gizmo handle) so that drags can continue indefinitely; see Defaults.

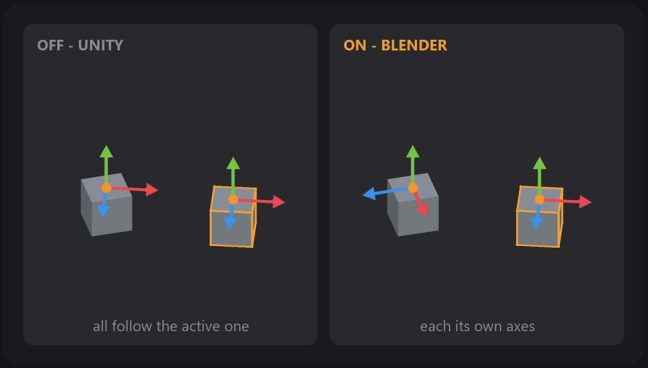


Figure 5.8 Per-object Local Axes: off, every object follows the active one's axes; on, each keeps its own.

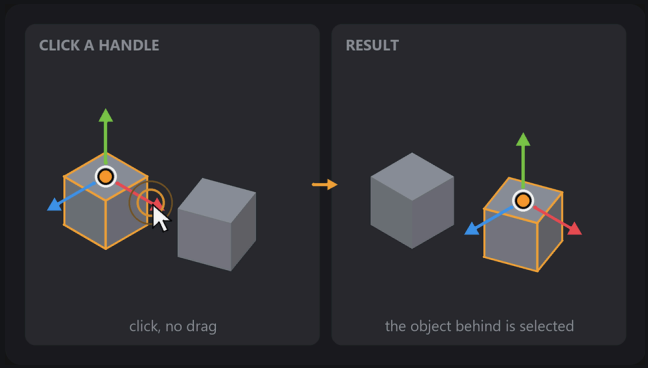


Figure 5.9 Click-select: a plain click on a handle selects the object behind it.

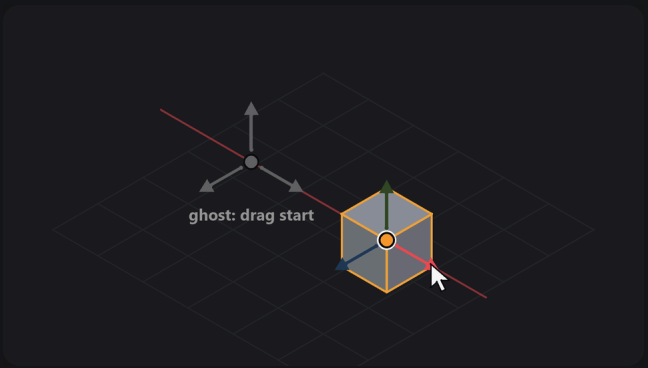


Figure 5.10 The drag ghost marks where the drag started.

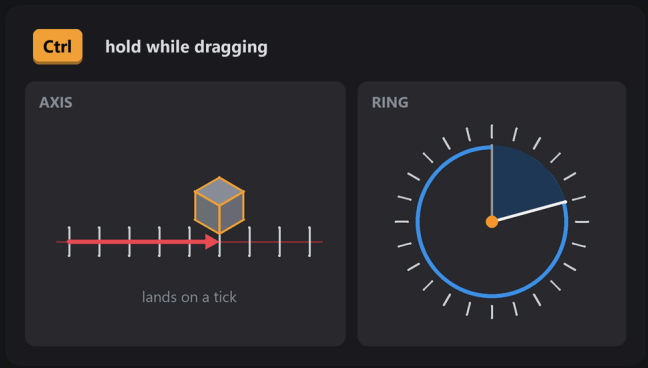


Figure 5.11 Snap Ticks: while Ctrl-snapping, ticks mark every increment along the handle.

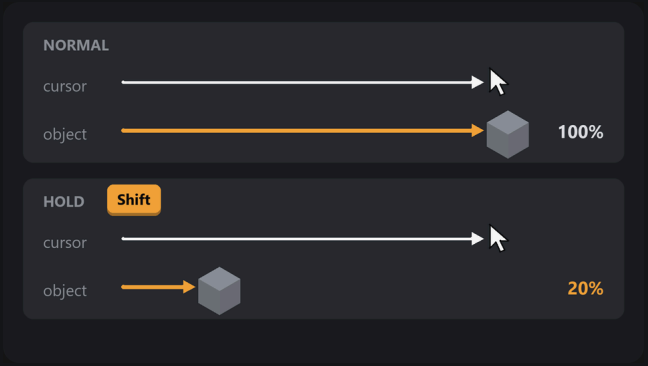


Figure 5.12 Precision Mode: hold Shift and the object moves a fifth as far as the cursor.

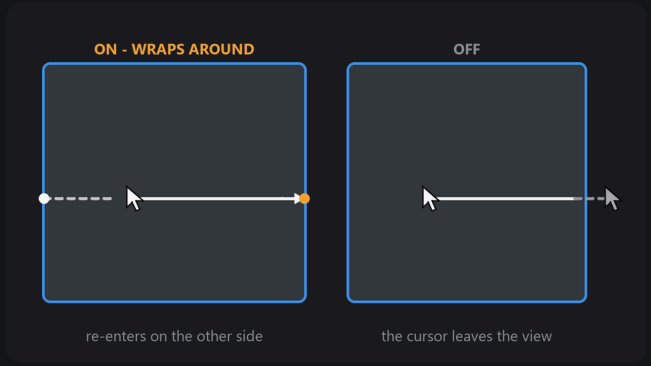


Figure 5.13 Cursor Wrap: a drag that reaches the edge carries on from the other side.

5.2 Pivot Point

Blender's pivot point decides what two or more selected objects turn and scale around. Blendon puts its four modes straight into Unity's own **Pivot/Center** dropdown in the Tool Settings overlay, so there is no new control to learn: click the dropdown and pick a mode.

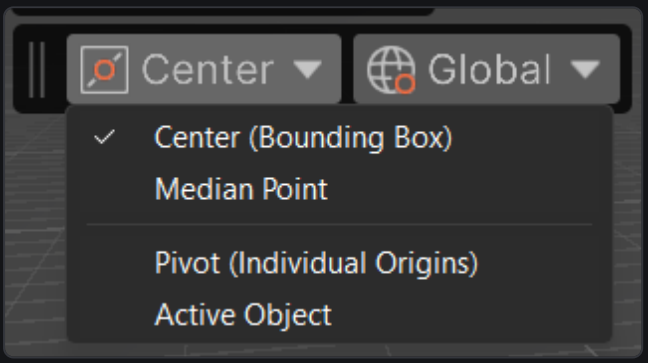


Figure 5.14 Unity's Pivot/Center dropdown with Pivot Point Menu on. The button shows the active mode, here Center (Bounding Box).

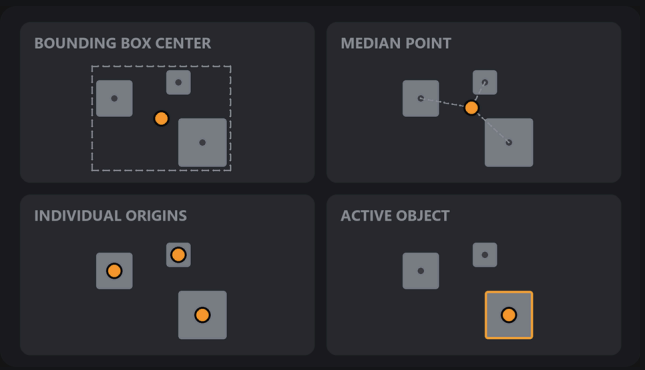


Figure 5.15 The four modes at a glance. The orange dot is where the selection turns and scales.

The four modes sit on the two sides of Unity's toggle, so the toolbar and the menu never disagree. Picking a mode flips the toolbar to its side, and clicking Pivot or Center on the toolbar brings back the mode last used on that side.

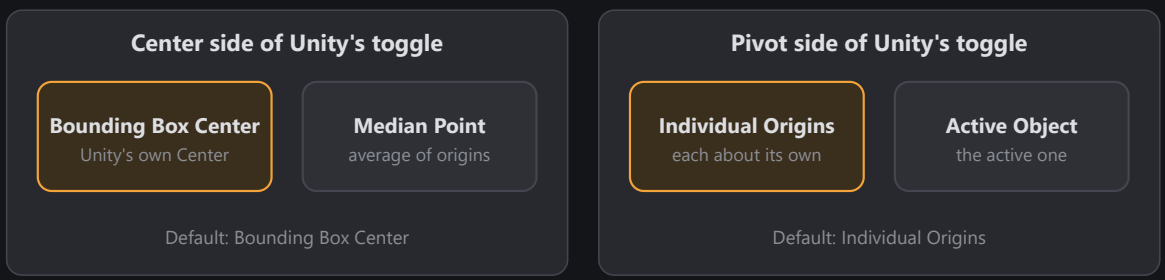


Figure 5.16 Which of Unity's two toggle states each Blender mode belongs to, and where each side starts.

Mode	Button label	Where the selection turns and scales
Bounding Box Center	Center	The center of the box around the whole selection. This is exactly Unity's own Center.
Median Point	Median	The average of the objects' origins, Blender's default.
Individual Origins	Pivot	Each object about its own origin, and in Local space along its own axes too.
Active Object	Active	The active object's origin. The rest of the selection turns and scales around it.

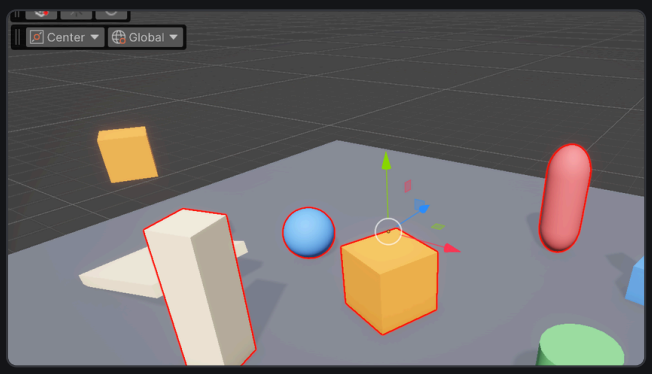


Figure 5.17 Bounding Box Center: the gizmo sits in the middle of the selection's bounds.

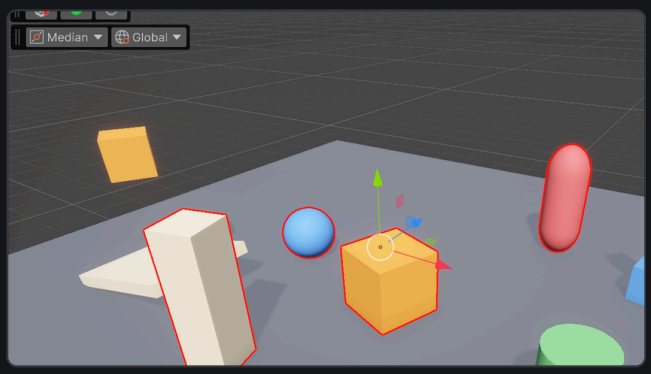


Figure 5.18 Median Point: the average of the four origins, pulled toward the cluster.

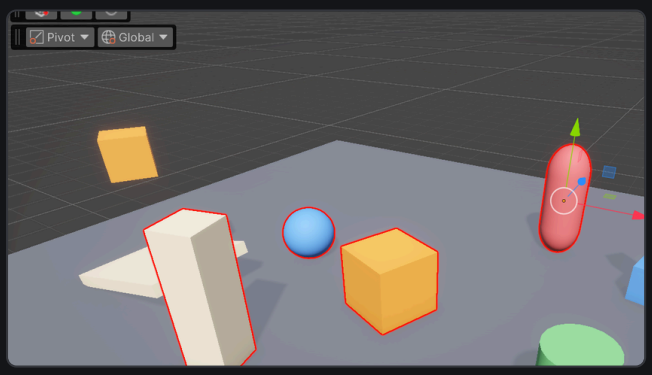


Figure 5.19 Individual Origins: every object turns about its own origin.

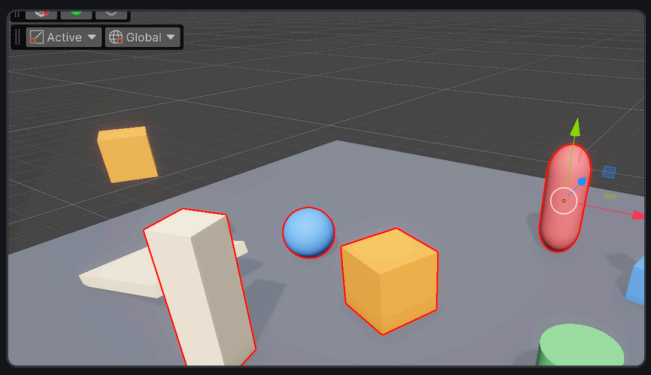


Figure 5.20 Active Object: the gizmo sits on the active object, the capsule.

- **Global/Local only changes the axes, never the pivot.** Individual Origins works in Global space too, and applies to screen-ring and trackball rotation as well.
- **Scaling about a shared pivot spreads positions.** With Bounding Box Center, Median Point or Active Object, a scale moves the objects away from the pivot as well as resizing them, and each object's scale is worked out along its own axes.
- **A single selection follows the toolbar alone**, as Unity's gizmos do.
- The defaults are **Bounding Box Center** on the Center side and **Individual Origins** on the Pivot side, which is Unity's own behavior until you pick something else.

- **Pivot Point Menu** on the All Tools page switches the whole feature off and gives the dropdown back to Unity. The Pivot Point pie offers the same four modes on the **.** key.

5.3 Move Tool - W

Three colored axis arrows, two-axis plane handles, a white screen-space ring for free dragging, and a center dot.

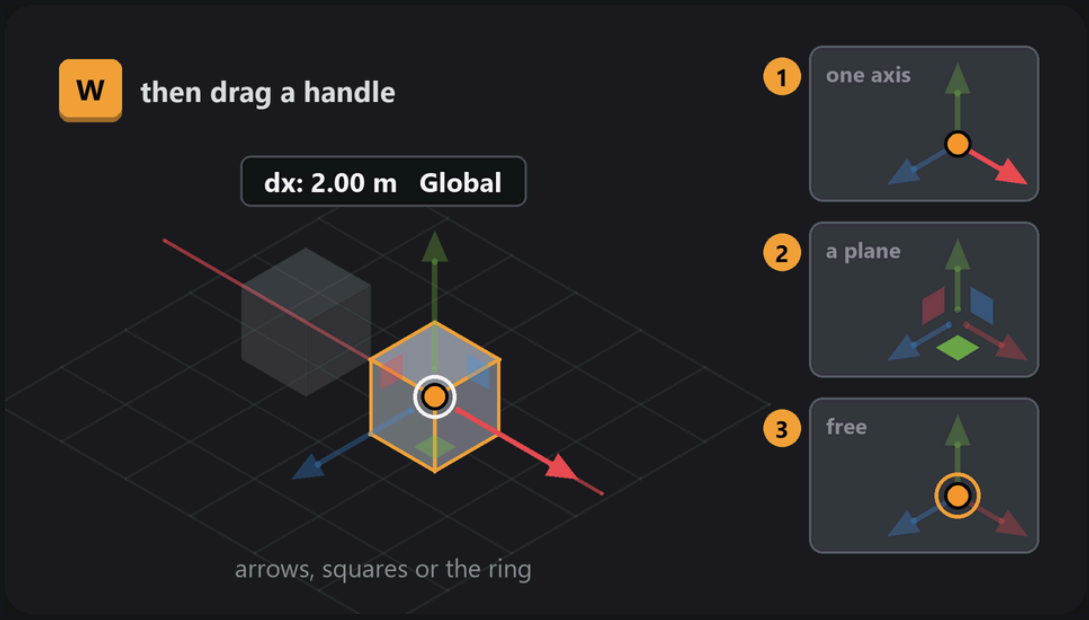


Figure 5.21 Press W and drag a handle: an arrow moves along one axis, a square across a plane, the ring freely on screen.

- Adjustable axis length, thickness and offset, and arrow-head size, with flat 2D or 3D arrow tips.
- Optional plane handles, with configurable shape, size, offset and outline.
- Optional screen-move ring with an adjustable radius.
- **Surface Snap.** Hold the aim modifier (**Alt**) while dragging to drop the selection onto the surface under the cursor, for placing objects on a floor, a wall or another object. A drag that is already running on an axis or plane keeps its constraint and only snaps as far along it as it can reach.
- **Grab.** The **G** key; see Grab.

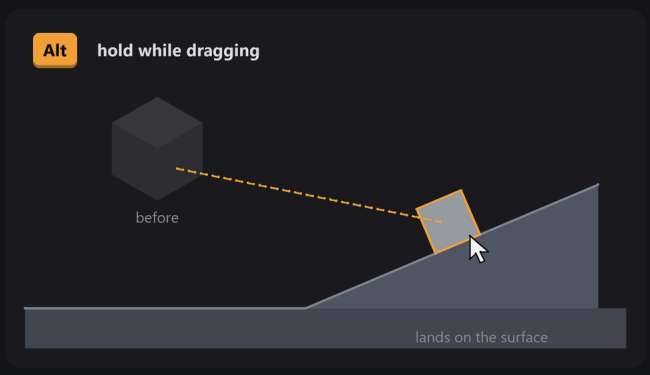


Figure 5.22 Surface Snap: hold Alt while dragging and the object lands on the surface under the cursor.

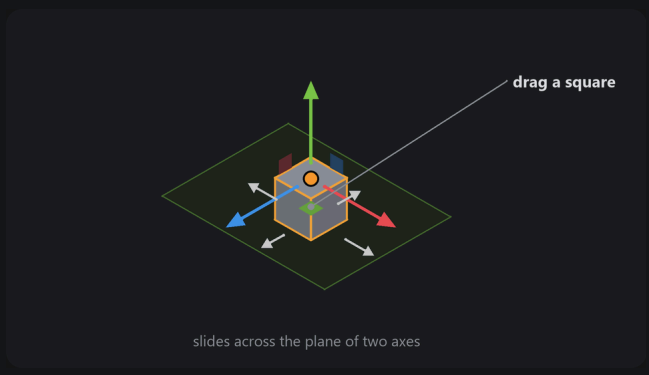


Figure 5.23 A plane handle slides the object across the plane of two axes.

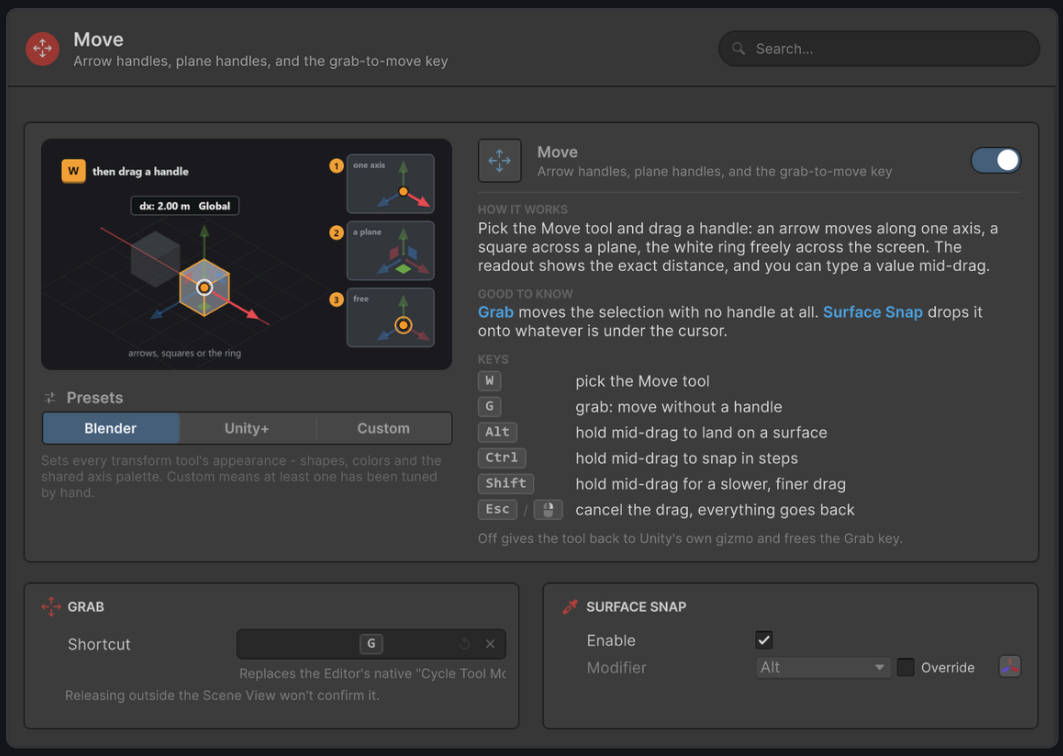


Figure 5.24 The Move page: the header card, the appearance presets, and the Grab and Surface Snap panels.

5.4 Rotate Tool - E

Three axis rings, an outer screen-space ring that rotates around the view axis, and an optional shaded trackball sphere for free rotation.

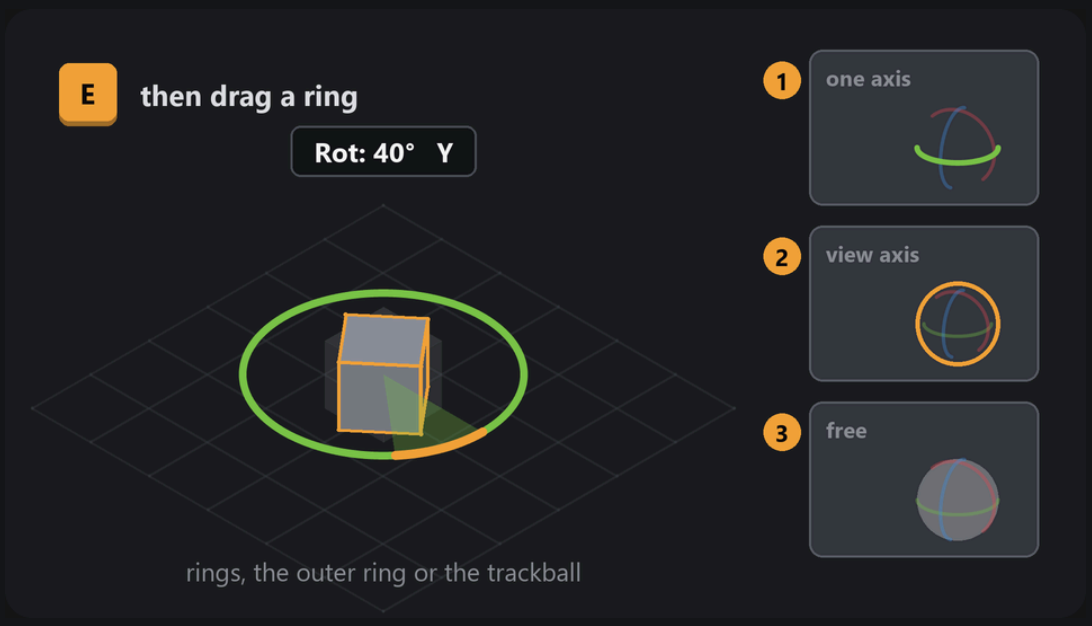


Figure 5.25 Press E and drag a ring: a colored ring turns around one axis, the outer ring around the view, the trackball freely.

- **Rotate To Cursor.** A Blender-style arc drag where the angle follows the cursor around the pivot. Turn it off for Unity's native behavior, which reads raw mouse movement along one screen axis.

- **Look At.** Hold the aim modifier (**Alt**) while dragging to aim the selection at the surface under the cursor, which is useful for orienting an object toward a camera or a player. The trackball turns the whole object; a single axis ring turns it only as far as that axis allows.
- **Depth Gradient.** Shades the dragged ring from the axis color on the near side to the constraint color on the far side, to suggest depth.
- **Angle arc.** A swept pie showing the rotation delta, with adjustable opacity and a **Dynamic Opacity** mode so that overlapping 360-degree laps do not wash out to white.
- Adjustable ring length and thickness, and trackball radius and opacity.

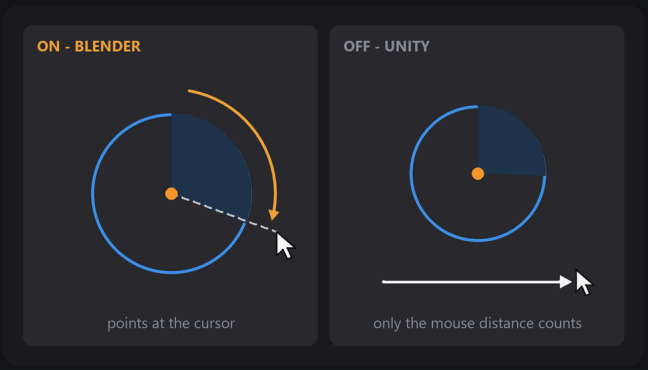


Figure 5.26 Rotate To Cursor: the angle follows the cursor around the pivot, as in Blender.



Figure 5.27 Look At: hold Alt and the selection turns to face the point under the cursor.



Figure 5.28 The trackball spins the object freely in any direction.



Figure 5.29 Dynamic Opacity keeps the angle arc even over several full turns.

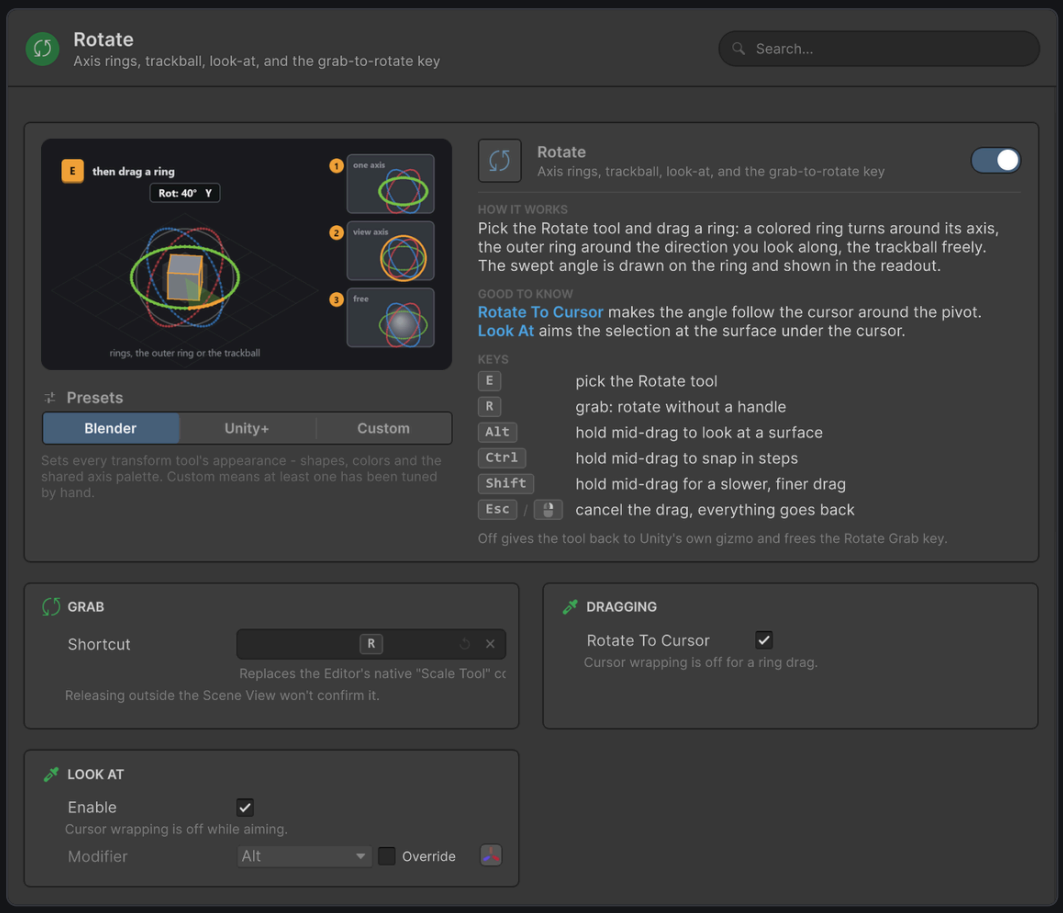


Figure 5.30 The Rotate page.

5.5 Scale Tool - R

Box-tipped axis handles, plane handles, an inner uniform-scale circle and an outer pick ring.

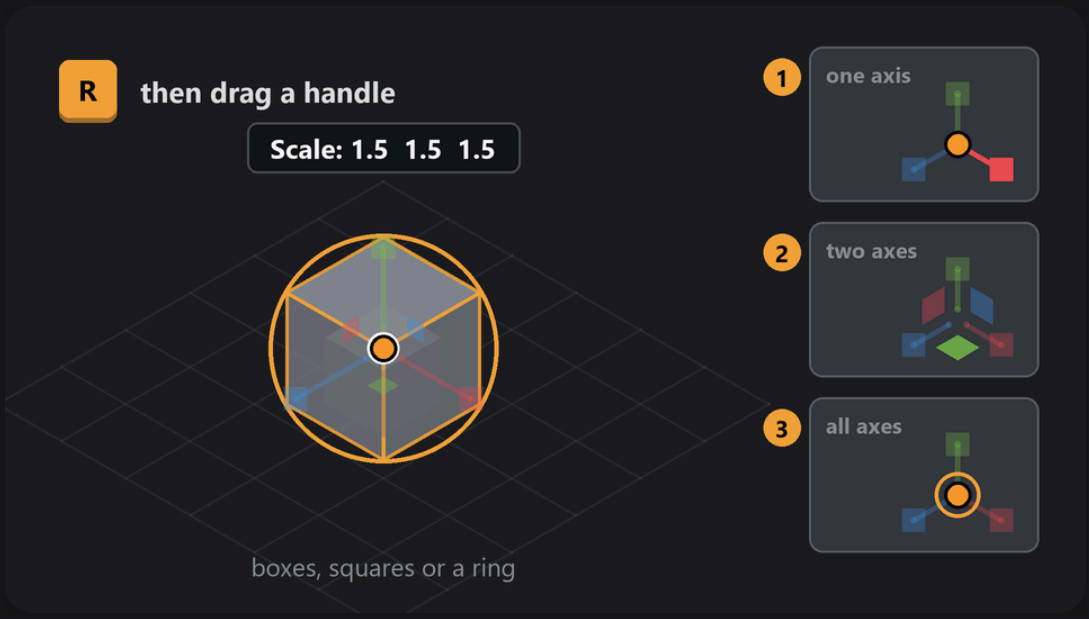


Figure 5.31 Press R and drag a handle: a box scales along one axis, a square along two, a ring along all three.

- **Force Local Orientation.** Unity-native behavior: always draw and scale along the object's local axes, ignoring the handle-rotation toggle. Off by default, which gives Blender-style Global and Local behavior that approximates scaling along the world axis you drag on a rotated object.
- Adjustable axis length, thickness and offset, and box size, with flat or 3D box tips.
- Optional plane handles and screen rings, with configurable inner and outer radii.

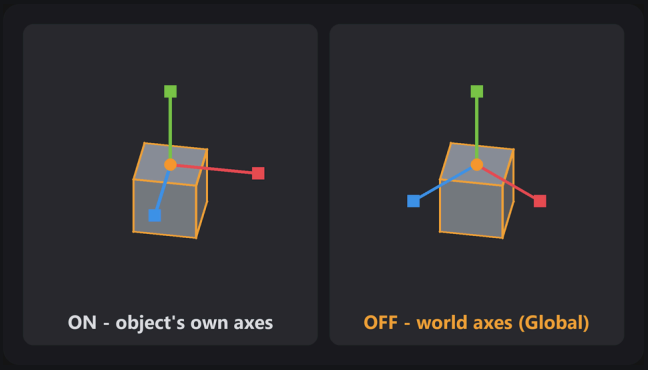


Figure 5.32 Force Local Orientation: on, always the object's own axes; off, the world axes in Global.

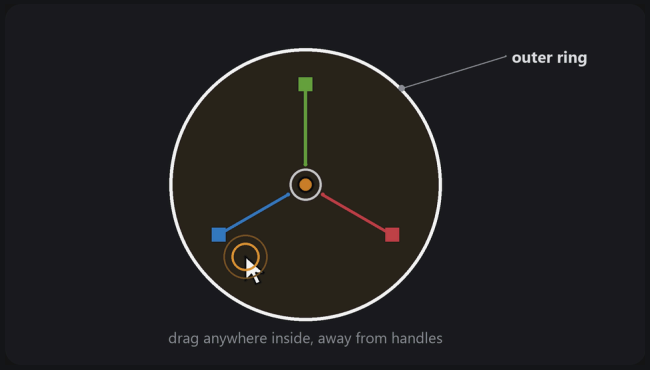


Figure 5.33 The outer ring: drag anywhere inside it, away from the handles, to scale evenly.

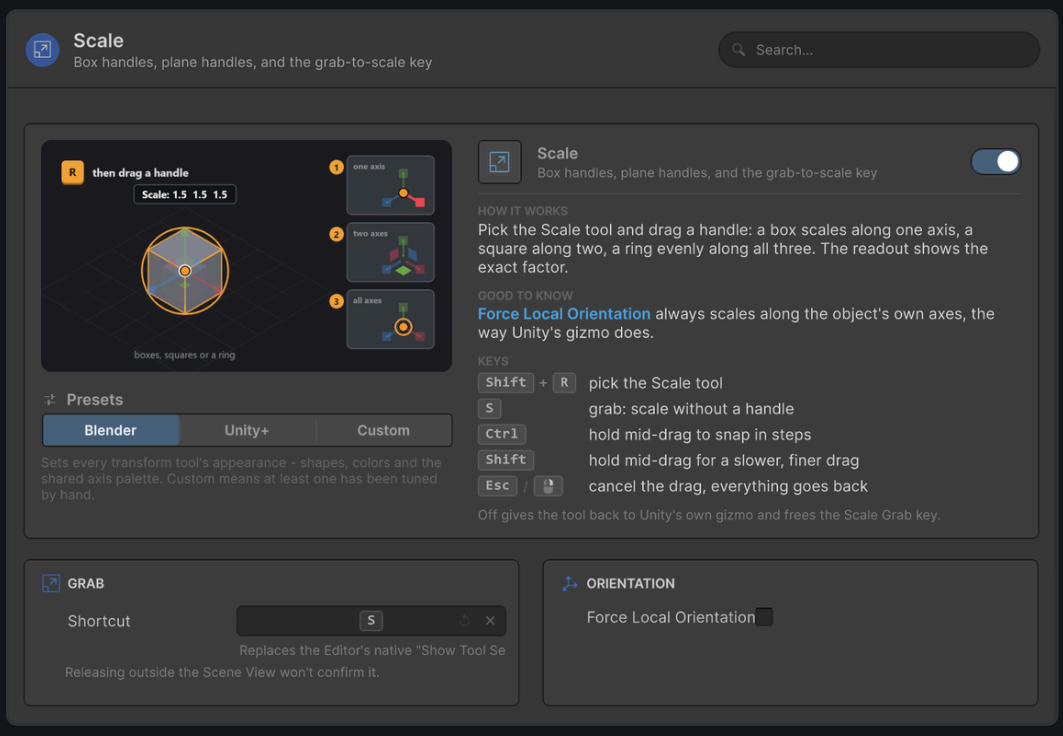


Figure 5.34 The Scale page.

5.6 Transform Tool - Y

All three gizmos composed into one and nested so that they do not fight each other: arrows outermost, rotate rings in the middle, scale boxes innermost.

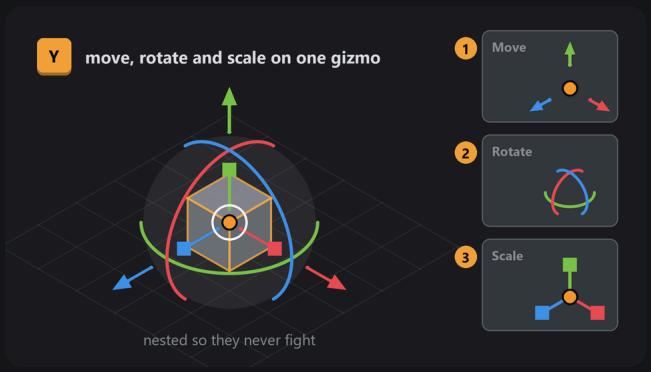


Figure 5.35 Press Y: move, rotate and scale on one gizmo.

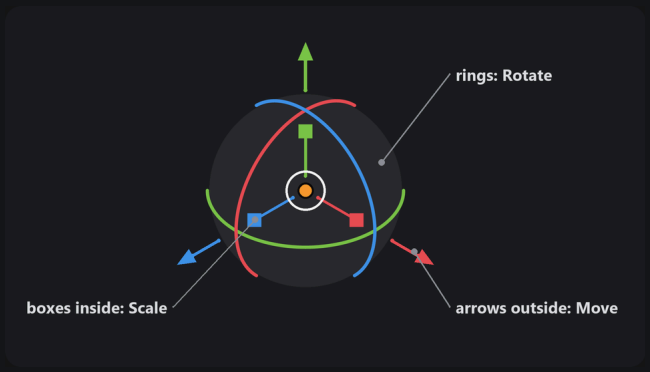


Figure 5.36 Arrows outside move, rings rotate, boxes inside scale.

- Each sub-gizmo has its own layout settings: length, offset, thickness, head size and screen rings, plus the rotate part's ring radius, trackball and ring color. Tuning a standalone tool never disturbs the nesting.
- Scale's outer pick ring is hard-disabled, since its click-anywhere fallback would swallow the other handles. Plane squares are off by default and can be switched on.
- The rotate trackball and the free-move screen ring are shown by default; either can be switched off.
- Grab keys, flat tips, surface snap, look-at, rotate-to-cursor, depth gradient, angle arc and force-local always follow the standalone tools' own values rather than storing duplicates.

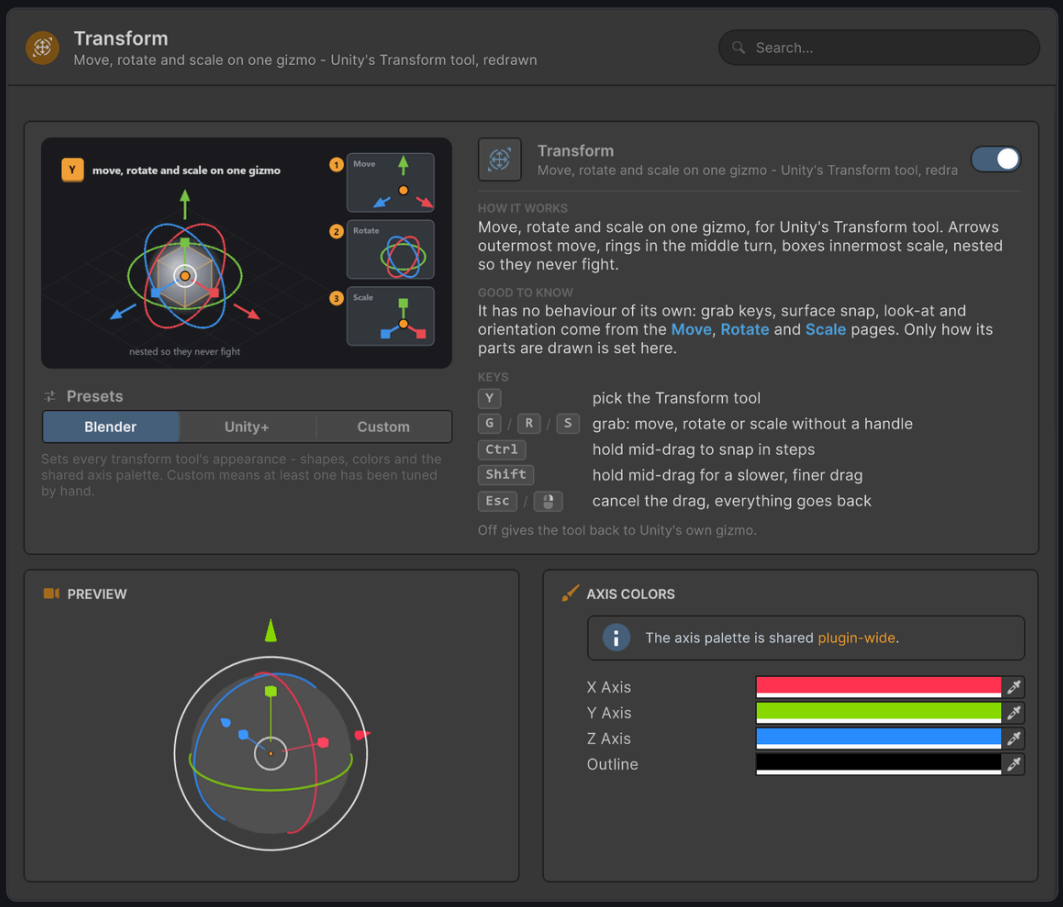


Figure 5.37 The Transform page. It has no behavior of its own, only how its parts are drawn.

5.7 Grab - G / R / S

Blender's modal grab. Press **G**, **R** or **S** anywhere in the Scene view and the selection follows the cursor immediately, with no handle to click, and whichever tool is currently active in the toolbar stays active. Confirm with a click or **Enter**, and cancel with a right-click or **Esc**, exactly as you would a handle drag.

- **Grab (G)** moves the selection across a camera-facing plane through the pivot.
- **Grab Rotate (R)** spins it around the view axis.
- **Grab Scale (S)** scales it uniformly by the cursor's distance from the pivot.

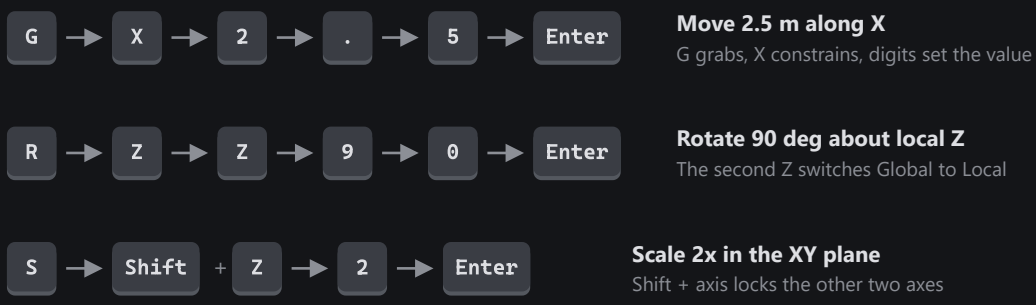


Figure 5.38 Three complete grabs, typed without touching the mouse.

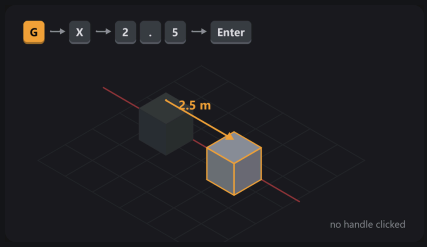


Figure 5.39 G, X, 2.5, Enter: moved 2.5 m along X, no handle clicked.

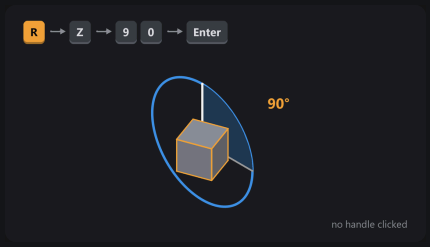


Figure 5.40 R, Z, 90, Enter: turned 90 degrees about Z.

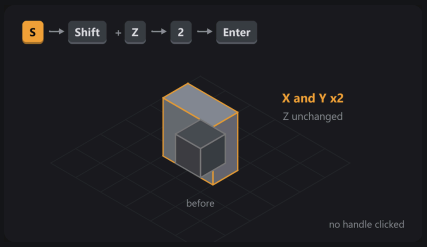


Figure 5.41 S, Shift + Z, 2, Enter: doubled in X and Y, Z unchanged.

A grab runs through the same modal machinery as a handle drag:

- **X**, **Y** and **Z** constrain it to an axis. **Shift** with an axis key locks a plane on Move and Scale, and turns about the excluded axis on Rotate. A third press of the same key drops the constraint again.
- Typed digits enter an exact value: a distance for **G**, degrees about the view axis for an unconstrained **R**, and a uniform factor for an unconstrained **S**.
- The precision modifier slows it down, and grid, rotation and scale snapping apply as usual.
- Grabs respect the shared Pivot Point, and Move's grab also honors **Surface Snap** when you hold the aim modifier.
- Every gizmo handle except the center dot is hidden while a grab is running.
- Each key is rebindable from its own tool's settings page (Move, Rotate or Scale), next to that tool's other options. There is no separate on/off switch: a grab is enabled together with its tool.
- By default **R** collides with Unity's own Scale-tool hotkey. See Keyboard presets for how Blendon settles that.

5.8 Vertex snapping - V / Shift + V

Blendon has its own vertex-snap mode on its own two keys. It applies to every gizmo handle and to the G, R and S grabs alike. The gizmo swaps to a square snap handle, a marker is drawn on each vertex the drag could land on, and the mesh edges leaving the vertex it currently sits on are traced out so that the chosen point stands apart from the rest.

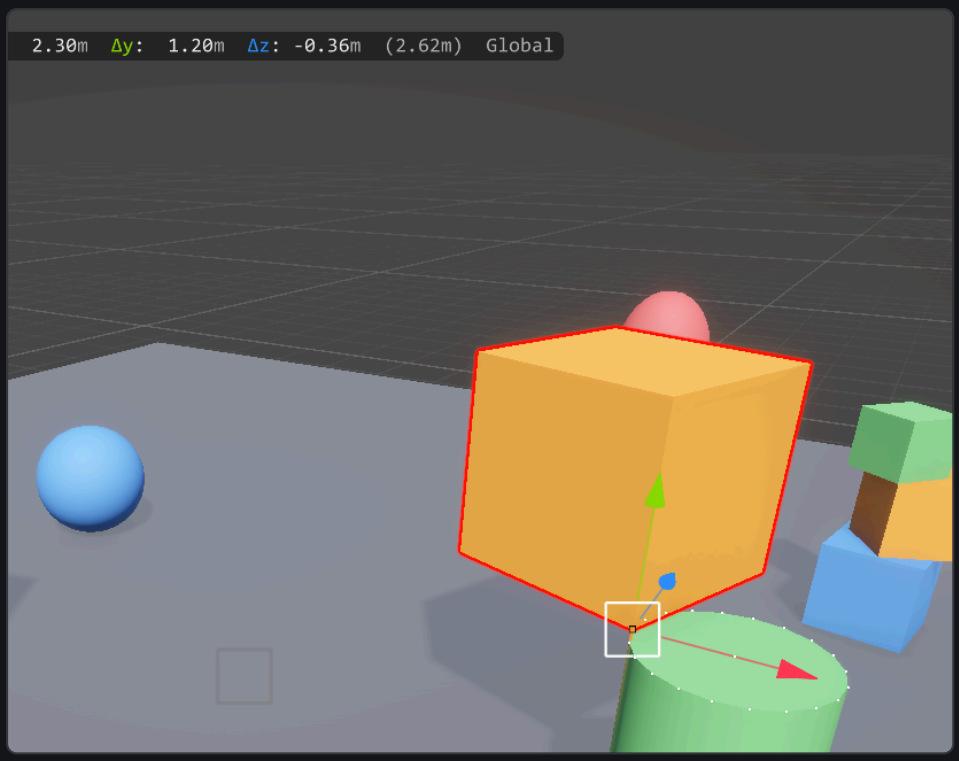


Figure 5.42 Hold V and drag: the crate's corner is carried onto a vertex on the barrel's rim. The small square at the bottom left marks where the drag started.

Hold V to snap for as long as the key is down. The selection is picked up by the vertex nearest the cursor and dropped onto the nearest vertex of another object.

Shift + V (Pick Virtual Pivot) opens a mode that stays on. It begins by ringing the vertex nearest the cursor; click to settle on it, and from then on drags run from that point instead of from the selection's own pivot, until the mode is switched off.

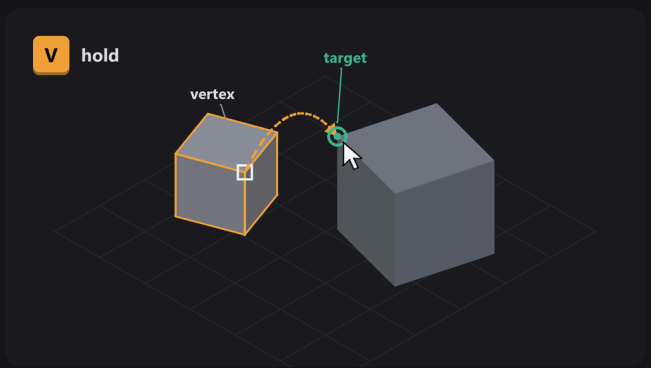


Figure 5.43 Hold V: the vertex under the cursor snaps onto a vertex of another object.

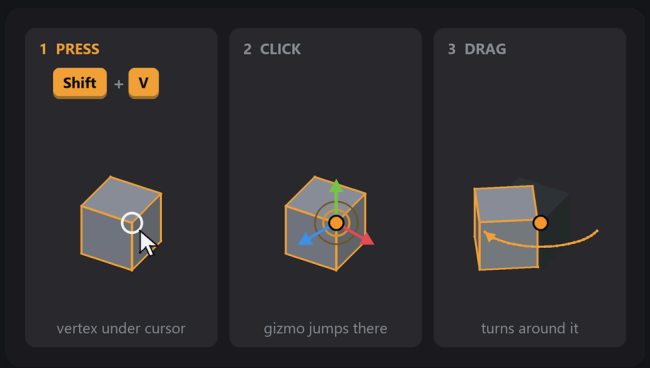


Figure 5.44 Shift + V: pick a vertex, and the gizmo moves there as a temporary pivot.

- Every entry starts the choice over, so the mode never silently reuses the vertex you picked last time.
- **Right-click** or **Esc** leaves the mode. A click past the gizmo drops just the settled vertex and reopens the choice, as does changing the selection.
- **Holding V** with nothing being manipulated reopens the choice for as long as the key is down. Release it over nothing and the old pick comes back.
- **Pressing V during a drag** toggles the pull toward other objects' vertices. Tapped, it toggles; held past a quarter of a second, it acts as a clutch and releasing it turns the pull back off.
- While a settled vertex is driving a drag, the other placement modes stand down: a drag that ends on one exact point leaves nothing for surface snap, grid snapping or the precision modifier to refine. Pressing **V** hands all three back.
- **Alt** (surface snap) suspends the pull for as long as it is driving the drag. The surface under the cursor and the vertex under it answer the same question, and Alt says which one you meant.

Skinned meshes are included. Their vertices are read from the posed mesh rather than from the bind pose, so a character snaps where it actually stands.

Engaging the mode never re-orients the gizmo. Only Scale's **Force Local Orientation** overrides Unity's handle-rotation toggle.

Both keys are rebindable from the All Tools page or from the Keyboard page.

6 Modal numeric input

While any gizmo drag or keyboard grab is active, you can type an exact value, exactly as you would in Blender.

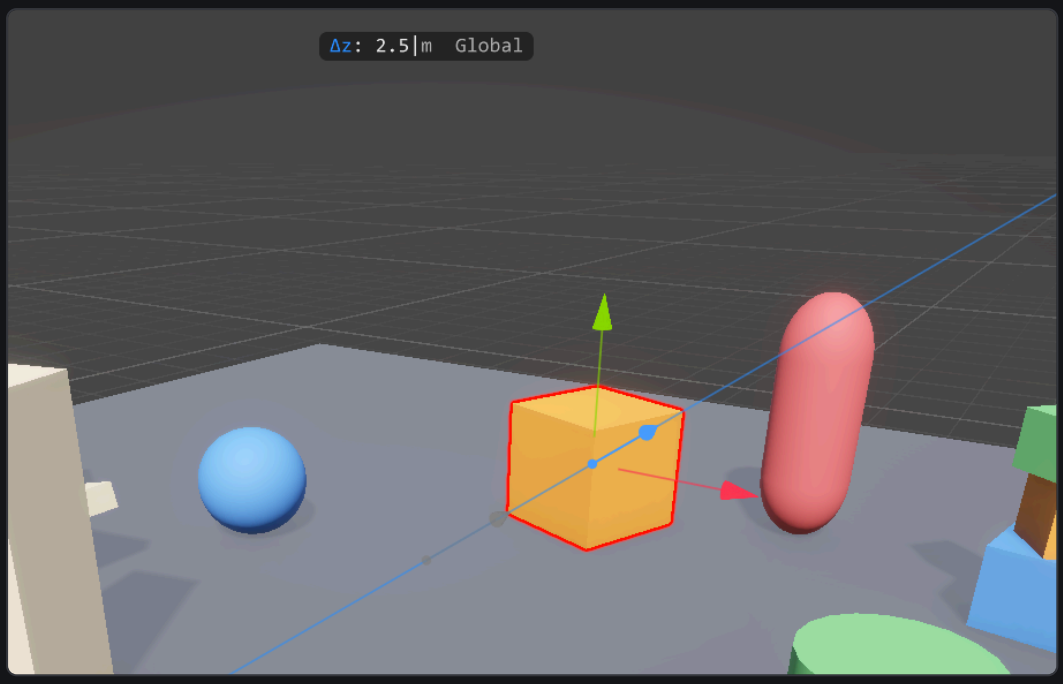


Figure 6.1 Typing 2.5 in the middle of a Move drag along Z. The mouse stops driving the drag, and the readout shows the value being typed, with its caret.

Key	Effect
`0` - `9`, `.`	Type the value. The first digit commits to numeric mode, and the mouse stops driving the transform.
X / **Y** / **Z**	Constrain to that axis. Whatever you have already typed transfers to the new axis.
The same axis key again	Switch to the other orientation, as in Blender; see the figure below.
Shift + **X** / **Y** / **Z**	Lock the opposite plane, excluding that axis instead of isolating it. It counts as the same letter for the orientation cycle, and on a rotation it turns about the excluded axis. The modifier is configurable.
`-`	Toggle the sign.
`/`	Toggle the reciprocal, `1/x`.
Backspace	Delete a character. Once the field is empty, the first press hands control back to the mouse and a second press exits numeric mode.
Enter or **left-click**	Confirm.
Esc or **right-click**	Cancel and restore the pre-drag transform.

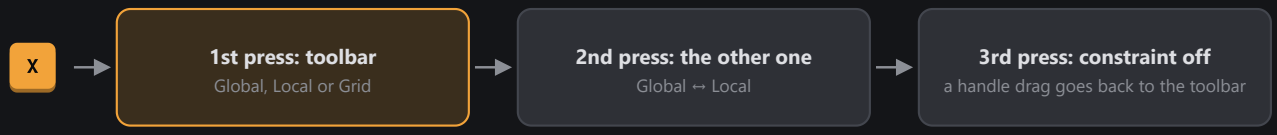


Figure 6.2 Pressing the same axis key repeatedly. The first press uses the toolbar's orientation, the second switches to the other one, and the third drops the constraint on a grab.

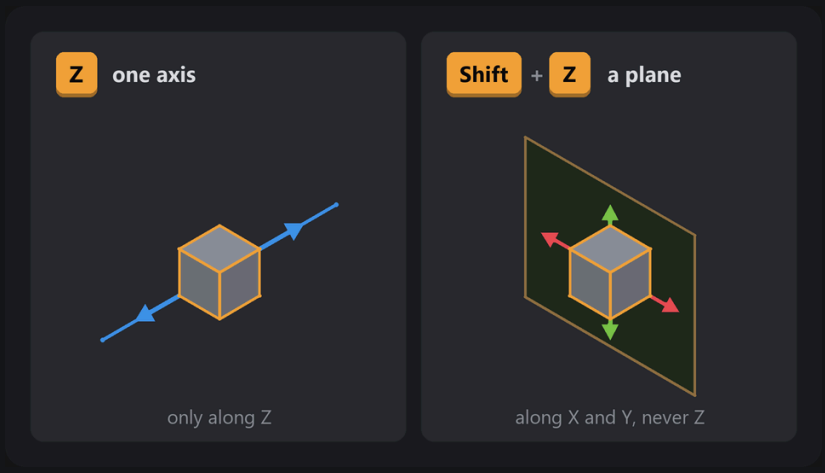


Figure 6.3 Z alone constrains to the Z axis. Shift + Z locks the plane of the other two, so the object slides along X and Y and never Z.

Pressing an axis key before typing anything does not enter numeric mode. The mouse keeps driving the drag, simply re-projected onto the newly selected axis or plane.

7 Scene utilities

7.1 Snap To Floor - End

Drops the selected objects onto whatever is below them. Blendon raycasts along the chosen direction, rests each object's bounds on the surface and ignores the object's own colliders. The operation is fully undoable.

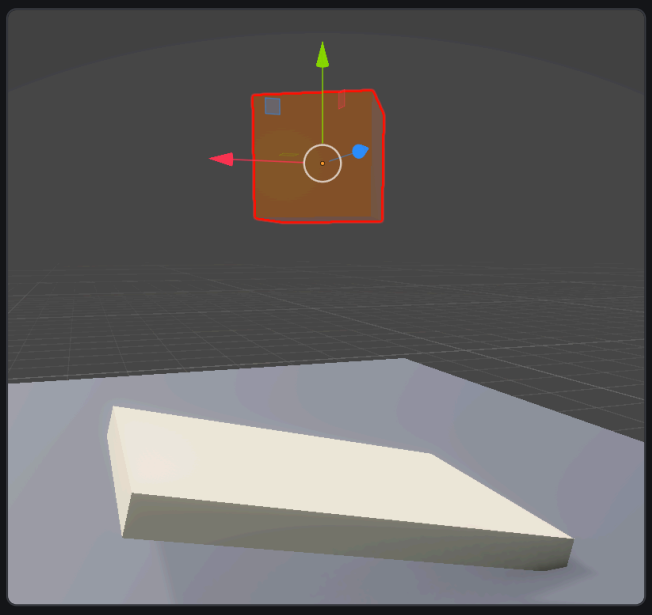


Figure 7.1 Before: a crate floating above a sloped ramp.

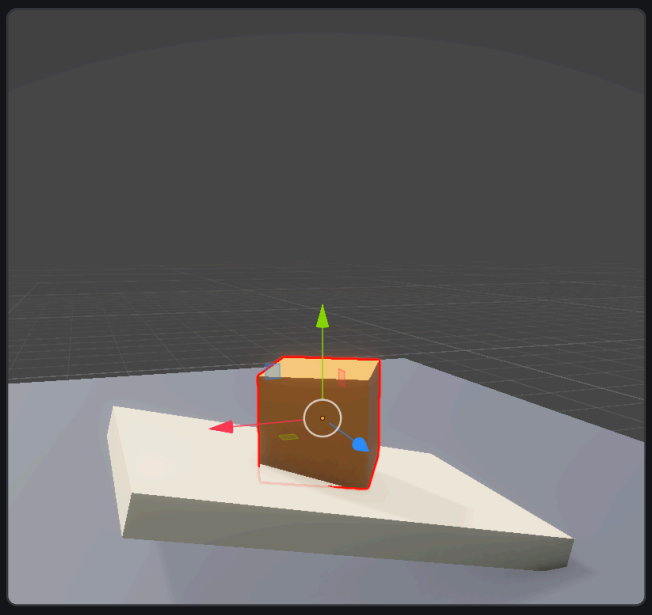


Figure 7.2 After pressing End: the crate rests on the ramp.

- **Direction.** Down onto floors, or Up onto ceilings.
- **Surface Layers.** Only snap onto surfaces on the chosen layers.
- **Align to Surface.** Also rotate each object so that its up axis matches the surface slope.
- **Surface Offset.** Leave a gap between the object's base and the surface.
- An optional bounce animation, with an adjustable duration.



Figure 7.3 Align to Surface: the object lies flush with the slope instead of staying upright.

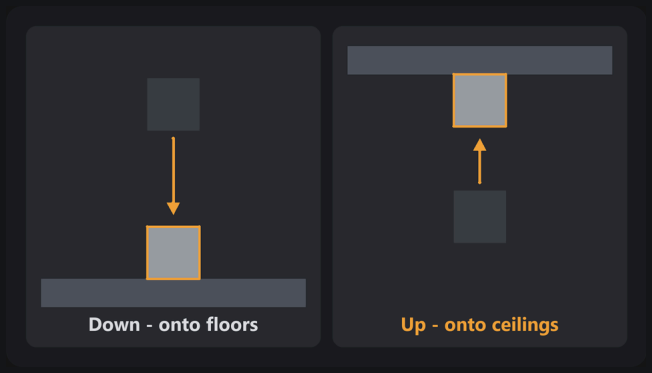


Figure 7.4 Direction: down onto floors, or up onto ceilings.

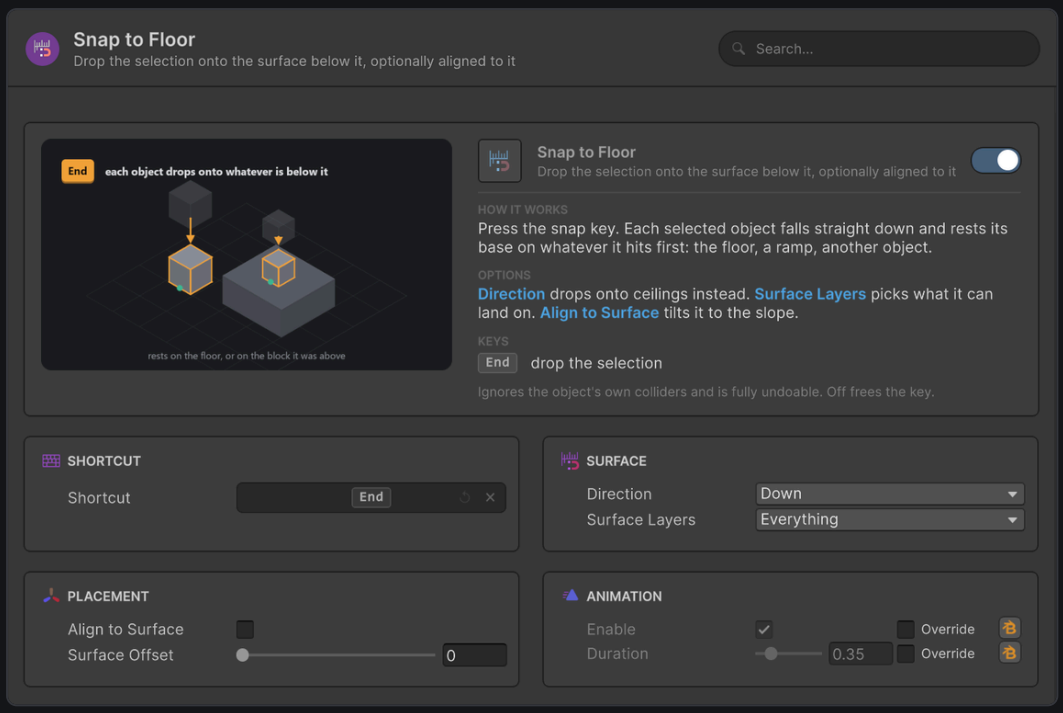


Figure 7.5 The Snap to Floor page.

7.2 Frame Selected - Numpad .

A cycling frame shortcut. Pressing it repeatedly steps through four framings and then starts over: the whole selection, the active object's pivot, a close-up of that pivot, and back to the view you started from. The Sequence table on the page sets which of these steps the key walks through and in what order.

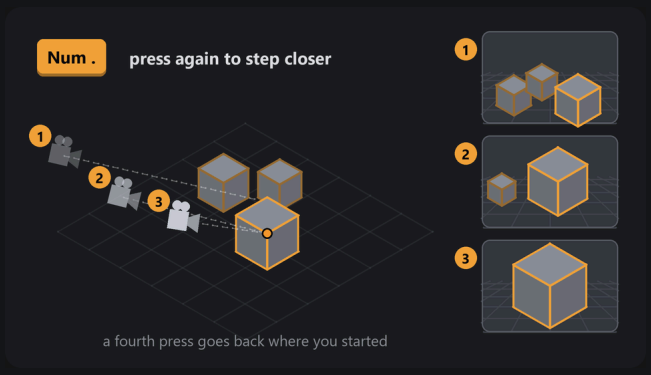


Figure 7.6 Each press steps the camera closer; a fourth press goes back where you started.

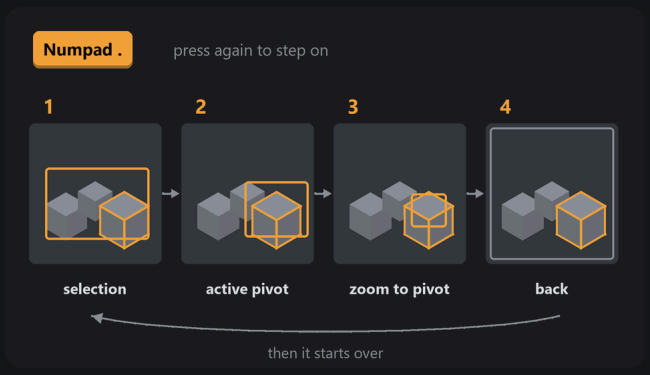


Figure 7.7 The four steps of the cycle, then it starts over.

The camera fly-to can be animated instead of snapping, with an adjustable duration.

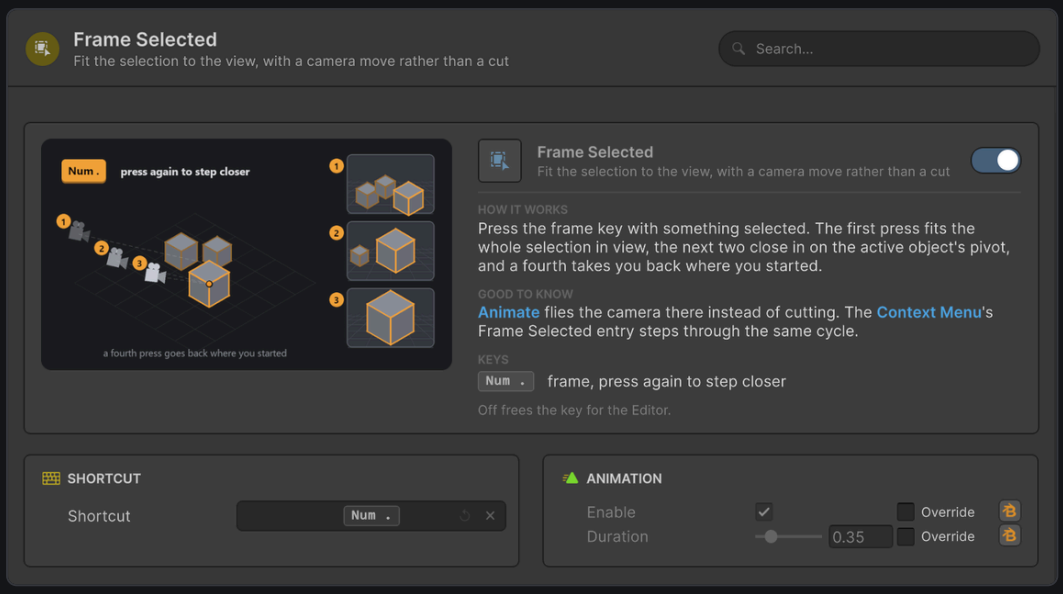


Figure 7.8 The Frame Selected page.

7.3 Isolate View - Numpad /

Blender-style local view. Blendon hides everything except the selection using Unity's Scene Visibility, and restores the scene when you press the key again.

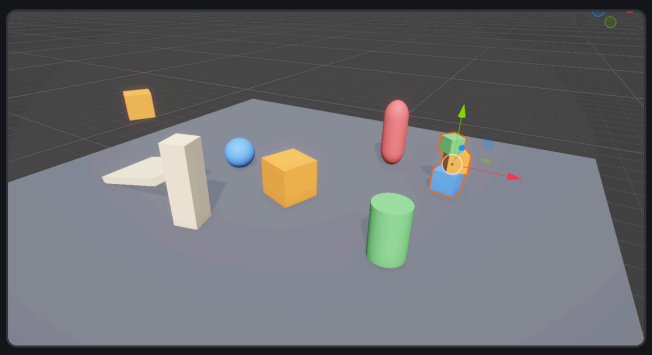


Figure 7.9 Before: the whole scene, with the stack of boxes selected.

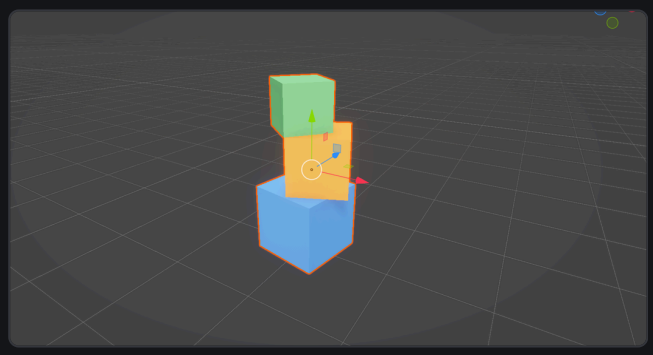


Figure 7.10 After Numpad /: only the stack remains, until the key is pressed again.

- **Include Children.** Isolate the whole hierarchy under each selected object. On by default.
- **Frame On Isolate.** Frame the selection when isolating. Off by default.



Figure 7.11 Include Children: on, the table keeps its cup and plate.

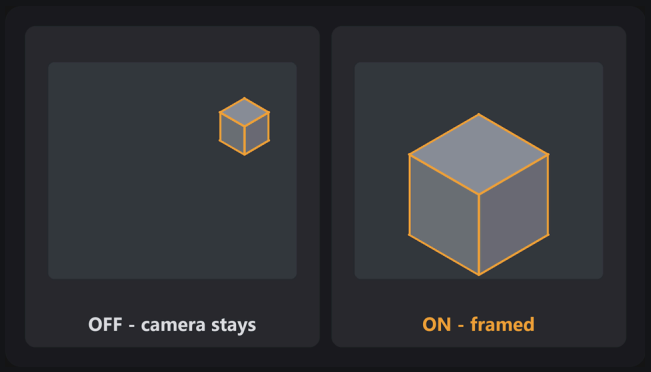


Figure 7.12 Frame On Isolate: on, the camera frames what is left.

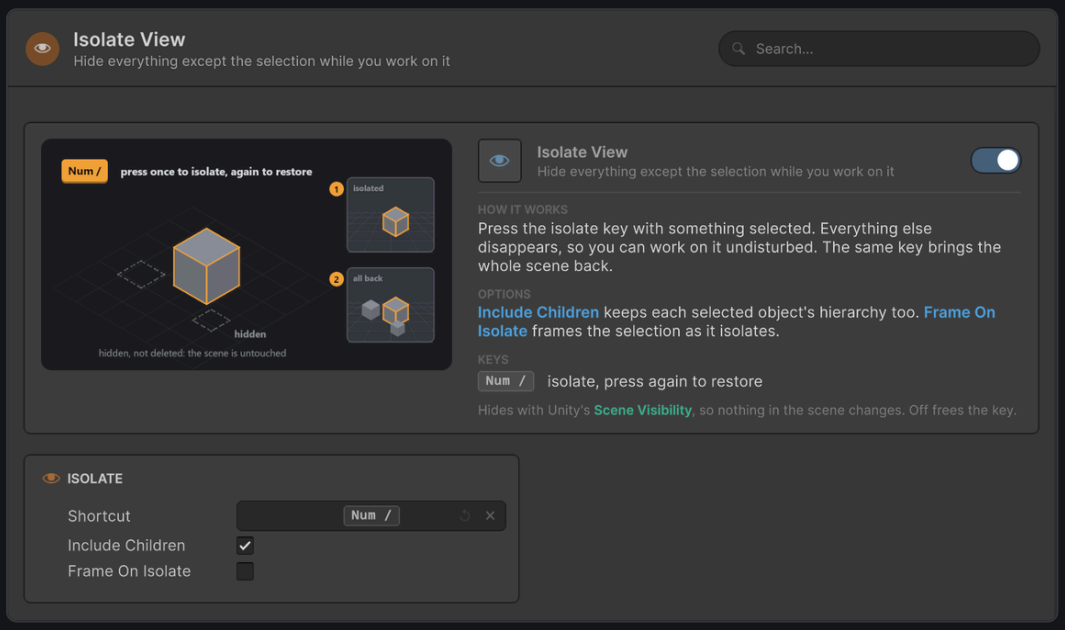


Figure 7.13 The Isolate View page.

7.4 Navigation History

Browser-style back and forward for both what you had selected and where you were looking, keeping up to 100 entries each.

- **Selection History.** Jump back and forward through previous selections. The default binding is **Mouse 3 / Mouse 4**, the side buttons on most mice.
- **View History.** Undo and redo Scene view camera positions. The default binding is **Shift + Mouse 3 / Shift + Mouse 4**, optionally animated.



Figure 7.14 Selection History: the mouse's side buttons step back and forward through what you had selected.

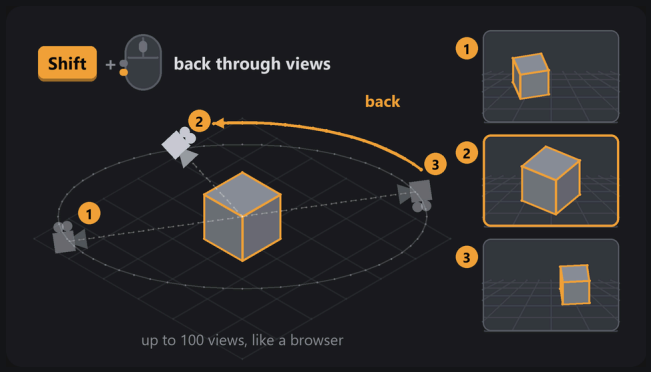


Figure 7.15 View History: with Shift, the same buttons step through camera views.



Figure 7.16 The Navigation History page. Each half can be switched off on its own.

7.5 Box Select - Left Mouse drag

Blender's box select. A box dragged over the Scene view selects every object it touches, not only the ones that fit wholly inside it as the Editor's own box does. Box Select ships switched off; switch it on from **Scene Tools -> Box Select** or from the Overview page.

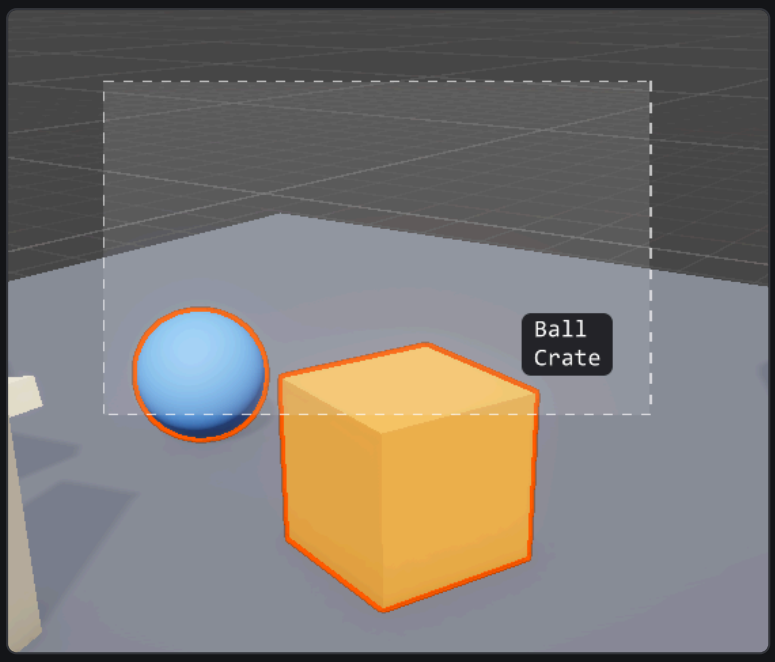


Figure 7.17 Only the top of the ball and a corner of the crate are inside the box, yet both are marked for selection. The readout at the box's corner names them.

- **Exact, not approximate.** The box is tested against each object's real triangles, sprite shape or line path, so a long diagonal object is selected only where the box actually crosses it, never just because its bounding box reaches in.
- **Visible Only.** On by default: the box only selects what you can actually see inside it, like Blender with X-Ray off. Press **C**, also in the middle of a drag, to switch it and select objects hidden behind others too. Sprites, lines and trails are always tested whole.
- **Add, remove and combine.** Hold **Shift** to add to the selection and **Ctrl** (**Cmd** on macOS) to remove from it. With both held, **Shift + Ctrl** does what its setting says: **Difference** (default) flips everything under the box, **Intersect** keeps only what was selected and is also under the box. These follow the Editor's own *Add Box Select* and *Invert Box Select* shortcuts, so rebinding those in Unity's Shortcuts window rebinds them here as well.
- **Move Box.** Hold **Space** while dragging to move the whole box instead of resizing it. Letting go carries on resizing from where the box now is.
- **While Dragging. Highlight** (default) marks the objects the release will select, with an outline or a fill. **Live** selects them for real as you go, Hierarchy and Inspector included. **On Release** shows nothing until the button comes up.
- **Cursor Readout.** Names the objects the release changes, or counts them, or both.
- **Fully Enclosed.** Off by default. On, only objects that fit entirely inside the box are selected, as in the Editor.
- **Select Prefab Roots** and **Respect Selection Base.** Touching any part of a prefab instance selects the instance, and touching part of an object under a parent marked `[SelectionBase]` selects that parent, the way a click does.
- **The Scene view's picking rules still apply.** Hidden objects, objects with picking disabled, locked layers and the Prefab Mode stage are respected, and in a LOD group only the level on screen counts.

- **Everything else stays the Editor's.** Handles, tool contexts such as Splines or Terrain painting, and Alt-orbit keep the box away exactly as they do without Blendon. Terrain, particle systems, UI and gizmo icons (lights, cameras) are selected when the box contains them wholly, as in the Editor.

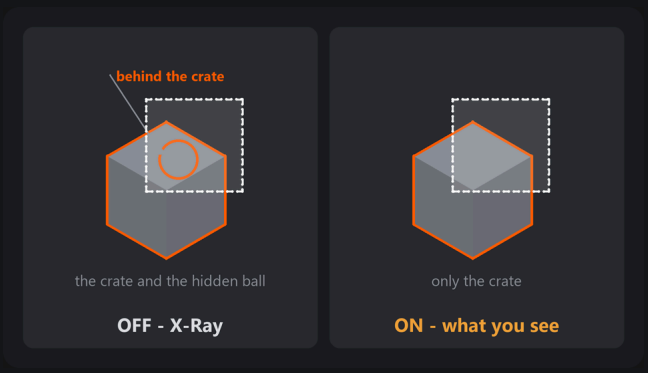


Figure 7.18 Visible Only: on, only what you can see; off, hidden objects behind others too.

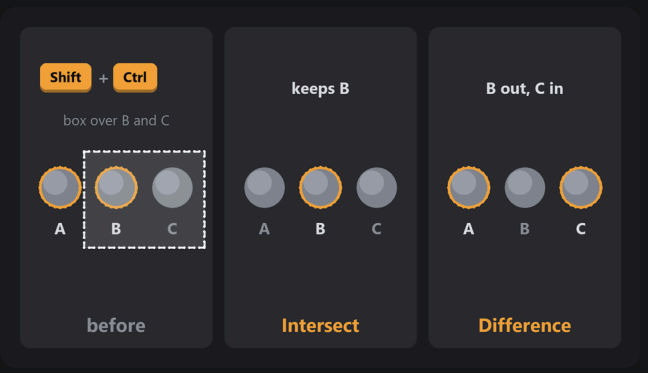


Figure 7.19 Shift + Ctrl: Difference flips what the box touches, Intersect keeps the overlap.

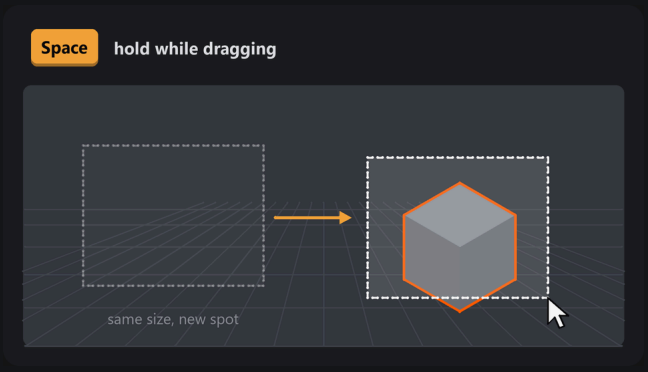


Figure 7.20 Hold Space while dragging to move the whole box.

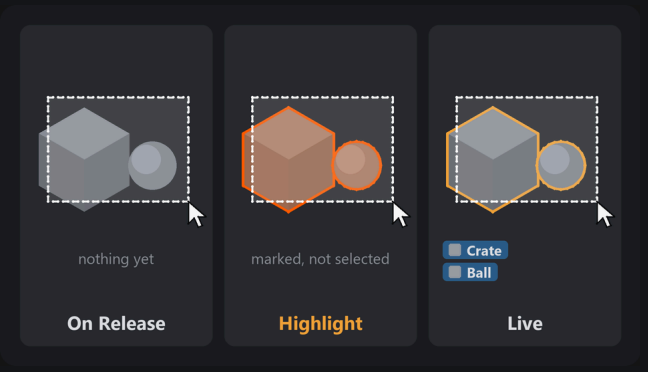


Figure 7.21 While Dragging: nothing until release, a highlight, or a live selection.

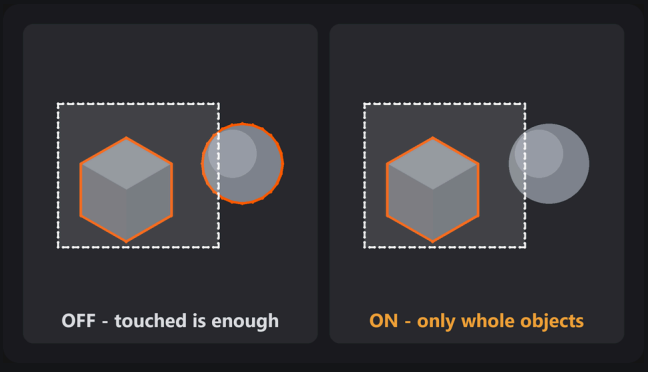


Figure 7.22 Fully Enclosed: off, touching is enough; on, only whole objects.

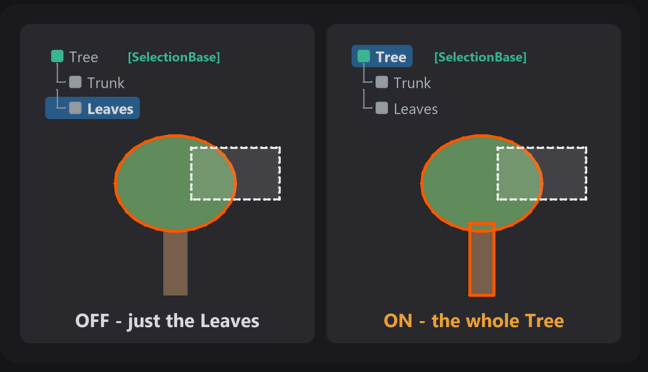


Figure 7.23 Respect Selection Base: touching the leaves selects the whole tree.

Each box is a single undo step. **Esc** or a right-click cancels a box mid-drag.

7.5.1 Click Selects Parent

On the same page, and on by default. Clicking the selected object again selects its parent, and each click after that the next parent up. Past the topmost one it goes to the object right under the cursor, so every level can be reached by clicking, including the parts under a `[SelectionBase]` or a prefab root. Clicks

with **Shift** or **Ctrl** are left as they are.

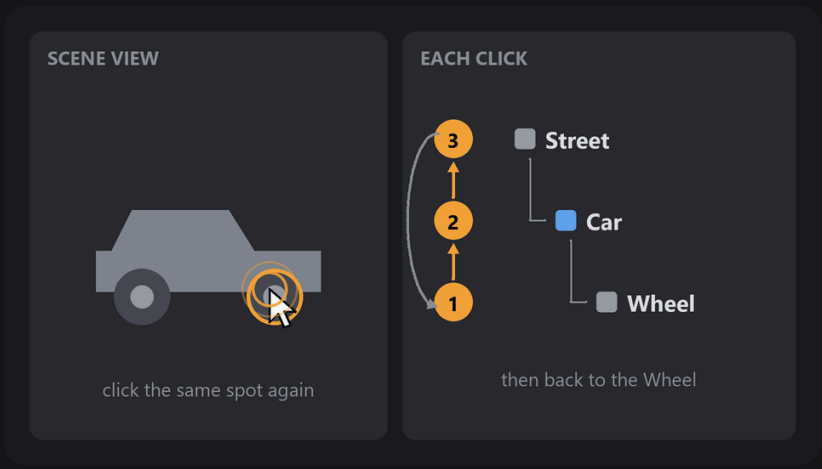


Figure 7.24 Click the same spot again, and the selection climbs to the parent, then the next one up.

7.5.2 Hierarchy keys

Blender's **[** and **]** walk the hierarchy from the Scene view. Objects with nowhere to go, with no parent or no children, stay selected.

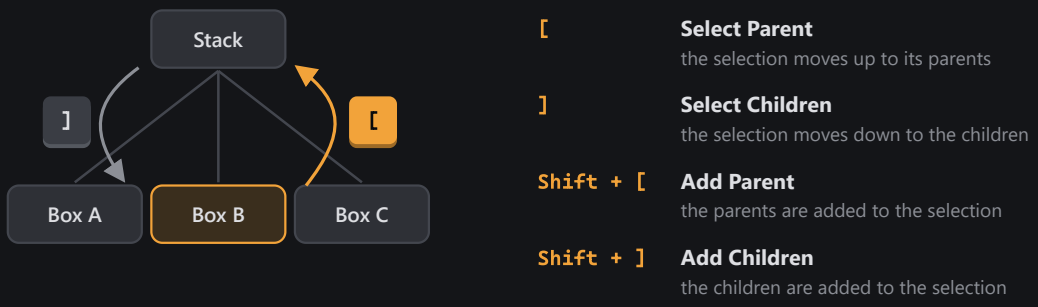


Figure 7.25 The hierarchy keys, shown on the stack from the pictures above.



Figure 7.26 The Box Select page: its header card, then the Selection, Visible Only and Hierarchy Keys panels.

8 Pie menus

Blender's radial menus, with both of its interaction styles on one key.

- **Flick.** Hold the key, move the mouse in a direction, then release. The item in that direction runs immediately.
- **Tap.** Press and release the key quickly without moving. The menu stays open, you can move freely, and a left-click runs an item.
- **Cancel.** Press **Esc**, right-click, left-click outside any item, or release the key while still inside the center deadzone.

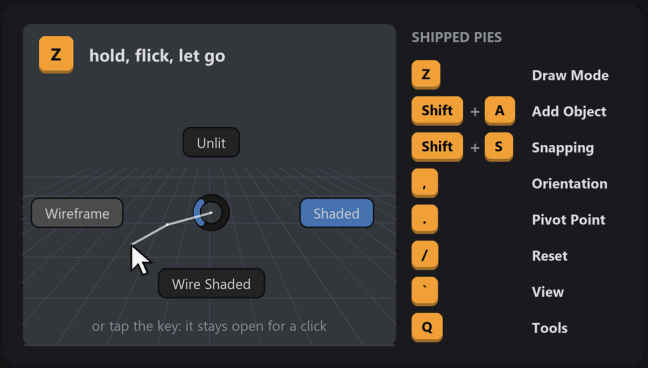


Figure 8.1 Hold a pie's key, flick toward an item, let go. A quick tap keeps the menu open for a click.

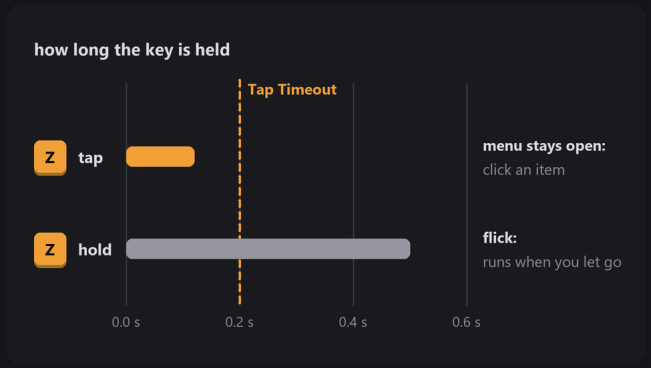
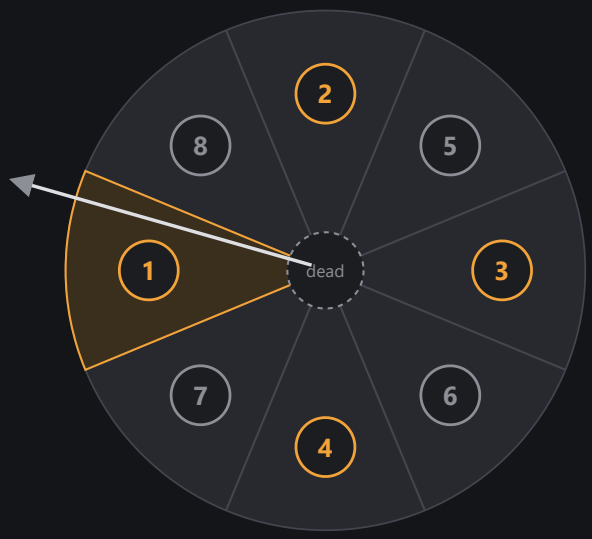


Figure 8.2 How long the key is held decides: a tap leaves the menu open, a hold flicks.



Slots 1-4
left, up, right, down

Slots 5-8
upper-right, lower-right, lower-left, upper-left

Selection by angle
only the direction counts, not the distance

Wedges
90 deg with four items, 45 deg with eight

Deadzone
release inside it and nothing runs

Figure 8.3 How a pie is laid out. The first four items take the sides, the next four the diagonals, and only the direction of the cursor counts.

Selection is by angle alone, measured from the menu's center, so you never have to land on a button: a flick 500 pixels past an item still picks it. A four-item menu gives each item a 90-degree wedge, and adding diagonals narrows every wedge to 45 degrees, exactly as Blender does.

While a menu is open it owns the Scene view. There is no marquee select, no orbit or zoom, no gizmo drag, and every Blendon shortcut stands down. Items are numbered, and their number keys, on the top row or the keypad, run them directly.

An item that matches the Editor's current state, such as the active tool, the draw mode or a snap toggle that is on, is tinted blue. The item under the cursor is lit. An item that cannot run right now, such as a Reset with nothing to reset or a tool whose package is not installed, is greyed out in place rather than removed, so the rest of the ring never shifts.

8.1 Built-in pies

Blendon ships eight pies. Each one can be switched off on its own, rebound, and edited exactly like a pie of your own; see Custom pie menus.

Pie	Default	Items
Draw Mode	Hold **Z**	Wireframe, Unlit, Shaded, Wireframe Shaded
Add Object	Hold **Shift + A**	Cube, Empty, Sphere, Camera, Cylinder, Light, Particle System, Plane
Snapping	Hold **Shift + S**	Incremental, Angle, Scale, Grid
Orientation	Hold **,**	Global, Grid, Local
Pivot Point	Hold **.**	Center, Median, Active, Pivot
Reset	Hold **/**	Position, Scale, Rotation, All
View	Hold `` ` ``	Left, Top, Right, Bottom, Front, Perspective / Orthographic, Camera, Back
Tools	Hold **Q**	Move, Scale, Rotate, Transform, Rect, Edit Box Collider, Create Spline, View

Items fill the ring in Blender's own order: the first four go **left, up, right and down**, and the next four go to the **upper-right, lower-right, lower-left and upper-left** diagonals. An empty item keeps its slot blank, which is how Orientation leaves Blender's own layout intact around the orientations Unity has no match for.

8.2 Draw Mode pie - Z

Blender's Z menu. Unity's Scene view has neither Blender's matcap Solid mode nor its HDRI Material Preview, so the pie switches between the four modes Unity does have: **Wireframe** to the left, **Unlit** above, **Shaded** to the right and **Wireframe Shaded** below. The mode the view is already in is tinted, and a mode the active render pipeline does not offer is greyed out.

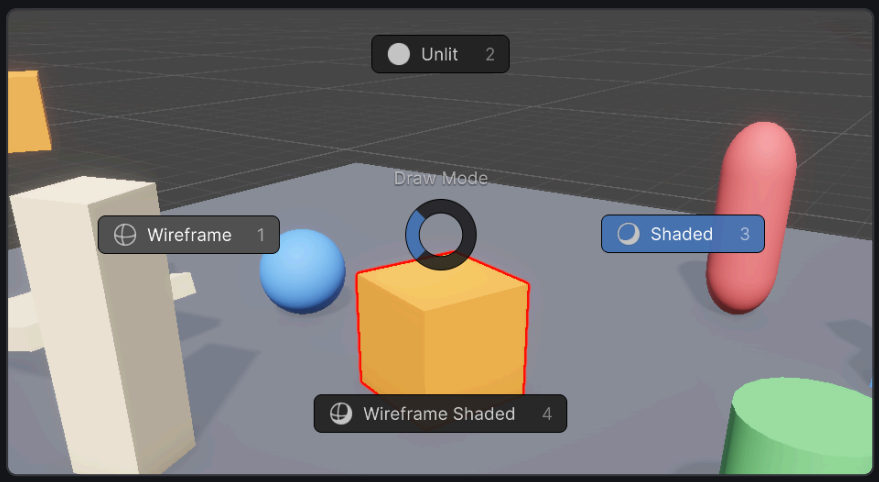


Figure 8.4 The Draw Mode pie. Shaded is tinted because the view is already in it; Wireframe is lit because the cursor points left.

Item	Unity equivalent
Wireframe	Wireframe draw mode; nothing else is touched
Unlit	Shaded draw mode with Scene Lighting off, so the view is lit from the camera; skybox, particles and visual effects on; post-processing, fog and flares off
Shaded	Shaded draw mode with scene lighting, skybox, post-processing, fog, flares, particles and visual effects on, and Always Refresh
Wireframe Shaded	Shaded Wireframe draw mode, with the same lighting and effects as Shaded

Unlit and Shaded are the same draw mode told apart by the Lighting toggle, so flipping that toggle by hand switches which one the menu considers current. Every other effect toggle you change is remembered per view and per mode for the rest of the Editor session, so leaving a mode and coming back restores your own toggles rather than the defaults.

In Unity's **Edit -> Shortcuts** window this pie is listed as `Blendon/Pie Menus/Shading`.

8.3 Toggle Wireframe - Shift + Z

The pie's other half, the way Blender pairs the two keys. **Shift + Z** drops the Scene view straight into Wireframe, and pressing it again returns to the mode you came from, or to Shaded if there is no earlier mode to go back to, such as in a fresh session or after a debug draw mode the pie does not name. What each mode remembers is unchanged, so the round trip keeps your own effect toggles.

Toggle Wireframe has a toggle of its own on the Pie Menus page, so switching every pie off leaves the key working. During a grab the key means Blender's plane lock instead, and the running manipulation receives it.

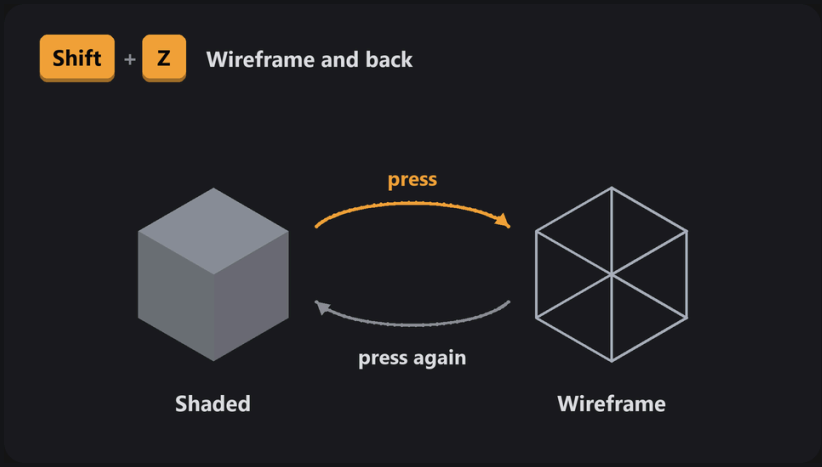


Figure 8.5 Shift + Z drops into Wireframe; pressing it again goes back.

8.4 Add Object pie - Shift + A

Blender's Shift + A menu: **Cube** to the left, **Empty** above, **Sphere** to the right, **Camera** below, then **Cylinder**, **Light** (a Point Light), **Particle System** and **Plane** on the diagonals.

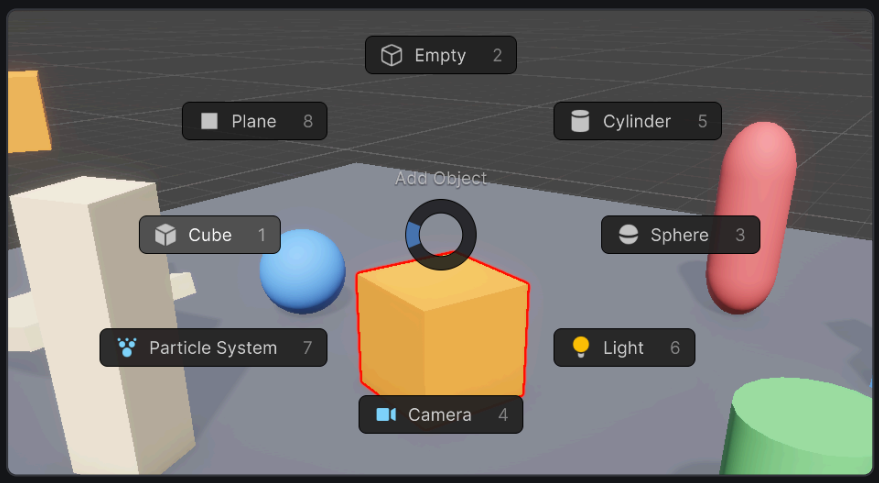


Figure 8.6 The Add Object pie, with eight items.

Each item runs Unity's own **GameObject** menu command, then moves the new object to where the pie was opened: onto the surface under the cursor, or, over empty space, onto the camera-facing plane through the view's pivot. With **Place On Floor** on, which is the default, an object with a renderer is lifted by half its height so that it rests on that point instead of sinking halfway into it. Creation and placement are a single undo step, and the new object is left selected.

The placement works for these items in any pie, not only in this one.



Figure 8.7 Place On Floor: on, a new object rests on the point instead of sinking halfway into it.

8.5 Snapping pie - Shift + S

Toggles Unity's own snap settings, the same state the Scene view toolbar shows and keeps: **Incremental** to the left, **Angle** above, **Scale** to the right and **Grid** below.

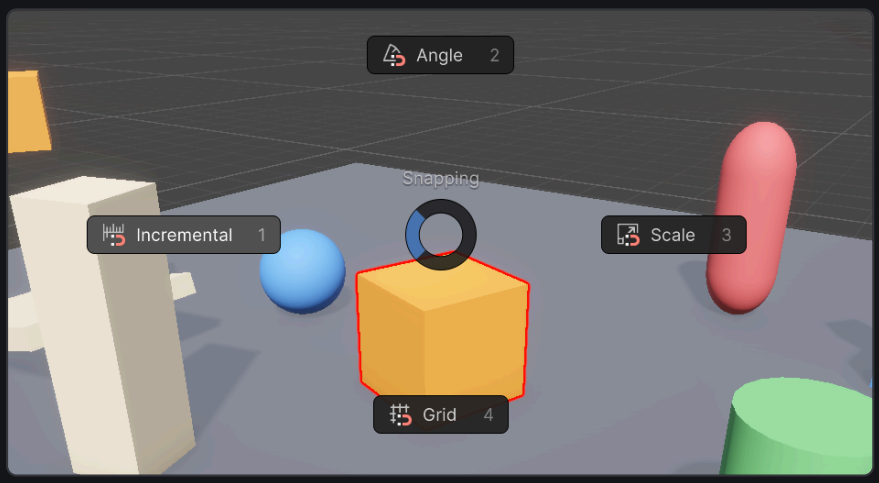


Figure 8.8 The Snapping pie.

- **Incremental.** Snap by the increment values from Unity's snap settings, relative to where the drag started. Turns Grid off.
- **Grid.** Snap to the Scene grid itself.
- **Angle and Scale.** Snap rotation and scale by their increments. Before Unity 6000.4 the Editor has a single toggle for all three increments, so there Incremental, Angle and Scale all flip that same toggle.

8.6 Orientation pie - Comma

Blender's Comma menu, set on Unity's own handle rotation toggle: **Global** to the left, **Local** to the right and **Grid** above, where Blender keeps Gimbal. Grid is greyed out on Unity versions whose toggle has no Grid, such as 6000.0.

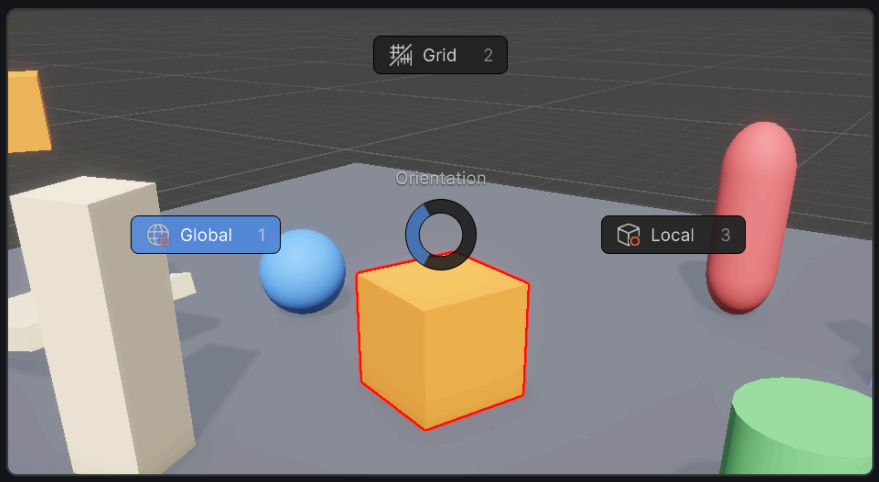


Figure 8.9 The Orientation pie. Global is tinted as the current orientation.

8.7 Pivot Point pie - Period

Blender's Period menu, laid out as Blender lays it out: **Median** (Median Point) above, **Center** (Bounding Box Center) to the left, **Active** (Active Object) to the right and **Pivot** (Individual Origins) below. The ring uses the toolbar button's short names; the action picker lists the full ones. Active sits where Blender keeps 3D Cursor, which Unity has no match for.

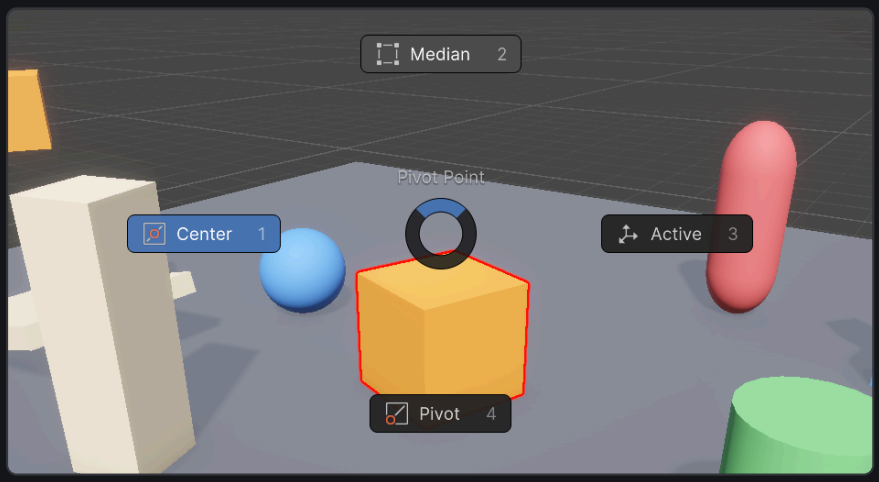


Figure 8.10 The Pivot Point pie. Center is tinted as the current mode.

Picking a mode flips the toolbar's Pivot/Center toggle to its side, the same as the Pivot Point menu. With that menu switched off, Median and Active are greyed out and only Unity's own Center and Pivot are left.

8.8 Reset pie - Slash

Blender's Alt + G, Alt + R and Alt + S, on the **/** key: **Position** to the left, **Scale** above, **Rotation** to the right and **All** below. The pie resets the *local* transform of every selected object, putting position to zero, rotation to none and scale to one, so a child returns to its parent's origin rather than the world's.

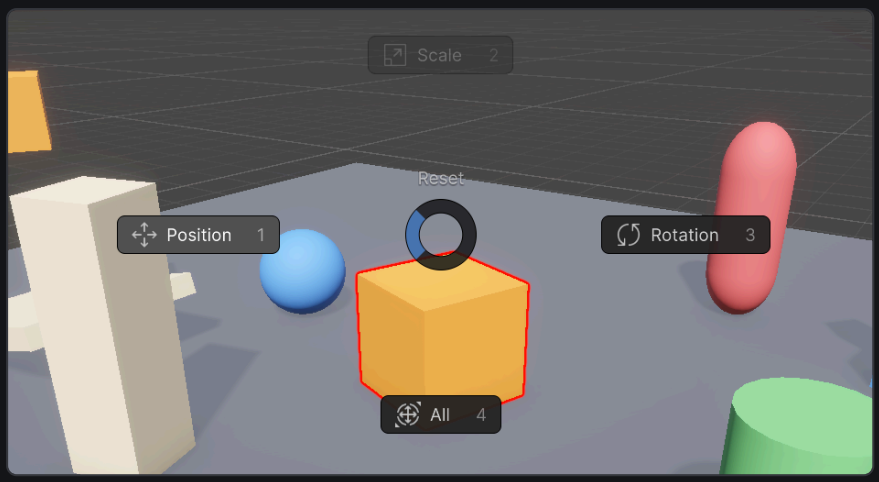


Figure 8.11 The Reset pie. Scale is greyed out because the selected crate is already at scale one.

It is fully undoable, and each item is greyed out while it has nothing to do: **Rotation** when every selected object already sits at zero, **Scale** when every one of them is at one, and the whole pie with nothing selected.

8.9 View pie - Backtick

The numpad's camera moves under the mouse, on the ` key: **Left**, **Top**, **Right** and **Bottom** on the four sides, **Front** upper-right, a projection toggle lower-right, **Camera** lower-left and **Back** upper-left. Each item runs the same move as its numpad key.

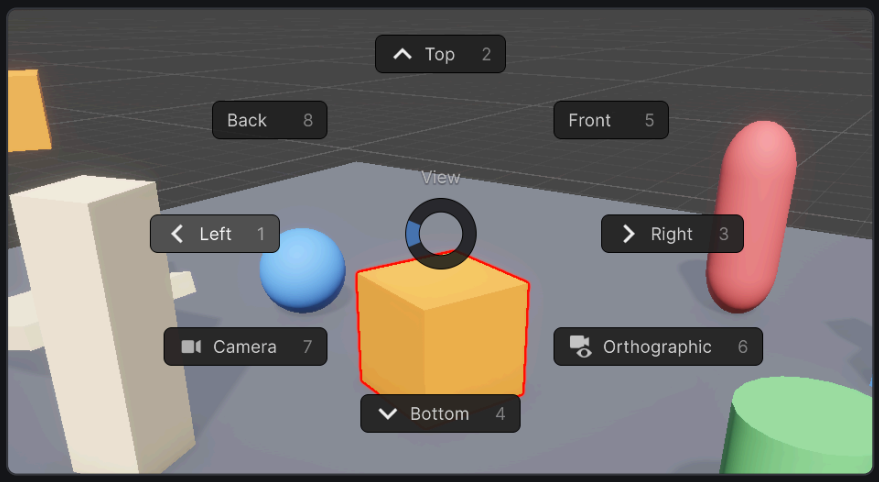


Figure 8.12 The View pie.

The projection toggle is labelled with the projection it switches *to*, **Perspective** or **Orthographic**. **Camera** aligns the view to the active camera like **Numpad 0**, and an axis item is tinted while the view is exactly aligned to it.

8.10 Tools pie - Q

Unity's own tools, without reaching for the toolbar: **Move** to the left, **Scale** above, **Rotate** to the right and **Transform** below, then **Rect**, **Edit Box Collider**, **Create Spline** and **View** on the diagonals. The active tool is tinted.

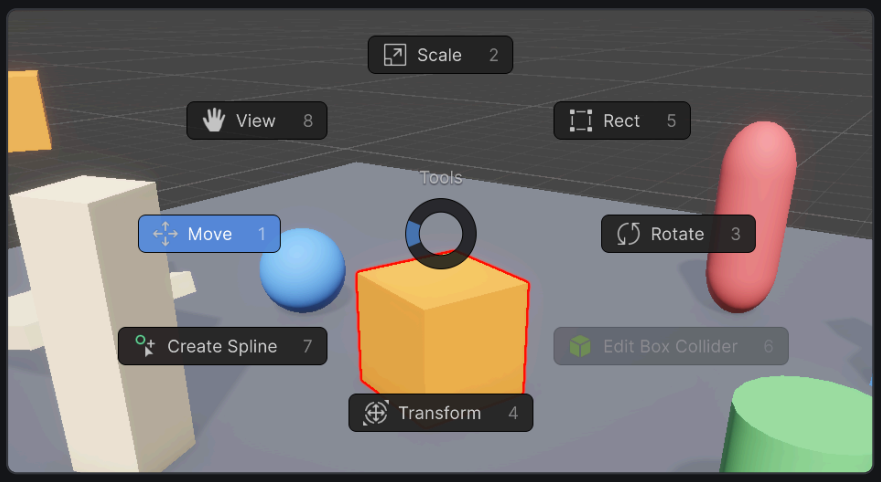


Figure 8.13 The Tools pie. Edit Box Collider is greyed out, for the reason given below.

- **Create Spline** needs the Splines package (`com.unity.splines`) and is greyed out without it. Picking it while it is already active drops back to Move.
- **Edit Box Collider** is greyed out in this version, because Unity's box-collider editing tool has no public entry point. Use the **Edit Collider** button on the Box Collider component instead, or put something else in that slot from the pie editor.

8.11 Custom pie menus

The **Pie Menus** settings page lists the built-in pies above your own.

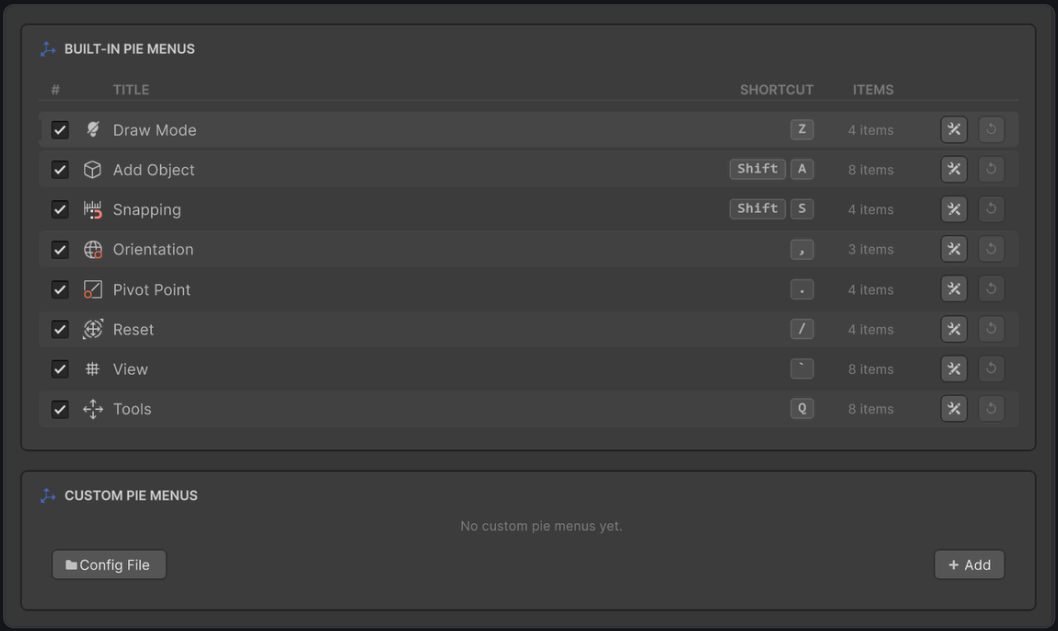


Figure 8.14 The pie lists on the Pie Menus page. Each row has a switch, the title, the key, how many slots run something, and Edit and Reset buttons.

Each row has:

- an on/off switch, which frees the pie's key for the rest of the Editor when off;
- the title, which opens the pie editor when clicked;
- the pie's key, highlighted if it clashes with another command, and how many of its slots run something;
- **Edit**, plus **Reset** for a built-in pie, which puts back the shipped title, items and key, or **Remove** for one of your own.

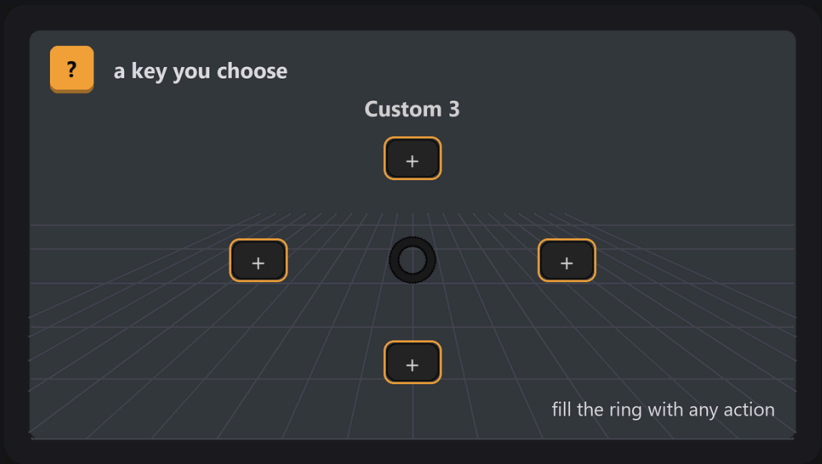


Figure 8.15 A pie of your own: any key, and any Blendon action or Unity menu command in its slots.

Add creates a new pie. There are eight custom slots, each with its own entry in Unity's Shortcut Manager (`Blendon/Pie Menus/Custom 1` to `Custom 8`), unbound until you give it a key. **Config File** reveals the file every pie on the page is saved in; see Data and files.

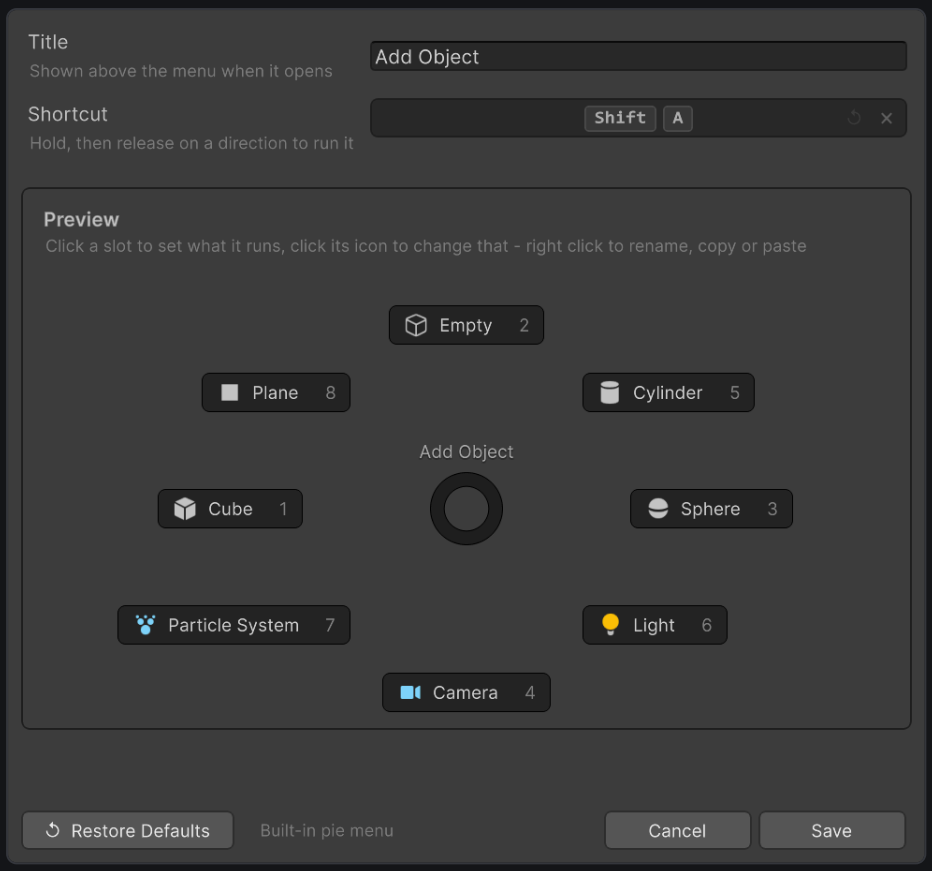


Figure 8.16 The pie editor, open on the built-in Add Object pie.

The pie editor edits one menu, built-in or custom:

- **Title.** Drawn over the center of the menu.
- **Shortcut.** Click the field and press the key to hold. A clash is flagged inline, naming the command it collides with. For a built-in pie this is the same binding that **Edit -> Shortcuts** shows.
- **The ring.** Both the preview and the editor, with all eight slots drawn in their real directions. Left-click a slot to choose what it runs, or click its icon to choose an icon. Right-click a slot for **Change Action**, **Set Icon**, **Rename**, **Copy**, **Paste** and **Clear Action**. A copied slot keeps its icon and name, and can be pasted into a different pie.
- **The action picker.** Lists every Blendon action, which is everything the built-in pies are made of, under a **Blendon** branch, followed by every command in Unity's main menu, including commands added by other packages and assets. Type to search, or use the picker's shortcut-capture button and press a key to find the menu command bound to it. A menu command greys out and tints in the ring exactly when the menu bar would grey it out or tick it.
- **Save and Cancel.** Save keeps the edits. Cancel, or closing the window, discards all of them, the shortcut included. A built-in pie also has **Restore Defaults**.

8.12 Shared pie settings

The top of the Pie Menus page tunes every pie at once.

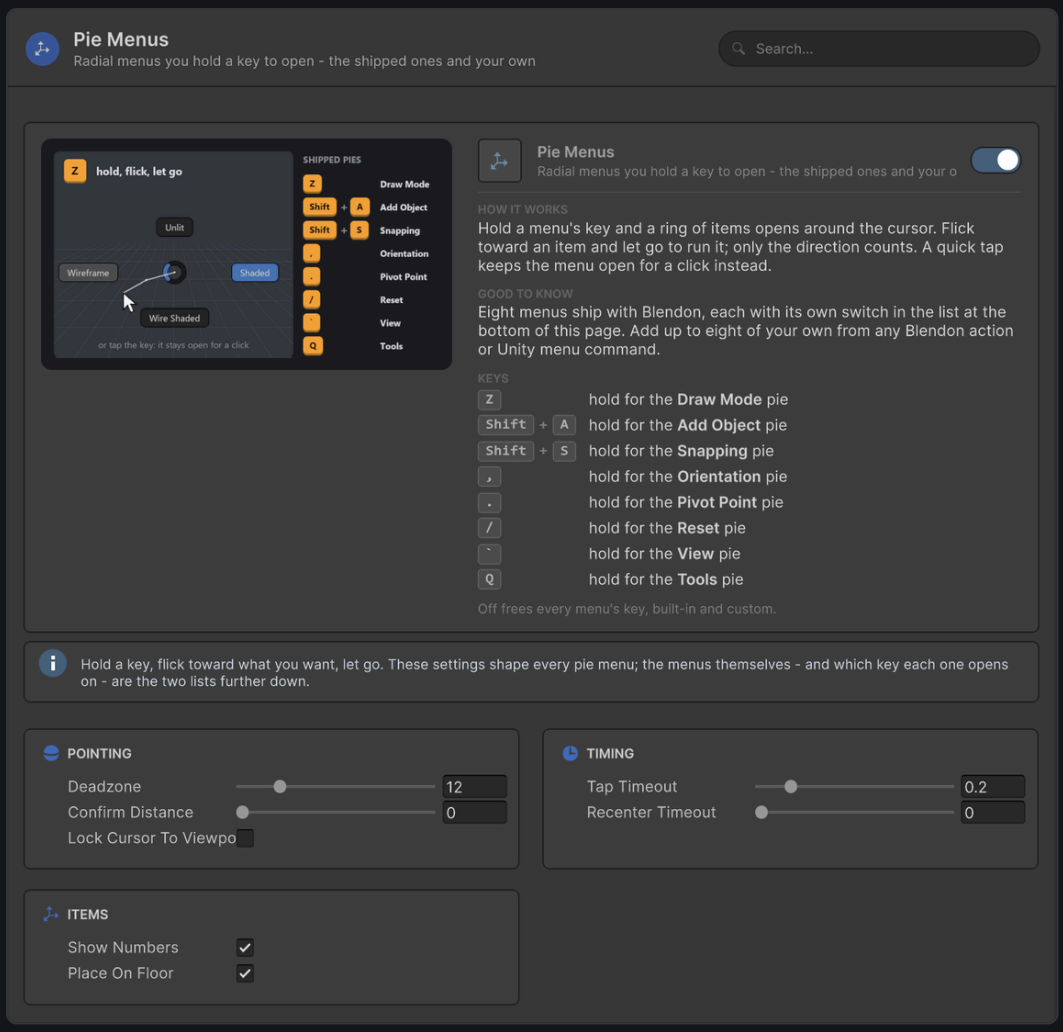


Figure 8.17 The top of the Pie Menus page: the header card with every shipped pie's key, then pointing, timing and items.

Setting	Default	Effect
Enable	On	Master switch. Off parks every pie's key, built-in and custom, handing the keys back to the Editor. Toggle Wireframe keeps its own switch.
Deadzone	12 px	How far the cursor has to move from the center before it points at an item.
Confirm Distance	0 (off)	Run the highlighted item once the cursor is this far past the deadzone and stops moving, without waiting for the key to be released.
Lock Cursor To Viewport	Off	Keep the cursor inside the Scene view while a pie is open.
Tap Timeout	0.2 s	Release the key faster than this without moving, and the menu stays open for a click.
Recenter Timeout	0 (off)	A menu opened near a view edge is nudged inward to fit. For this long afterward, directions are still measured from where the cursor really was.

Setting	Default	Effect
Show Numbers	On	Number the items so that their digit keys run them. The keypad digits need NumLock on.
Place On Floor	On	Lift an object created from a pie by half its height, so that it rests on the point instead of sinking into it.
Radius	100 px	How far the items sit from the center. Under **Appearance**.
Unfold Time	0.06 s	How long the items take to spring out from the center; 0 shows them in place at once. Under **Appearance**.
Colors	-	Item, Highlight, Current, Text and Center Ring. Under **Appearance**.

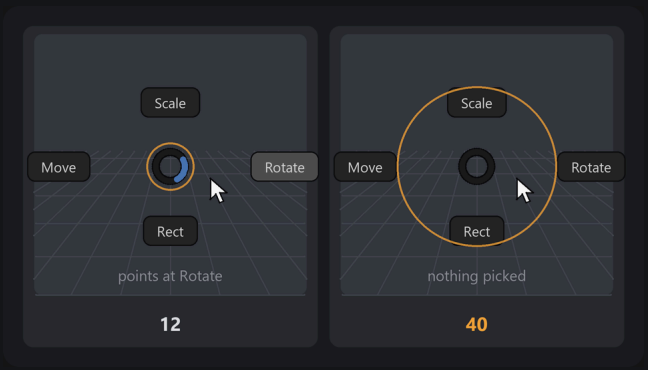


Figure 8.18 Deadzone: how far the cursor moves before it points at an item.

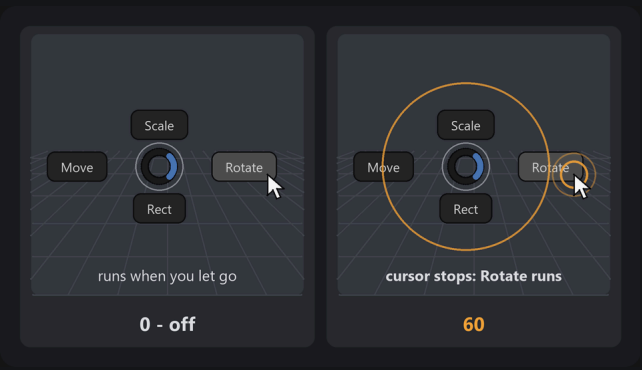


Figure 8.19 Confirm Distance: stop past this distance and the item runs without letting go.

9 Context menu

Right-clicking the Scene view opens Blendon's context menu instead of the Editor's: a compact, searchable panel drawn directly in the viewport, the way Blender draws its menus.

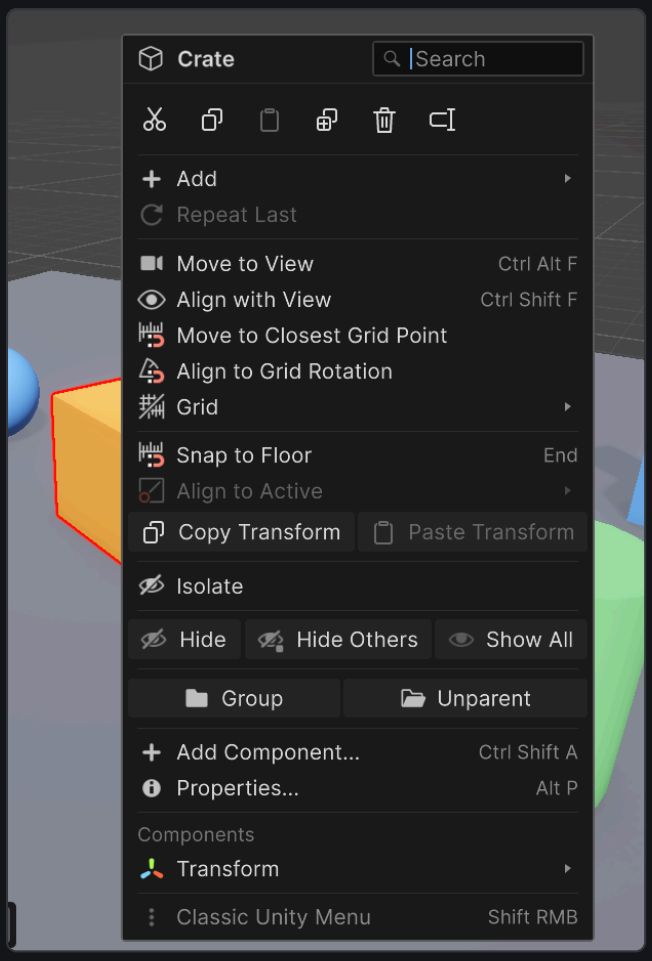


Figure 9.1 The context menu for a single object: the header with its name and search field, the quick-action icons, then the grouped entries.

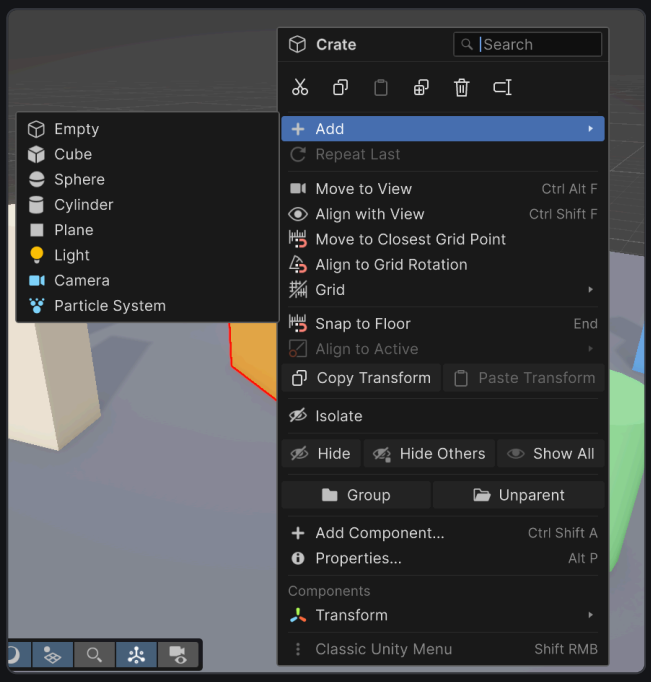


Figure 9.2 Hovering Add opens its submenu. Near the right edge of the view it flips to the left.

- **Every entry.** The menu is built from exactly the entries Unity's own right-click menu would show for the current selection and active tool, including anything that other packages and assets add to it. Blendon only arranges and decorates them: an entry it does not recognise still appears, under **Extensions** or **More**, in the Editor's own order. Greyed-out and ticked states match the Editor's menu as well.
- **Quick actions.** Cut, Copy, Paste, Paste As Child, Duplicate, Rename and Delete sit in a single row of icon buttons along the top, whichever of them Unity offers for the selection, with Blendon's own **Rename** closing the row. Hover over one to see its name and key. Set **Edit Actions** to **List** to show them as ordinary rows instead.
- **Sections.** The remaining entries are grouped: aligning to the view and the grid, isolation, object commands, prefab commands, the selection's components, and whatever the active tool or tool context adds. Rows show an icon and, where the command has one, its key.
- **Submenus.** A submenu opens beside its row as you hover over it, and flips to the left near the edge of the view.
- **Search.** The header names the selection and holds a search field that already has the caret. Start typing to filter every entry, submenus included, then press **Enter** to run the highlighted match. While you type, the Editor's own shortcuts stand down, so letters land in the field instead of framing the view or switching tools.

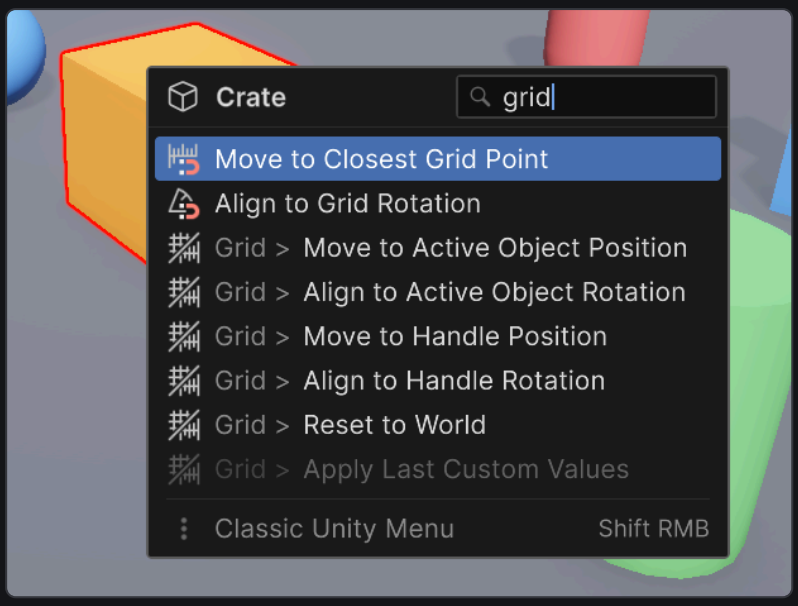


Figure 9.3 Typing "grid" filters every entry, submenus included. Entries from a submenu show their path in grey.

- **Rename.** Click the name in the header, or the **Rename** entry in the edit row, and the rest of the header line, beside the selection's icon, becomes a text field holding the current name, selected and ready to be typed over. **Enter** takes the new name as a single undo step, **Esc** leaves it as it was; either way the menu stays open and the header goes back to showing the name. It is there whenever one object in the scene is selected.
- **Extra entries.** Blendon can add entries the Editor's menu does not carry; see the table below. Each joins the group it belongs to rather than piling up at the end.
- **Keyboard. Up** and **Down** move the highlight, **Right** opens a submenu or steps along the quick actions, **Left** closes the submenu, **Enter** runs the highlighted entry, and **Esc** closes a submenu, then clears the search, then closes the menu.
- **Closing.** Running an entry, clicking outside the menu, changing the selection or switching to another window closes it.

9.1 The Editor’s own menu

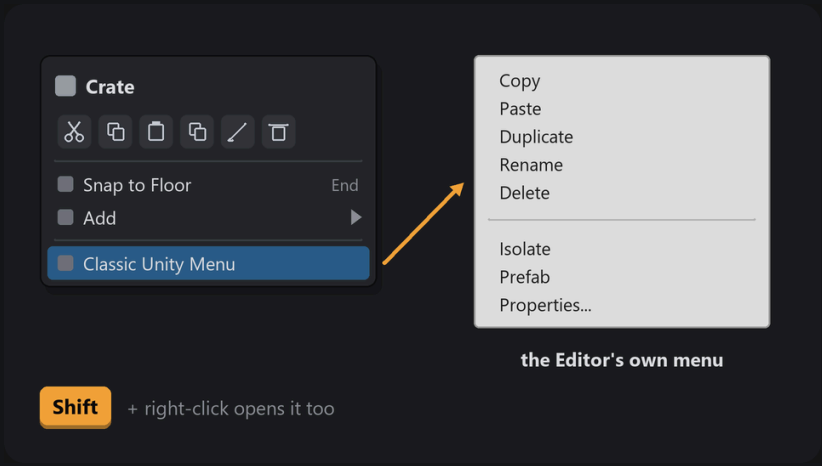


Figure 9.4 The Editor’s own menu is one row away, or Shift + right-click.

The Editor’s own menu is always one step away. **Shift + right-click** opens it instead, and so does the **Classic Unity Menu** row at the bottom of Blendon’s menu, unless **Classic Menu Row** is switched off. Both hold the very same entries.

A right-click that drags is left alone, so holding the right mouse button to fly through the scene works exactly as before. A right-click with **Ctrl**, **Alt** or **Cmd** held also goes to the Editor.

9.2 Context menu settings

The **Context Menu** page, under **Keys & Menus**, holds the menu’s shortcut, four appearance settings and a table of extra entries.

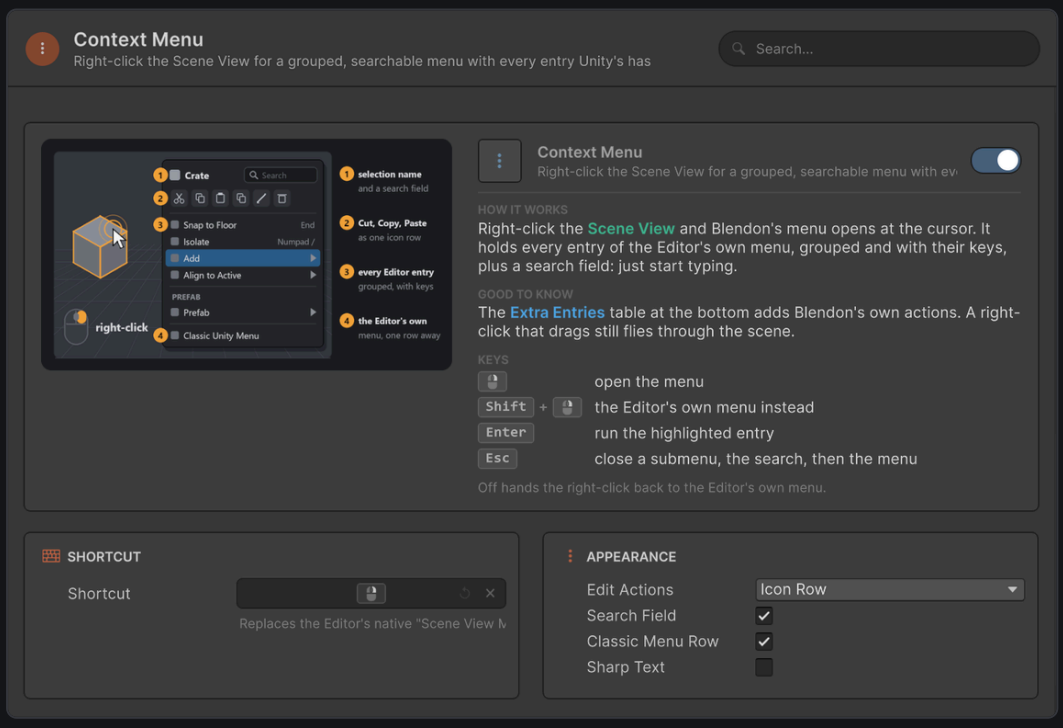


Figure 9.5 The Context Menu page: the header card, the shortcut and the four appearance settings.

Setting	Default	Effect
Enable	On	Off hands the right-click back to the Editor's own menu.
Shortcut	Right-click	The button that opens the menu, `Blendon/Context Menu` in Unity's Shortcut Manager. It replaces the Editor's own Scene View menu binding, and handing it back is a matter of switching the page off.
Edit Actions	Icon Row	**Icon Row** puts Cut, Copy, Paste, Duplicate, Rename and Delete in one row of buttons along the top. **List** shows them as ordinary rows with their names and keys.
Search Field	On	Shows the search box in the header. Off leaves the whole header to the selection's name; the Editor's shortcuts still stand down while the menu is open.
Classic Menu Row	On	Shows the **Classic Unity Menu** row at the bottom. **Shift + right-click** opens the Editor's own menu either way.
Sharp Text	Off	Draws the menu's text on whole screen pixels. Off draws it the way the Editor's own panels do, which is softer. Icons stay sharp either way.

Each extra entry adds something of Blendon's own. An entry the selection cannot take is greyed out rather than hidden.

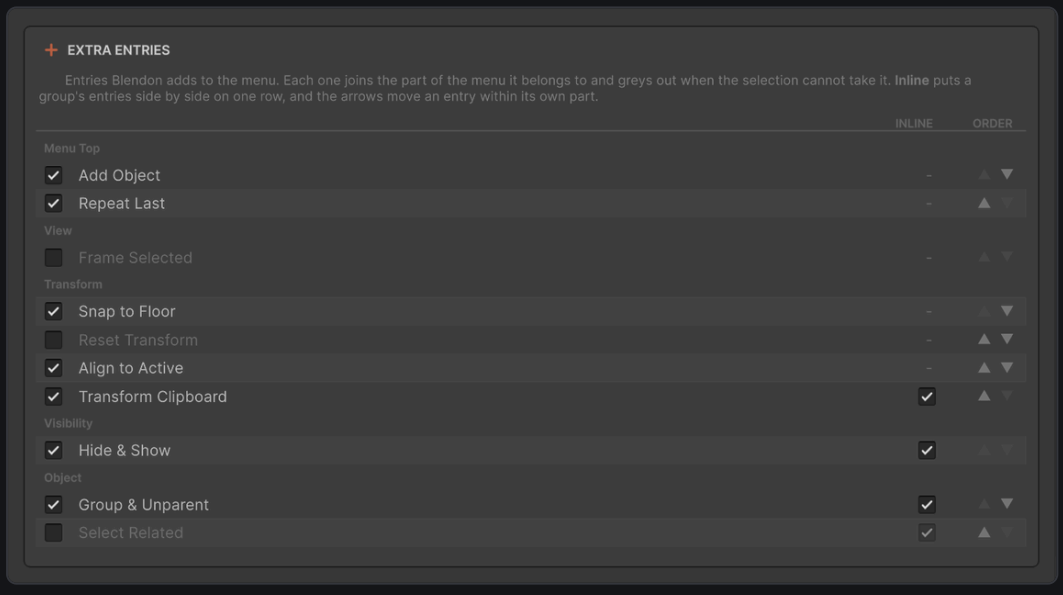


Figure 9.6 The Extra Entries table. The Inline column puts a group's entries side by side on one row, and the arrows reorder an entry within its group.

Extra Entry	Default	What it adds
Add Object	On	An **Add** submenu with the usual objects: empty, primitives, light, camera, particles. Whatever it creates lands at the point you right-clicked, on the surface under the cursor or on the grid, the way the Add pie menu places things.

Extra Entry	Default	What it adds
Repeat Last	On	Repeats the entry you ran last, Blendon's own as well as the Editor's, and names it on the row.
Frame Selected	Off	Frames the selection, through Blendon's own framing cycle where that feature is on and the Editor's otherwise.
Snap to Floor	On	Drops the selection onto whatever is below it, the same action as Blendon's **Snap to Floor** shortcut.
Reset Transform	Off	A **Reset** submenu for position, rotation, scale, or all three at once. A channel already at rest, rotation at zero or scale at one, is greyed out, the same as in the Reset pie.
Align to Active	On	Puts the rest of the selection where the active object stands: its position, its rotation, or both.
Transform Clipboard	On	**Copy Transform** takes the active object's position, rotation and scale. **Paste Transform** hands them to whatever is selected next. Inline by default.
Hide & Show	On	**Hide**, **Hide Others** and **Show All**, over the Editor's own scene visibility, the same thing the eye column in the Hierarchy toggles. Inline by default.
Select Related	Off	A **Select** submenu that walks the hierarchy from the Scene view: the children, the whole subtree, or the parent.
Group & Unparent	Off	**Group** puts the selection under a new empty parent. **Unparent** moves it back out to the top of the scene. Inline by default.

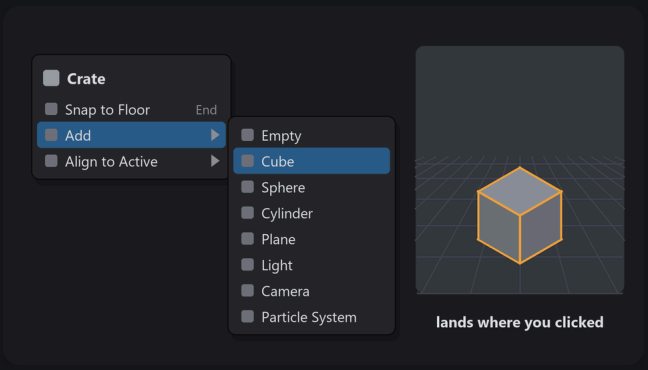


Figure 9.7 Add: the new object lands where you right-clicked.



Figure 9.8 Align to Active: the others move onto the active one.

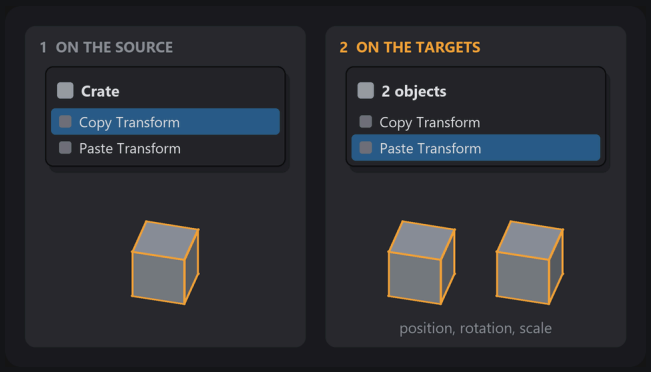


Figure 9.9 Copy Transform on the source, Paste Transform on the targets.

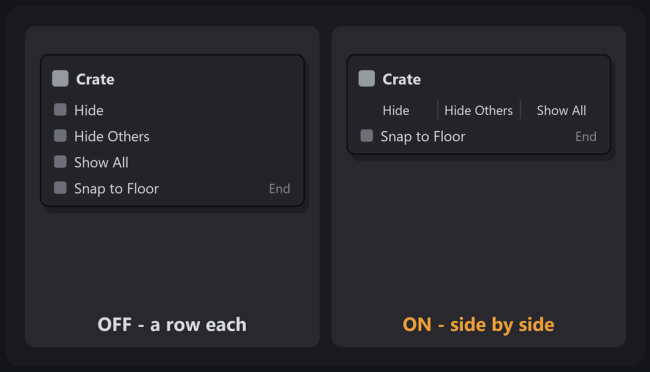


Figure 9.10 Inline: a group's entries side by side on one row.

The **Essentials** and **Tools Only** presets switch the context menu off, and **Full** switches it on.

10 Scene tutorial

A floating checklist card walks new users through Blendon's gestures directly in the Scene view. It can be dragged anywhere by its header, collapsed to a small pill, or dismissed; dismissing asks for confirmation and can be undone later from the Overview settings page.

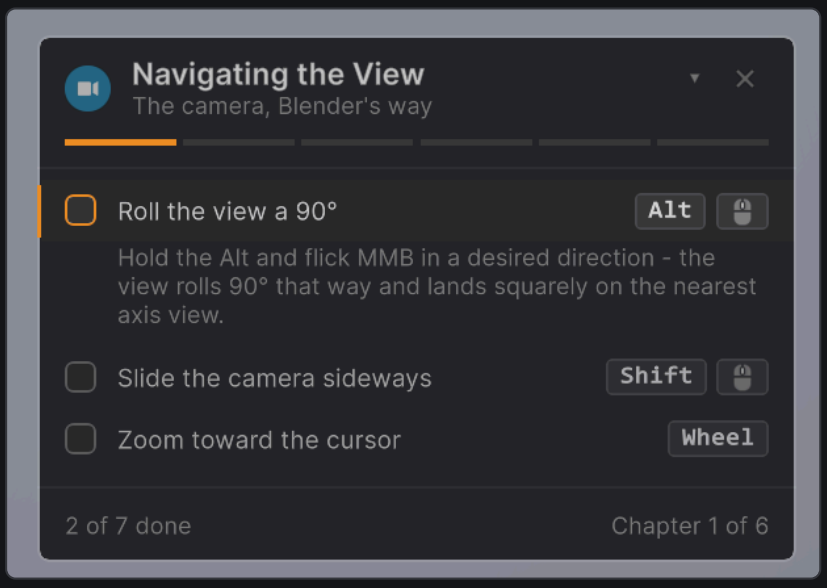


Figure 10.1 The tutorial card on its first chapter, with two of seven tasks done. The current task is highlighted and explained, and its keys are shown on the right.

A task ticks off only once you actually perform the gesture, not by reading about it. The current task is highlighted, and a progress bar tracks how far you have come. Three tasks are shown at a time: a finished one sits still for a moment, fades, and the list scrolls up to bring in whatever was waiting below, so a chapter of eight tasks takes no more room than one of four.

There are six chapters and 33 tasks, in order:

1. **Navigating the View.** Orbit, axis snap, quick roll, pan, zoom to cursor, frame selected and numpad axis views.
2. **Grab, Rotate, Scale.** Grab-move, confirm, cancel, grab-rotate and grab-scale.
3. **Handles & Snapping.** Axis and plane handles, rotation rings, surface snap, look-at, vertex snapping, virtual pivot and snap to floor.
4. **Precision.** Axis and plane constraints, typed numeric input, increment snapping and precision drag.
5. **Scene Tools.** Isolate view, selection history, view history and the orientation gizmo.
6. **Pie Menus & Setup.** Flicking and tapping a pie menu, running an item by number and opening settings.

Progress is kept per machine rather than per project, so it survives across projects, and it is left untouched by the settings window's **Reset All Pages**. The tutorial can be restarted at any time with **Start Over** on the Overview page.

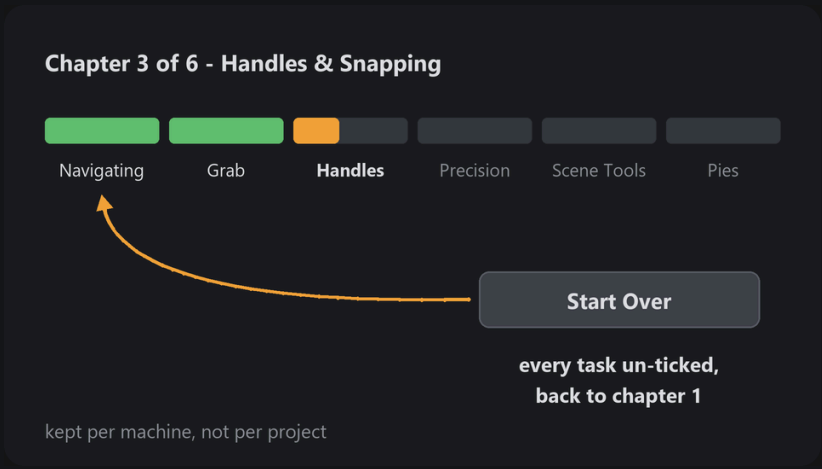


Figure 10.2 Start Over un-ticks every task and goes back to chapter 1.

10.1 Shortcut tips

The first time you press one of Blendon's keys that took over an Editor command, a card appears in the same corner of the Scene view. It names the command that stepped aside and where it lives now, with a button to open each one.

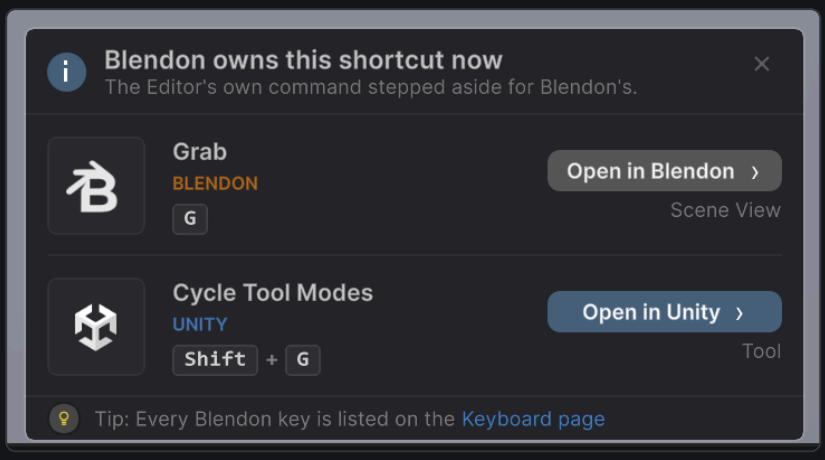


Figure 10.3 A shortcut tip after the first press of G: Blendon's Grab now owns the key, and the Editor's Cycle Tool Modes moved to Shift + G.

Each card is shown once per key per Editor session, and armed again whenever a binding changes. While the tutorial is on screen it owns the corner, so no tip interrupts it. **Shortcut Tips** on the Overview page switches the cards off, and **Show Again** arms every one of them again.

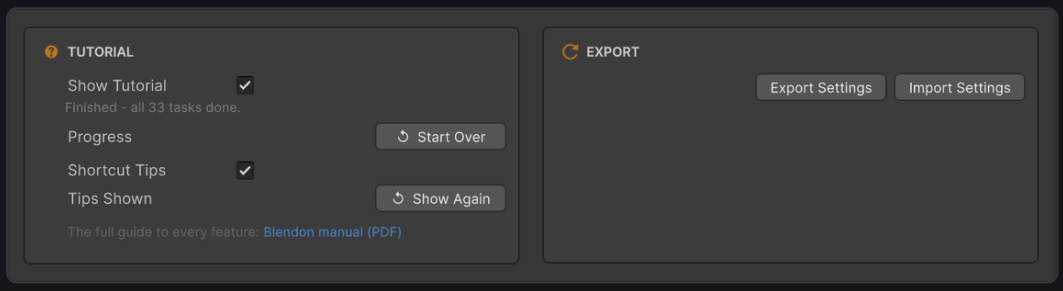


Figure 10.4 The Tutorial and Export panels on the Overview page.

11 Modifier keys

Blendon's hold-to-engage gestures have no entry in Unity's Shortcut Manager, so they are configured as modifier dropdowns in the settings window. All of them are rebindable, and each can be set to **None** to switch the gesture off.

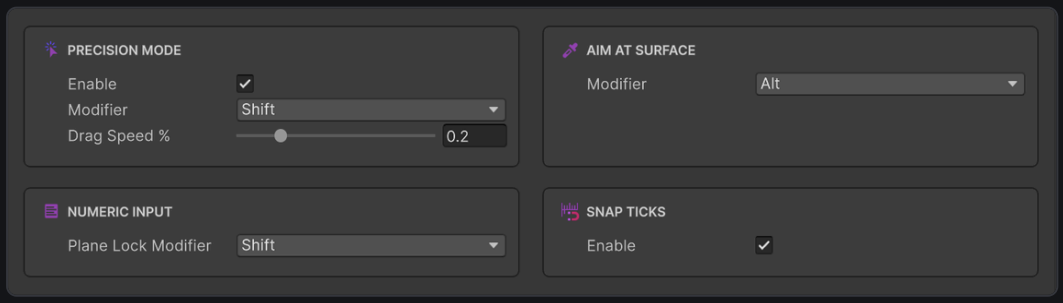


Figure 11.1 The modifier dropdowns on the All Tools page.

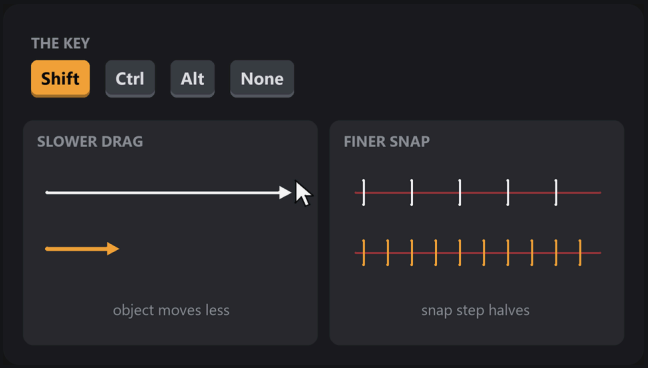


Figure 11.2 The precision modifier slows the drag and halves the snap step.

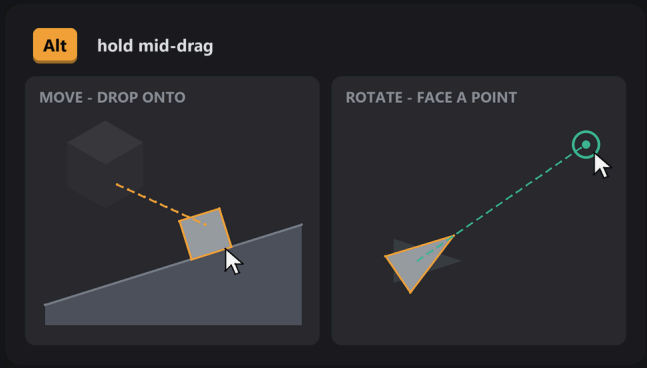


Figure 11.3 The aim modifier: Move drops onto a surface, Rotate faces a point.

Gesture	Default	Where to change it
Precision drag (slow drag, half snap increment)	**Shift**	All Tools -> Precision Mode
Aim at surface (Move surface snap, Rotate look-at)	**Alt**	All Tools -> Aim at Surface
Plane lock during numeric input	**Shift**	All Tools -> Numeric Input
Orbit axis snap	**Alt**	Orbit Selected -> Snap View to World Axis
Quick Roll clutch	**Alt** + Middle Mouse	Orbit Selected -> Quick Roll
Box Select add / remove	**Shift** / **Ctrl**	Unity's Shortcuts window: *Scene View/Add Box Select* and *Invert Box Select*

Labels are platform-correct: macOS users see Cmd and Option rather than Ctrl and Alt.

Unity's **Ctrl** for grid snapping works throughout. Vertex snapping is Blendon's own, on **V** (hold) and **Shift + V** (pick a virtual pivot). The Editor ships its own pair on those same keys, so this is a contested pair like the others. On the **Blendon** preset the Editor's two are switched off outright rather than moved aside, because Blendon's is the same mode with the Editor's limits removed, and a second key for the lesser of two vertex snaps teaches nothing. On the **Unity** preset Blendon's two move instead, to **Alt + V** and **Shift + Alt + V**. The Overview page switches between the two.

12 Default shortcuts

Action	Default binding
Orbit Selected	Hold Middle Mouse
Quick Roll	Alt + Middle Mouse, flick a direction

Action	Default binding
Camera Pan	Shift + Middle Mouse
Zoom	Scroll Wheel
Zoom In / Out	Numpad + / Numpad -
Move Gizmo	W (Move tool)
Rotate Gizmo	E (Rotate tool)
Scale Gizmo	R (Scale tool)
Transform Gizmo	Y (Transform tool)
Grab / Grab Rotate / Grab Scale	G / R / S
Vertex Snap	Hold V
Pick Virtual Pivot	Shift + V
Frame Selected	Numpad .
Snap To Floor	End
Isolate View	Numpad /
Box Select	Left Mouse drag (off until switched on)
Box Select: Visible Only	C
Box Select: Move Box	Hold Space while dragging
Select Parent / Children	[/]
Add Parent / Children to selection	Shift + [/ Shift +]
Draw Mode Pie	Hold Z, flick a direction
Toggle Wireframe	Shift + Z
Add Object Pie	Hold Shift + A
Snapping Pie	Hold Shift + S
Orientation Pie	Hold ,
Pivot Point Pie	Hold .
Reset Pie	Hold /
View Pie	Hold `
Tools Pie	Hold Q

Action	Default binding
Custom Pies 1 to 8	Unbound
Context Menu	Right-click, without dragging
Editor’s own context menu	Shift + Right-click
Selection History Back / Forward	Mouse 3 / Mouse 4
View History Back / Forward	Shift + Mouse 3 / Shift + Mouse 4
View Front / Back	Numpad 1 / Ctrl + Numpad 1
View Right / Left	Numpad 3 / Ctrl + Numpad 3
View Top / Bottom	Numpad 7 / Ctrl + Numpad 7
Toggle Perspective/Ortho	Numpad 5
Orbit Up / Down / Left / Right	Numpad 8 / 2 / 4 / 6
Reverse View	Numpad 9
Align To Camera	Numpad 0

All of these except Shift + right-click are rebindable, either from Blendon’s settings window or from Unity’s own **Edit -> Shortcuts** window under the `Blendon/...` entries. Disabling a feature clears its bindings from the Shortcut Manager so that they cannot collide with anything else, and restores them when you re-enable it. Any Editor command that one of those keys had displaced gets its own key back while the feature is off, and moves aside again when the feature comes back on.

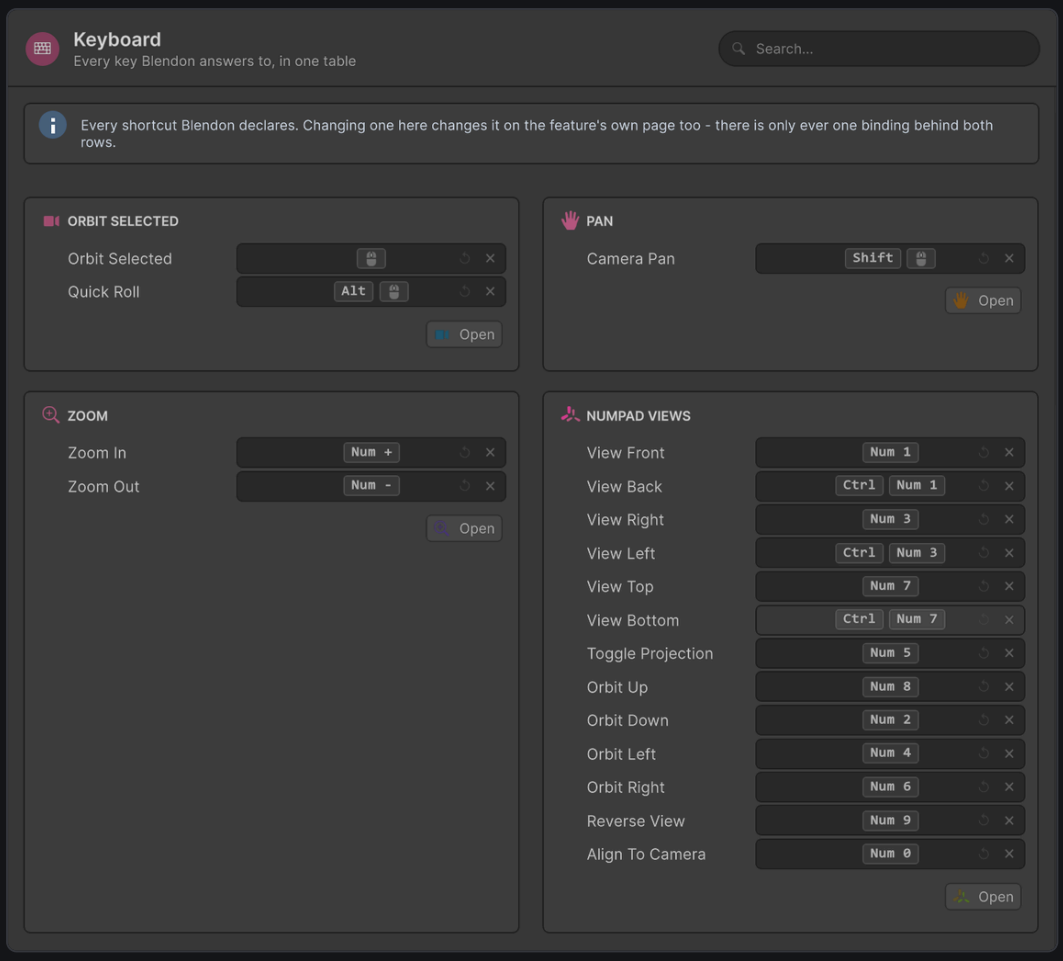


Figure 12.1 The Keyboard page lists every shortcut Blendon declares, grouped by feature.

13 Settings window

Open the settings window from **Tools -> Blendon**.

The sidebar groups every feature into **Get Started**, **Navigating**, **Transform Tools**, **Scene Tools** and **Keys & Menus**. Each feature page can enable or disable its feature, rebind its shortcut and tune its options. The search field at the top finds a specific setting, and a per-page **Reset to Defaults** button restores defaults, including clearing shortcut overrides. Changes apply immediately and persist across sessions in `EditorPrefs`.

Sidebar group	Pages
Get Started	Blendon (the Overview page)
Navigating	Orbit Selected, Pan, Zoom, Numpad Views, Orientation Gizmo
Transform Tools	All Tools, Move, Rotate, Scale, Transform
Scene Tools	Frame Selected, Isolate View, Snap to Floor, Navigation History, Box Select

Sidebar group	Pages
Keys & Menus	Keyboard, Pie Menus, Context Menu

13.1 Overview page

This is the page the window opens on, shown in the setup chapter.

- **Master switch.** Turns every Blendon feature off at once, parks all of its shortcuts and hands the Editor back every command Blendon had displaced, so that the Editor behaves as if the plugin were not installed rather than as if it had been half-removed. Nothing is lost: every setting is exactly where you left it when you switch back on. Unity rebuilds its whole shortcut directory for each binding that moves, so the keys are handed over across the following frames instead of freezing the Editor, and the page reports how far that has progressed while it runs.
- **Presets.** Three presets for a fresh install: **Full**, **Essentials** and **Tools Only**. See Choosing a starting preset.
- **Keyboard Presets.** See Keyboard presets.
- **Feature list.** One row per feature, grouped by category, with the pie menus and the context menu together in a **Menus** group. Each row has an on/off toggle and its current key, highlighted if it clashes with a Unity command. A feature that answers to more than one key lists them all underneath its row, as the pie menus do. Clicking a feature's name jumps to its settings page.
- **Tutorial.** Shows how far the Scene tutorial has progressed, with **Start Over**, the **Shortcut Tips** switch and **Show Again**.
- **Export Settings / Import Settings.** Save every Blendon setting, shortcut binding and pie menu to a single JSON file, or load one back, either as a backup or to carry your setup to another machine.
- **Defaults.** The values the rest of Blendon falls back to; see Defaults.
- **Reset All Pages.** Puts every setting back to what the plugin ships with.

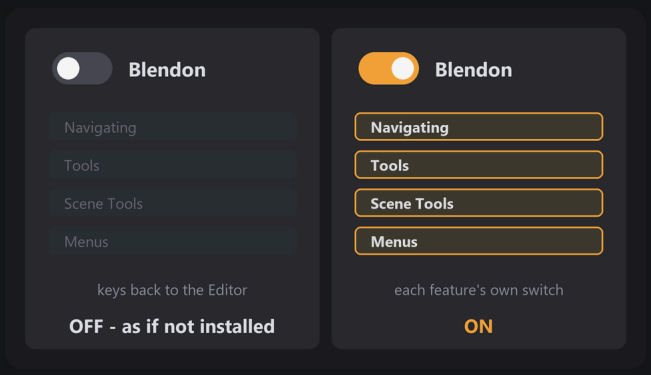


Figure 13.1 The master switch: off hands every key back, as if Blendon were not installed.



Figure 13.2 Export Settings: settings, shortcuts and pie menus in one file.

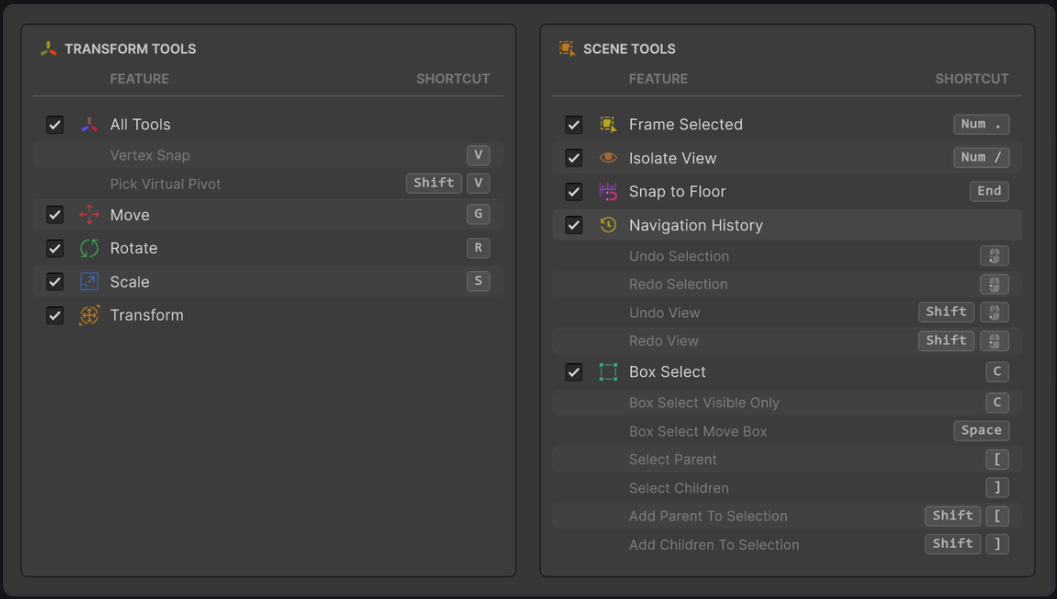


Figure 13.3 The feature list on the Overview page. Rows under a feature list its extra keys.

13.2 Defaults and overrides

A few settings matter to many features at once: animation, dragging, cursor wrap and the axis colors. They are set once, in the **Defaults** panel at the bottom of the Overview page, and every page that carries the same setting shows the default greyed out, with an **Override** box beside it. Tick **Override** to give that one page its own value; changing a default changes it everywhere that has not been told otherwise.

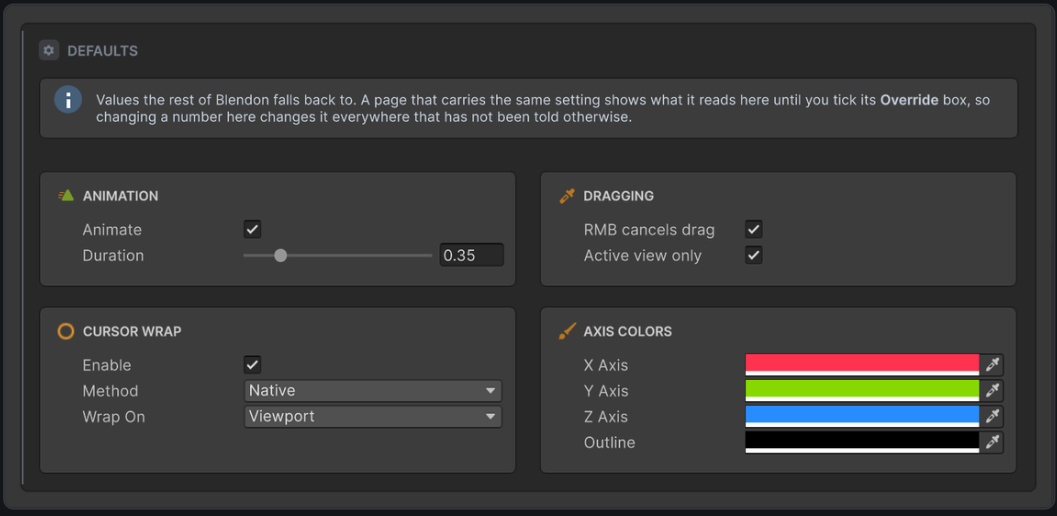


Figure 13.4 The Defaults panel. Changing a value here changes it on every page that does not override it.

Default	Value	Effect
Animate	On	Camera moves (numpad views, the orientation gizmo, frame, history) fly instead of cutting.
Duration	0.35 s	How long each of those flights takes.

Default	Value	Effect
RMB cancels drag	On	A right-click during any Blendon drag abandons it and restores the selection.
Active view only	On	A drag stays in the Scene view it started in instead of being mirrored into every open Scene view.
Cursor Wrap	On	Lets drags run past the edge of the view.
Method	OS Native	**OS Native**: Blendon teleports the cursor to the opposite edge itself. **Unity**: the Editor's own edge jumping. **Clamp To View**: no teleport, and the drag stops at the border.
Wrap On	Scene View Viewport	Where OS Native wraps: the Scene view's 3D area, the whole **Unity Window**, or the **Screen Boundary**.
Axis Colors	Blender palette	X, Y, Z and the outline, shared by every gizmo and the orientation gizmo.

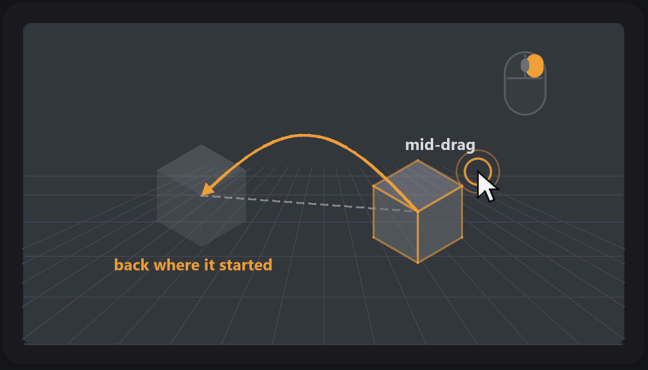


Figure 13.5 RMB cancels drag: a right-click puts the object back where it started.

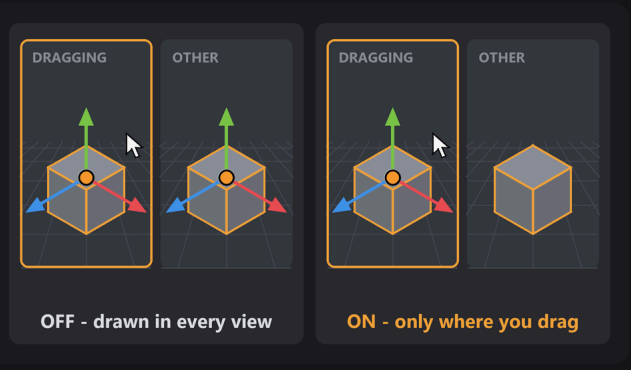


Figure 13.6 Active view only: the gizmo is drawn only in the view you drag in.

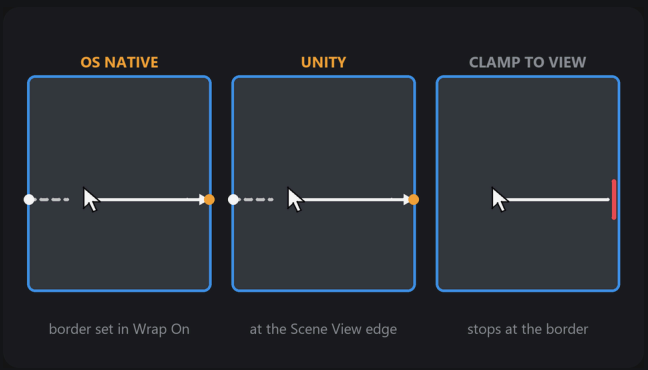


Figure 13.7 The three wrap methods: OS Native, Unity and Clamp To View.

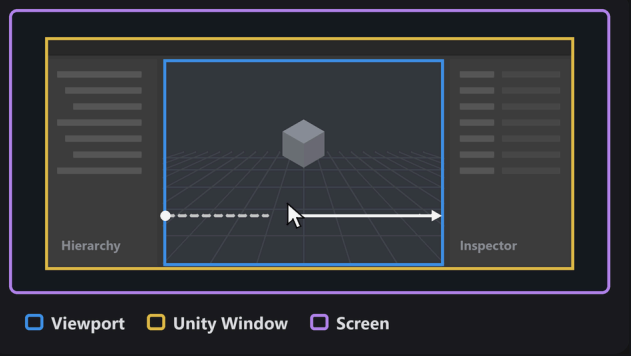


Figure 13.8 Where OS Native wraps: the viewport, the Unity window or the screen.

13.3 Keyboard page

Every shortcut Blendon declares, in one table grouped by the feature that owns it; see Default shortcuts. Each row edits the very same Shortcut Manager entry that the feature's own page edits: there is one binding behind both rows, never a copy. A count of how many bindings clash with something else sits at the top, and a switched-off feature collapses to a single line saying that its keys are parked and belong to the rest of the Editor until it comes back on.

13.4 Keyboard presets

Blendon and Unity both want some of the same keys: G, R, S, Z, Q, V, the backtick, the numpad views and the middle mouse button. The Overview page's **Keyboard Presets** picker settles that, Editor-wide:

Option	Effect
Blendon (default)	Blendon keeps its Blender-style keys. Editor commands that wanted the same ones move to a free binding instead, or, where the Blendon command is the same thing without the Editor's limits (vertex snapping, middle-mouse pan), are switched off rather than duplicated.
Unity	The Editor keeps every key untouched and Blendon's gestures move instead: orbit on **Ctrl + MMB**, quick roll on **Ctrl + Shift + MMB**, pan on **Alt + MMB**, vertex snapping on **Alt + V**, and so on.

Each option is a real Unity Shortcut Manager profile, so settings tuned on either side persist independently when you switch back and forth. The picker reads back as **Custom** whenever the Editor is on some other profile.

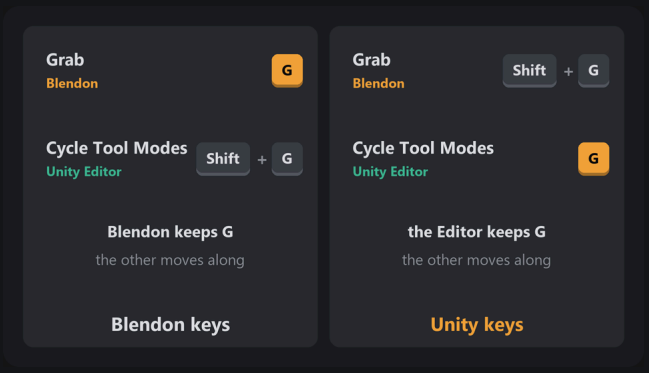


Figure 13.9 One contested key: Blendon or the Editor keeps it, and the other moves along.

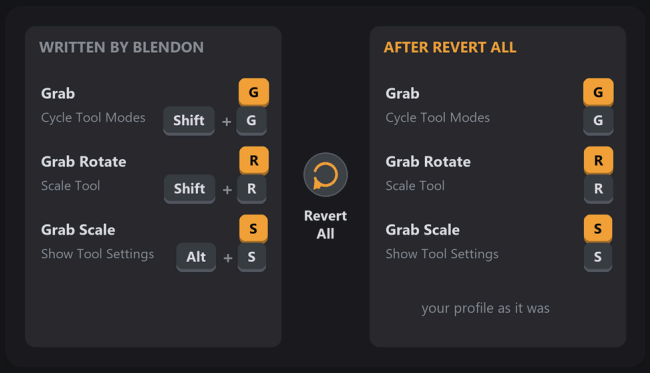


Figure 13.10 Revert All puts your profile back as it was.

Below the picker, the contested-keys table lists every clashing pair side by side, as shown in Step 2: which command currently holds the key and which was moved elsewhere, with a per-row reset button to hand back just that one key. An **Apply Here** option settles the clashes on whatever profile the Editor is already using, instead of switching to one of Blendon's own two, and **Revert All** undoes every key Blendon has written into the active profile.

If one of Blendon's shortcuts clashes with something outside this system, for example after a binding has been hand-edited, the settings window still flags it inline wherever that shortcut appears, naming the conflicting command and offering to open Unity's Shortcuts window. Unity offers no public way to filter that window, so Blendon copies the command's name to the clipboard and says so; paste it into the window's search box.

14 Data and files

Blendon is Editor-only and adds nothing to player builds. It writes in three places, all of them yours to delete.

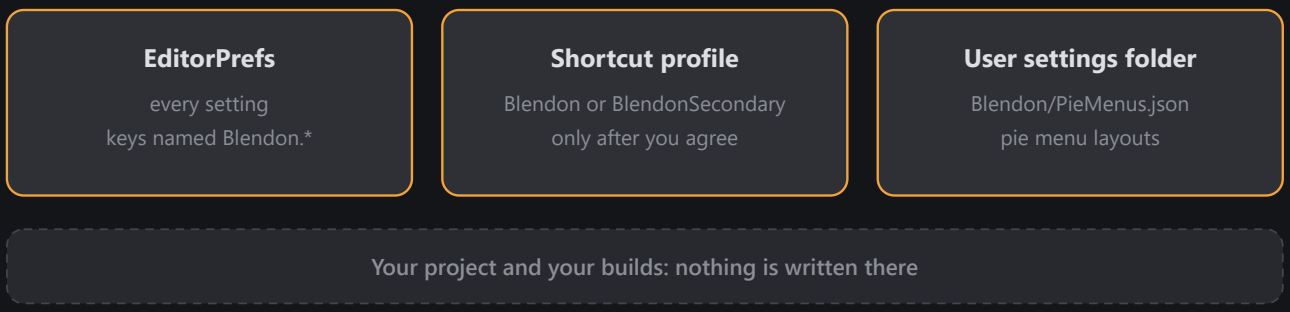


Figure 14.1 Everything Blendon writes, and where.

- **EditorPrefs.** Every setting in the settings window, under `Blendon.*` keys.
- **Unity's shortcut profiles.** The `Blendon` profile, or `BlendonSecondary`, and only after you have agreed to it. Switching back to `Default` in **Edit -> Shortcuts** puts everything back.
- **Your user settings folder.** Pie menu layouts, in `Blendon/PieMenus.json` under `%APPDATA%` on Windows, `~/Library/Application Support` on macOS, and `$XDG_CONFIG_HOME` (or `~/.config`) on Linux. These are kept outside the project on purpose: a menu edited once is edited for every project on the machine, which is the same scope as the EditorPrefs the rest of the settings use.

Blendon makes no network requests, collects no analytics and installs no packages.

15 Troubleshooting

A shortcut does nothing. Open **Tools -> Blendon** and check the feature's row on the Overview page. If the row is switched off, its keys are parked and belong to the rest of the Editor, including any Editor command they had displaced. If the key is highlighted, it clashes with another command; see Keyboard presets.

The numpad viewport shortcuts do nothing. NumLock must be on. The Numpad Views settings page warns inline when it is off.

Pressing R scales instead of starting a grab rotate. R is contested between Blendon's Grab Rotate and Unity's Scale tool. Select the **Blendon** keyboard preset on the Overview page, or rebind either command from the Keyboard page.

Unity's own gizmo is drawn instead of Blendon's. That tool is disabled. A disabled transform tool deliberately falls back to Unity's own gizmo. Switch it back on from its settings page or from the Overview list.

The Pivot/Center dropdown shows only Unity's two entries. Pivot Point Menu is off on the All Tools page, or every Blendon transform tool is switched off. The four pivot modes only matter to Blendon's gizmos, so Unity keeps its own menu in both cases.

Right-clicking the Scene view opens Unity's menu instead of Blendon's. Check the **Context Menu** row on the Overview page, and that the master switch is on. A right-click with **Shift**, **Ctrl**, **Alt** or **Cmd** held always goes to the Editor's own menu. If the `Blendon/Context Menu` binding was changed in Unity's Shortcuts window, the plain right-click belongs to the Editor again; press **Reset to Defaults** on the Context Menu page to restore it.

An entry is missing from the context menu. Blendon's menu holds exactly what the Editor's own menu holds for the current selection and active tool. Select the object the entry belongs to, or switch to the tool that adds it, and right-click again.

Dragging over the Scene view draws Unity's box, not Blendon's. Box Select ships switched off; check its row on the Overview page. Terrain, particle systems, UI and gizmo icons are selected only when the box contains them wholly, as in the Editor.

Box Select misses an object that is behind another one. Visible Only is on, which is the default. Press **C** while dragging to include hidden objects, or switch it off on the Box Select page.

Vertex snapping finds no vertices. Check that the objects you are snapping to have a MeshRenderer or SkinnedMeshRenderer and are not hidden by Scene Visibility or excluded by Isolate View.

I want everything back the way it was. Use the **Master switch** on the Overview page. It parks every Blendon shortcut and hands back every command Blendon had displaced, without losing any of your settings.

Something is still wrong. See Support, and include your Unity version and your platform.

16 Uninstalling

Blendon writes nothing into your project, so removing it is a matter of deleting its folder. Three things outlive the folder, though, and all three are undone from inside Blendon, so do this first, while it is still installed:

1. **Shortcuts.** Open **Edit -> Shortcuts** and select the **Default** profile. Every key Blendon moved goes back at once. This still works after the folder is gone, but the `Blendon` profile is then left behind holding bindings that answer to nothing.

2. **Settings.** Open **Tools -> Blendon**, go to the Overview page and click **Reset All Pages**. This puts every setting back to what the plugin ships with. Settings live in `EditorPrefs`, which is machine-wide rather than per-project, so a leftover set would otherwise be picked up by the next project you install Blendon in.
3. **Pie menus.** Delete `Blendon/PieMenus.json`, and the `.bak` file beside it, from your user settings folder: `%APPDATA%` on Windows, `~/Library/Application Support` on macOS, or `$XDG_CONFIG_HOME` (or `~/.config`) on Linux. This file is kept outside the project on purpose, so that a reinstall finds your own menus where you left them; delete it only if you want them gone for good.

Nothing else is left anywhere: no files inside the project, no registry entries, and nothing that was ever part of a player build.

17 Support

Questions, bug reports and feature requests: blendon@drygiel.com

Please include your Unity version and your Editor platform. Most of what Blendon touches (shortcut profiles, the Scene view overlay set, the native cursor and the NumLock read) differs between Windows, macOS and Linux, and between Unity 6.0 and the current release.